CHAPTER 12

Architecture Development Method



This chapter is part of the PINK, GREEN, and BLUE reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

The Architecture Development Method is a key component of the TOGAF Standard. It is the heart of the framework as it provides a comprehensive and iterative approach for developing and managing Enterprise Architectures. It is considered a method that enables Enterprise Architects to develop and implement Enterprise Architecture.

The Architecture Development Method delivers an iterative approach by providing guidance for creating and maintaining architecture artifacts and ensuring alignment with business goals. But don't be fooled by what seems to be a waterfall approach at first sight.

The Architecture Development Method describes all the actions one can take, not what must be taken. It also does not prescribe the order of these steps. The TOGAF Standard explicitly states that the Architecture Development Method is to be tailored so that it (referring to the steps and its content) fits an organization's needs.

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12.1. The Heart of the TOGAF Standard

All the theory described in the TOGAF Standard can be put into practice by executing the Architecture Development Method. This method – still too often mistaken for a process – provides tools for creating and developing a sound Enterprise Architecture.

The Architecture Development Method (see Figure 12-1) consists of ten phases, each of which has its own characteristics and provides a step-by-step approach to implementing or approaching a particular part of the architecture work. It should be noted that the activities described for each phase are *in no way prescriptive*. All steps described should be considered as *possible steps* to be taken. Depending on the situation or activities, steps may be skipped or adapted as the situation requires. The Architecture Development Method is therefore highly configurable and adaptable and can be used in almost any situation.

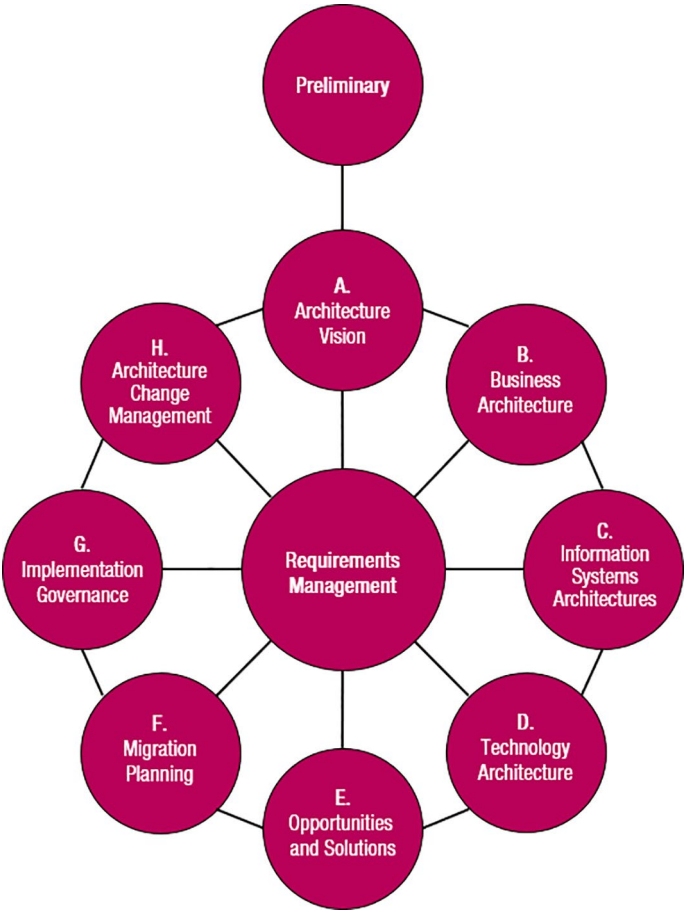


Figure 12-1. *The Architecture Development Method*

The ten phases of the Architecture Development Method each have their own distinct objectives:

- **Preliminary Phase:** The first phase of the Architecture Development Method has two unique objectives. One is to *determine the Architecture Capability* that the organization believes it needs, and the other is to *establish it* in that form.

- **Phase A (Architecture Vision):** The main goal of the Architecture Vision Phase is to *establish a high-level vision of the architecture* and to *ensure stakeholder alignment*. It focuses on creating a shared understanding of the architecture's goals and ensuring that the business requirements are clearly articulated and agreed upon by all stakeholders.
- **Phase B (Business Architecture), Phase C (Information Systems Architectures), Phase D (Technology Architecture):** The objectives of phases B, C, and D are twofold. The first one is to *craft the Target Architecture* to outline how the organization should function to meet business objectives. The second objective is to *identify potential components for the Architecture Roadmap*, pinpointing the gaps between Baseline and Target Architectures that must be bridged.
- **Phase E (Opportunities and Solutions):** This phase contains three key objectives. The first is to *identify the work packages* required to realize the Target Architecture. The second objective is to *determine whether an incremental approach is required* to achieve the first objective. The third and final key objective is to *define the building blocks* needed to complete the Target Architecture.
- **Phase F (Migration Planning):** The primary objective of Phase F (Migration Planning) is to *create a detailed plan* for the implementation of the architecture.

- **Phase G (Implementation Governance):** The primary objectives of Phase G include *ensuring adherence to the established architecture*, *managing any changes* during implementation, and *continually assessing progress* to make necessary adjustments to meet the organization's objectives.
- **Phase H (Architecture Change Management):** The key objectives of the Architecture Change Management Phase are to *continuously assess changes* in the organizational strategy, business environment, and technology landscape that may impact the architecture.
- **Requirements Management Phase:** Although not considered a separate phase, Requirements Management is a continuous activity that runs throughout the Architecture Development Method. It involves *managing and maintaining requirements* and ensuring that they are properly addressed during each phase.

During the development of architecture, it is important to be able to visualize certain components. As the saying goes, *a picture is worth a thousand words*, and visualizations are often an effective way to convey a message. Typically, an image immediately clarifies the intended change or implementation.

Artifacts are used to visualize the architecture. Several of these artifacts are created during the phases of the Architecture Development Method. It is important to remember that these artifacts must add *value* to the organization, the project, or architecture in general.

The artifacts shown in this chapter are examples based on the fictional healthcare organization that was described in Chapter 5.

The artifacts need to be adapted to your own organization, situation, and environment.

Each phase of the Architecture Development Method consists of a set of fixed components:

- Key objectives per phase
- Steps to be performed during the phase
- Outputs generated by the phase
- Approach to performing the phase

The subject matter of the phases largely determines the amount and form of information that will serve as input to the next phase.

12.2. Architecture Document Creation

During the implementation of the Architecture Development Method, a number of architecture documents are created. These documents identify the various phases of the work and describe the agreements and outline the architecture work. They do not contain specific details of the architecture or the work done in specific phases of the Architecture Development Method:

- **Request for Architecture Work:** The initiation of an architecture development cycle requires a formal document from the sponsoring organization to the architecture team. It's important that this document maintain a high-level perspective, providing high-level guidance without delving into granular details.

- **Statement of Architecture Work:** Defining the scope and approach of an architecture development cycle is critical. The Statement of Architecture Work does just that. It serves as a benchmark for measuring the success of the architecture project and can even form the basis of a contractual agreement between the provider and consumer of architecture services. It typically includes the architecture project description and scope, an overview of the architecture vision, and acceptance criteria and procedures. In addition, the Statement of Architecture Work may include the architecture project plan and schedule and formal approvals.
- **Architecture Definition Document:** The Architecture Definition Document serves as a comprehensive repository for all key architectural artifacts generated throughout a project, encompassing essential related information. It spans all architecture domains (Business, Data, Application, and Technology), while also addressing all relevant architectural states (Baseline, Transition, and Target).
- **Architecture Contract:** Architecture Contracts serve as formal agreements between development partners and sponsors that describe the deliverables, quality standards, and overall suitability of an architecture. Their successful execution depends on robust architecture governance mechanisms.

Figure 12-2 visualizes the creation of these documents.

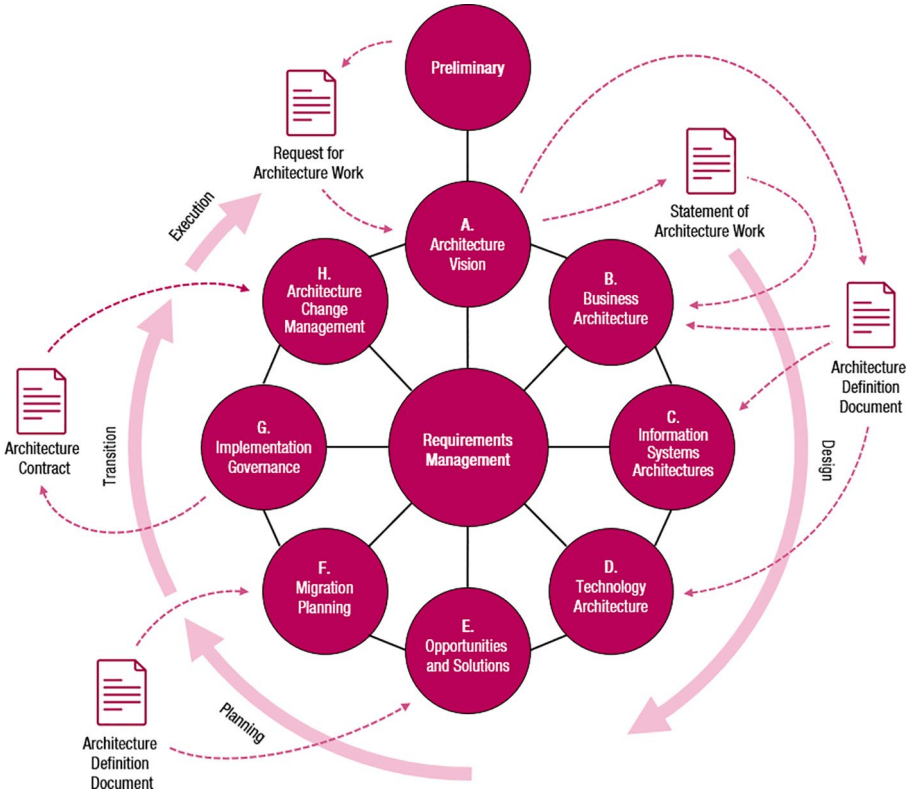


Figure 12-2. *Document creation during the execution of the ADM*

The Preliminary Phase initiates the architecture work and captures it in a *Request for Architecture Work* document, to the extent that it can be identified at this stage.

The Request for Architecture Work document is sent from the sponsoring organization to the architecture organization to trigger the start of an architecture development cycle [11].

During Phase A (Architecture Vision), a high-level model of the candidate building blocks is created. These are captured in a *Statement of*

Architecture Work document, which serves as input to the *design phases*, leading to the creation of the Baseline and Target Architectures. Phases B, C, and D, the Business Architecture, Information Systems Architectures, and Technology Architecture, respectively, are then completed.

The Statement of Architecture Work is created as a deliverable of Phase A and is effectively a contract between the architecting organization and the sponsor of the architecture project [11].

Reference models, viewpoints, and tools are selected, and a Baseline Architecture Description is developed for each architecture domain. This is a high-level model of the existing building blocks, reusing definitions from the Architecture Repository where available. During phases B, C and D, the *Architecture Definition Document* is created and supplemented by each Baseline Architecture Definition from the corresponding phase.

The Architecture Definition Document is the deliverable container for the core architectural artifacts created during a project and for important related information [11].

After the Baseline Architecture Description is developed, the Target Architecture Description is created. This is a more extensive process because it involves defining and developing an architecture that does not yet exist.

A view of the required building blocks is developed by creating artifacts such as catalogs, matrices, diagrams, and maps. Each building block is documented as comprehensively as possible, accompanied by a clear rationale. The decisions that led to that building block are explained. All required building blocks are identified and validated against those in the Architecture Repository. As much as possible is reused.

The reference models from the different continua are used to select standards.

Finally, the definitive building blocks selected or to be created are recorded in the Architecture Landscape. Building blocks that will serve as new standards in the future – after the Target Architecture has been implemented – are recorded in the Architecture Repository.

After the desired Target Architecture is determined, a gap analysis is performed. This determines which building blocks can be reused, which need to be built from scratch, and which will be removed from the Architecture Repository because they are no longer needed. Following the gap analysis, the Architecture Roadmap is created based on the components identified. Any deviations in the Architecture Landscape are addressed and resolved at this point.

All steps and building blocks to be created are coordinated with stakeholders through formal stakeholder reviews, after which the architecture is finalized. The *Architecture Definition Document* is now completed. In Phase E, the document is formally approved for the defined Target Architectures and includes draft Transition Architectures. The latter are approved in Phase F, where the document is finalized.

At the start of Phase E (Opportunities and Solutions), an overview of all the missing building blocks is produced. These are linked to work packages that will ultimately ensure that the missing building blocks are created. This is also when the *planning phase* begins, during which the proposed solutions are further developed. The final solution selected is documented in an *Architecture Contract*. The creation of this document initiates the *transition phase* and the governance process.

Architecture Contracts are the joint agreements between development partners and sponsors of the deliverables, quality, and fit for purpose of an architecture [11].

The *Architecture Contract* is also used during the *execution phase* where the change process is initiated. If the change process results in the need for (major) changes to the already developed architecture, a new *Request for Architecture Work* document must be created. This starts a new iteration of the Architecture Development Method.

The above description of the implementation of the Architecture Development Method is brief and concise. The following sections describe each phase in more detail.

12.3. Preliminary Phase

The Preliminary Phase is the first phase of the Architecture Development Method. It is all about getting the organization ready for architecture development. This is done by *aligning stakeholders*, *defining principles*, and *tailoring the TOGAF Standard*. The Preliminary Phase sets up the foundation for the architecture development process, by ensuring that the necessary frameworks, tools, and governance structures are in place before progressing into the more detailed architectural phases.

Skipping or inadequately executing this phase could result in misalignment between the architecture and business goals or poor management of the architecture initiative.

The Preliminary Phase is a crucial phase for ensuring that the architecture initiative is positioned for success right from the start.

12.3.1 Determining and Establishing the Architecture Capability

The Preliminary Phase of the Architecture Development Method has two distinct objectives. One is to *determine the Architecture Capability* that the organization believes it needs, and the other is to *establish it* in that form.

Essential to these objectives is to determine the organization's current architecture maturity and the desired target maturity of the capability.

It also includes assessing the organization's readiness for architectural changes and improvements.

The *first objective*, which entails the definition process, includes determining the scope of the architecture. This could be organization-wide, focused on a particular business unit, or addressing a specific concern. Part of defining the Architecture Capability is to identify the stakeholders and determine their requirements. This will inform how the architecture will be developed to meet those requirements. Tables 12-1 and 12-2 list the steps to perform the Preliminary Phase.

Table 12-1. *Steps to perform the Preliminary Phase*

Step to Perform	Description of Step	Output Generated by Step
Define the enterprise	By defining the organization, the scope of the impacted organization is determined, allowing architects to create an understanding of the organization and its structure.	Scope of impacted organization(s)
Identify key drivers and organizational context	Pinpointing the critical factors that influence the architecture, such as key drivers and the organizational context.	Baseline of objectives, goals, and drivers
Manage relationships between frameworks	Understanding how various management frameworks interact with each other and creating a framework to manage the architecture activities.	Enterprise Architecture Capability's structure Architecture Governance Framework
Assess Enterprise Architecture maturity	Evaluating where the organization stands in its architectural development.	Maturity assessments, identified gaps, and strategies to address them

Define the Enterprise: By defining the organization, the scope of the impacted organization is determined, allowing architects to create an understanding of the organization and its structure. Scoping the organization entails identifying stakeholders. By stakeholders, the TOGAF Standard refers to those who are most affected by and achieve most value from the architecture work, those who will see change to their work (but are otherwise not directly affected by the architecture effort), and external stakeholders outside the organization.

Make an initial assessment of the stakeholders who are or are likely to be involved in the work. Bring them together by scheduling a meeting and discussing the scope of the work during the meeting. This often leads to the recruitment of additional stakeholders. After the consultation, contact the new stakeholders and review the work and scope of the impacted organization with them as well. Over time, an overview of the actual stakeholders involved will emerge. For this particular step, it is important to understand the concept of Stakeholder Management. Since this is an essential topic that will find its way into other steps of the Architecture Development Method, it is described in more detail in Chapter 14, Section 14.3.



Scope the Organization: Start by figuring out which part of the organization will be affected. This helps architects understand how the organization works and its structure. Identify who the key players are.

Identify Stakeholders: Begin by assessing the stakeholders involved in the project. Schedule a meeting with them to discuss the work and its scope. Afterward, reach out to them and review the project scope.

Manage Stakeholder Involvement: Understand the importance of managing the stakeholders effectively. Keep communication flowing as the project evolves, and stay flexible to ensure all relevant voices are heard and accounted for.

Identify Key Drivers and Organizational Context: Since this is the first phase of the Architecture Development Method, there are no other phases that can provide input. However, the Preliminary Phase does take input from the TOGAF Library (see Chapter 3, Section 3.4) and other architecture frameworks as appropriate.

Organizational strategies and business plans can also provide important input. These strategies may include a business and/or IT strategy, business principles and objectives, goals, and drivers. Scan these types of documents for the presence of any concrete drivers, goals, or objectives. Distill them from the documents and verify them with stakeholders.



Understand the Organizational Needs: In this first phase, there are no previous phases to provide guidance. However, useful information can be pulled from resources like the TOGAF Library and other frameworks. Also, look at the organization's business and IT strategies, goals, and plans to find important factors like drivers or objectives.

Gather and Analyze Documents: Scan business documents for key points about the organization's needs and goals. Look for concrete goals or objectives, and carefully go through them with stakeholders. Make sure to understand the true focus of the business.

Verify with Stakeholders: Once these key drivers are identified, verify and clarify them by discussing them with the relevant people in the organization. This ensures everyone is on the same page and supports the findings.

Some time ago, I worked for a hospital. This healthcare organization had done little with Enterprise Architecture prior to my arrival. There was a strategy, but it consisted mostly of stacks of documents trying to interpret the hospital's strategic direction. The pages of documents were full of information – much of it irrelevant – and were not easy for the hospital's staff to read.

I manually went through all of these strategic documents, looking for clues that said something about drivers, goals, and objectives. I spent many hours interpreting the motivational elements and summarizing them so that they could be used in the context of the architecture work.

In the end, I was able to distill a set of drivers, goals, and objectives from the strategically colored documents and visualize them in a diagram (see Figure 12-3).

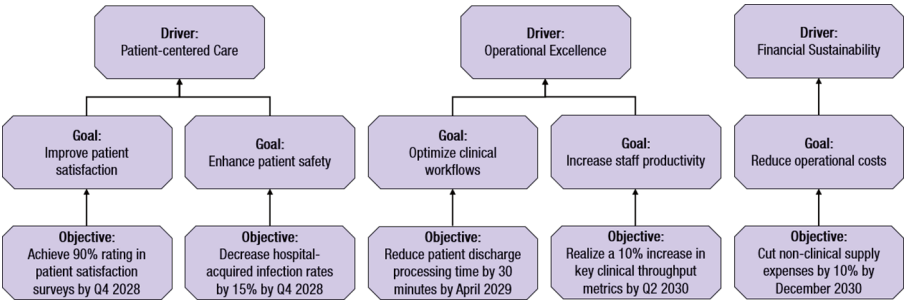


Figure 12-3. *Driver/Goal/Objective Diagram*

Later, I coordinated the drivers, goals, and objectives with the stakeholders (consisting mainly of senior management) to add the requirements and concerns of these stakeholders to the previously created drivers, goals, and objectives diagram. All the elements in the diagram were also given descriptions so that there would be no ambiguity as to what they meant. Over time, a Solution Concept Diagram emerged.

Today, some organizations are experimenting with using AI to do this manual and especially time-consuming work. AI tools can scan these strategic documents for keywords and then categorize them incredibly quickly. This makes it easy to find and translate an organization's strategy.

Manage Relationships Between Frameworks: In addition to the reference materials from the TOGAF Library, several types of pre-existing models for operating Enterprise Architecture can be used as a baseline for the Preliminary Phase. This may include existing maturity assessments, defined roles and responsibilities for architecture teams, and established budget requirements. Existing architecture frameworks may consist of methods or content, tools, principles, and even (parts of) a repository. It is important to go through the existing material with a fine-toothed comb and delete what is outdated or obsolete. It is not intended to be a one-to-one copy of all existing material. Other relevant frameworks, such as those used for project and portfolio management, also provide valuable insight for the Preliminary Phase.

In order to manage the architecture activities, an Architecture Governance Framework needs to be established. This framework ensures that architecture activities are aligned with organizational goals. In the Preliminary Phase, a Governance Framework focuses on how architectural materials (such as standards, models, and reports) are managed. This includes defining repository requirements and identifying governance processes (compliance, retirement, etc.). To implement this framework, existing governance models may need to be modified. Stakeholders need to evaluate current models and understand potential impacts, ensuring that all touch points and changes are identified and aligned across the organization. Bring the models to stakeholder meetings and agree on who will review the models for completeness and accuracy and when. After receiving feedback, adjust the models and have them reviewed again. Have stakeholders re-verify the accuracy of the models.

It is often forgotten that the stakeholders play a significant role in evaluating the architecture.

The decisions to be made in architecture are not the responsibility of the architect, but of the stakeholders. It's the stakeholders who make the decisions based on the architect's input.

Involving stakeholders in a timely and appropriate manner creates support for the architectural work. At this point, too, Stakeholder Management is crucial.



Review Existing Frameworks: Start by reviewing any current frameworks or models that the organization already uses, like maturity assessments, architecture roles, or budget requirements. Don't just copy these materials – scrutinize them closely and remove outdated or irrelevant parts.

Use Relevant Models: Consider other frameworks, such as those for project management, as valuable input during the Preliminary Phase.

Set Up Architecture Governance: Establish a Governance Framework that aligns architecture activities with the organization's goals. This includes managing architecture materials (standards, models, reports) and setting repository and governance requirements.

Evaluate and Adjust Governance Models: Assess the existing governance models. Involve stakeholders to identify any necessary changes and impacts, making sure everything is aligned. Discuss these changes in meetings, assign review responsibilities, and set deadlines for revising and validating the models.

Assess Enterprise Architecture Maturity: Evaluate where the organization stands in its architectural development. It is important to assess the current organization and business capabilities that are needed to understand how to work with architecture. A maturity assessment can help accomplish this.

In the "Architecture Maturity Models" Series Guide (see also the *Further Reading* information box at the end of this section), the TOGAF Standard refers to two applicable methods for measuring architecture maturity. These are the Architecture Capability Maturity Model (ACMM) framework and Capability Maturity Model Integration (CMMI) framework.

The ACMM framework has a structure in which the core components of an architecture process are represented and measured, often in relation to the organization's IT. The CMMI framework, on the other hand, was originally a standardized model for assessing and improving the processes

used in software development and other engineering disciplines. Today, the framework is also regularly used to indicate the maturity of an Enterprise Architecture because of its focus on achieving business goals.

Of course, there are alternatives to both ACMM and CMMI frameworks, and it is advisable to choose an approach that best suits the organization. Chapter 15 further details the use of architecture maturity models in combination with the TOGAF Standard.



Assess Maturity: Look at where the organization is in terms of its architecture. This means understanding what the organization is currently capable of and how ready it is to work with architecture. Knowing this helps to figure out what needs to be improved.

Use the Right Tools: The TOGAF Standard suggests using either the Architecture Capability Maturity Model framework or the Capability Maturity Model Integration (CMMI) framework. CMMI is a more appropriate framework for assessing Enterprise Architecture because of its emphasis on achieving business goals.

Choose the Best Fit: While CMMI is one option, there are other ways to assess the organization's maturity. It's important to pick the approach that works best for the situation.

Another method that helps evaluate where the organization stands in its architectural development is to understand the required readiness for change. A critical success factor for coordinating change requests lies with informing stakeholders of current architecture efforts, assessing the impact on their plans, and identifying and resolving common interests and conflicts. Chapter 14, Section 14.7, details how to apply a Business Transformation Readiness Assessment. This technique is used for evaluating and quantifying an organization's readiness to undergo change.

Returning to the objectives for the Preliminary Phase, the *second objective*, establishing the Architecture Capability, involves defining the contours of the Enterprise Architecture. This will be accomplished through the following focus areas.

Table 12-2. *Steps to perform the Preliminary Phase*

Step to Perform	Description of Step	Output Generated by Step
Establish the Enterprise Architecture team	Establishing an architecture team to perform the architecture work.	Architecture team Defined roles and responsibilities for architecture teams Constraints for architecture efforts
Set Architecture Principles	Defining the principles that will guide all architecture-related efforts.	Architecture Principles
Choose and tailor the architecture framework	Identifying and tailoring the TOGAF Standard, Architecture Development Method, and the Content Framework.	Tailored framework Tailored Architecture Development Method Tailored Content Framework

Establish the Enterprise Architecture Team: Establishing an Enterprise Architecture team is one of the first things that needs to be done to meet the requirements of working with architecture. The best results are achieved by creating a team of dedicated members. An architecture team does not have to consist only of architects. Project managers, senior management, or even other people from the organization can be part of the architecture team. However, it is important that all members are able to grasp the essence of architecture before joining the team.

The composition of the architecture team is largely related to the positioning of the Architecture Capability in the organization. If the Architecture Capability – as is still the case in many organizations – is positioned within the IT department, then the architecture team will mainly consist of one or more architecture functions, supported by technical experts. On the other hand, if the Architecture Capability is

positioned more strategically, for example, under the board of directors or a (strategic) executive, then the team may consist more of an Enterprise Architect, supplemented by members of senior management, project and portfolio managers, and other management functions.

"Architecture teams" are different from "teams consisting of people with the job title architect". An architecture team can consist of anyone involved in architecture-related work.

Regardless of the composition of the architecture team, the roles and responsibilities of all members must be defined so that there is no ambiguity. The most effective way to do this is to use a RACI matrix (see Table 12-3).

Table 12-3. *Example RACI Matrix for an Enterprise Architecture Team*

Role / Activity ▼	Enterprise Architect	Segment Architect	Solution Architect	Project Manager
Create architecture vision	AR	C	C	C
Develop Segment Architectures (Baseline and Target)	A	R	C	C
Define opportunities and solutions	A	R	R	C
Create migration plan	A	R	R	
Implement governance	AR	C	C	C
Manage architecture changes	A	R	R	C
Manage requirements	A	R	R	I

The RACI matrix shown in Table 12-3 is an example of how a matrix *might* look like when it is being set up for an architecture team. It lists the *activities* (or processes) in the left column and shows the *roles* involved in those activities in the top row. The cells of the table contain the ways in which each of the roles is involved in the activities. Chapter 8, Section 8.2, describes the structure of a RACI matrix in more detail.



Build the Architecture Team: Start by setting up the Enterprise Architecture team. This is a crucial first step when planning for architecture. The team doesn't have to consist only of architects. It could also include project managers, senior management, or anyone else in the company who gets the basics of architecture.

Define the Team Composition: The way the team is formed depends on where the Architecture Capability sits in the organization. If it's within the IT department, the team may focus on technical roles, like architects and experts. But if it's a broader, strategic initiative, the team could include senior managers and project leaders.

Clarify Roles and Responsibilities: Ensure that every team member knows their role. This removes confusion and makes things run smoothly. A great tool for this is the RACI matrix, which clarifies who's responsible, accountable, consulted, and informed.

Set Architecture Principles: Define the principles that will guide all architecture-related efforts. Architecture Principles need to be established to ensure consistency and alignment with strategic goals throughout the Architecture Development Method process. After thoroughly understanding the organizational context, it's crucial to define a tailored set of Architecture Principles that aligns with the organization's unique needs. This ensures a strong, adaptable framework that supports both governance and strategy. Defining Architecture Principles – and how they relate to and differ from Enterprise Principles – is described in more detail in Chapter 14, Section 14.2.4.



Understand the Big Picture: Before diving into setting principles, take a moment to understand the organization's goals and context. Knowing where the company is headed helps to align the principles with the overall mission.

Define Clear Architecture Principles: These principles will guide every architectural decision. Think of them as the rules of the game, ensuring that everything that's done supports the company's strategy and vision.

Customize for the Organization: Every company is unique, so make sure the principles reflect the organization's specific needs.

Choose and Tailor the Architecture Framework: Identify and tailor the framework that will be used. To be able to create an architecture in the first place, an architecture framework and appropriate tools must be decided upon. Because each organization has unique needs, the architecture framework must be tailored to those needs, and tailoring the TOGAF Standard is therefore an absolute must.

Tailoring is done by assessing how the framework can be adapted to the organization. This includes refining the terminology so that it's clearly understood throughout the organization. To do this, an agreed-upon glossary must be created to ensure consistency in architecture discussions. Publishing the glossary so that it is always available to all stakeholders will greatly assist in achieving and maintaining consensus.

The Enterprise Metamodel must then be adapted to accurately represent the organization. This is done by defining new entities, deliverables, and artifacts that are specific to the organization.

Next, the Architecture Development Method needs to be adapted. Start by removing redundant tasks, adding organization-specific checkpoints, or integrating it with other operational processes. Finally, the Content Framework must be customized to support the organization's unique needs. This is accomplished by customizing the structure of the framework.

Having an effective tools strategy is another necessity. Creating a plan that addresses the use of both general-purpose and specialized architecture tools allows for a focus on flexibility and scalability. Techniques such as business modeling need to be incorporated to meet the diverse requirements. Be sure to formalize change control to maintain consistency across all architectural activities.



Pick the Right Framework: First, decide which architecture framework to use. A framework and tools are necessary to get started. Choose one that fits the organization's needs and customize it.

Tailor the Framework: Adjust the framework to the organization. This involves changing the terminology so everyone understands it. Create and share a glossary to keep things consistent.

Adapt the Enterprise Metamodel: Modify the Enterprise Metamodel to fit the organization by defining new entities and removing unused ones.

Modify the Architecture Development Method: Simplify or add steps to the Architecture Development Method. Remove unnecessary tasks and integrate it with existing processes.

Customize the Content Framework: Change the structure of the Content Framework to better support the organization.

Choose the Right Tools: Plan how to use both general and specialized tools. Focus on flexibility and scalability, and include techniques like business modeling.

Control Changes: Make sure there's a formal process for managing changes to keep everything consistent.

Enterprise Architecture provides a strategic view of an organization from the top down and helps different stakeholders work together effectively. It's important to recognize that Enterprise Architecture is not an

isolated effort. Enterprise Architects must consider how their frameworks interact with broader business frameworks to ensure interoperability and alignment across the organization.

The Preliminary Phase must consider business goals and ensure that the architecture adapts to different objectives. In addition, Enterprise Architecture must operate at different levels within the organization to reflect this flexibility.

Depending on the size of the organization and its budget for Enterprise Architecture, different approaches may be used to partition the architecture teams, processes, and deliverables. At times, it may be necessary to revisit the Preliminary Phase from the Architecture Vision Phase to ensure that the organization's Architecture Capability is equipped to address a specific architectural issue.

In essence, the Preliminary Phase is about building the architecture needed to support the Architecture Capability.



Further Reading

For more information on establishing and using maturity models, see the TOGAF Series Guide "Architecture Maturity Models", document number G203, and "The TOGAF® Leader's Guide to Establishing and Evolving an EA Capability" (Appendix B), document number G168.

For detailed guidance on establishing the Architecture Capability, refer to "The TOGAF® Leader's Guide to Establishing and Evolving an EA Capability", document number G168.

12.3.2 Preliminary Phase Artifacts

Table 12-4 lists the artifacts that can be created during the execution of the Preliminary Phase.

Table 12-4. *Artifacts created during the Preliminary Phase*

ADM Phase	Artifacts	Used Entities
Preliminary Phase	Principles Catalog	Principle

Principles Catalog: Captures Enterprise and Architecture Principles of the organization. Enterprise Principles are high-level, general guidelines that shape the overall strategy, culture, and decision-making framework of an organization. Architecture Principles describe what a *good* solution or architecture should look like.

Principles are used to evaluate and agree on an outcome for architecture decision points. Principles are also used as a tool to assist in architectural governance of change initiatives. An example of a Principles Catalog is shown in Table 12-5.

Table 12-5. *Principles Catalog*

Name	Statement	Rationale	Implications
Comply with laws and regulations	Information management processes comply with all relevant laws, policies, and regulations.	To comply with laws and regulations, relevant laws and regulations are considered in the development, procurement, and design of systems, products, and services.	• The organization must be mindful to comply with laws, regulations, and external policies regarding the collection, retention, and management of data. • Efficiency, need, and common sense are not the only drivers. Changes in the law and changes in regulations may drive changes in our processes or applications.
Unambiguous ways of working	Consistent ways of performing tasks and activities are used.	The best way of working is chosen so that work is done more efficiently. This ensures that information management is under control.	• Achieving maximum organization-wide benefit will require changes in the way we plan and manage information. • Some departments may have to concede their own preferences for the greater benefit of the entire organization. • Information management initiatives should be conducted in accordance with the organizational plan.

(*continued*)

Table 12-5. (*continued*)

Name	Statement	Rationale	Implications
Store information once and use it more often	Information is available from a single source.	Used information is recorded once and in one place so it can be used more often.	• To enable data sharing we must develop and abide by a common set of policies, procedures, and standards governing data management and access. • Common methods and tools for creating, maintaining, and accessing the data shared across the organization must be adopted. • The principle of data sharing will continually "bump up against" the principle of data security – under no circumstances will the data sharing principle cause confidential data to be compromised.

Table 12-5. (*continued*)

Name	Statement	Rationale	Implications
Standard products and services	Products and services are selected based on the functionality required for the task.	Duplicative capability is expensive and proliferates conflicting data.	• A capability which does not serve the entire organization must change over to the replacement organization-wide capability. • It will not be allowed to develop capabilities that are similar/duplicative of organization-wide capabilities. • Data and information used to support organizational decision-making will be standardized.

12.4. Phase A: Architecture Vision

The Architecture Vision Phase is considered the first actionable phase in the Architecture Development Method cycle. This phase plays a crucial role in setting the groundwork for subsequent architecture work, as it ensures a clear understanding of the objectives, scope, and expected outcomes of the architecture project. It builds on the initial input gathered during the previous phase, the Preliminary Phase.

12.4.1 Creating the Architecture Vision

The main goal of the Architecture Vision Phase is to *establish a high-level vision of the architecture* and to *ensure stakeholder alignment*. It focuses on creating a shared understanding of the architecture's goals and ensuring that the business requirements are clearly articulated and agreed upon by all stakeholders. It also involves obtaining initial (stakeholder) approval to proceed with creating and implementing the architecture. The end result of this work is recorded in an architecture vision document.

Important inputs to the Architecture Vision Phase are the reference materials gathered during the Preliminary Phase. These reference materials can be supplemented with existing business principles, goals, and drivers. The existing architectural documentation (framework description, architectural descriptions, baseline descriptions, etc.) is then used to populate an initial version of the Architecture Repository. For example, the capabilities needed to start the architecture work are documented as essential Architecture Building Blocks in the Architecture Landscape part of the repository. The various roles and responsibilities, preferably represented in a kind of RACI matrix, are also stored in the Architecture Repository, more specifically in the Architecture Capability section. Table 12-6 lists the steps to perform Phase A (Architecture Vision).

Table 12-6. *Steps to perform Phase A (Architecture Vision)*

Step to Perform	Description of Step	Output Generated by Step
Establish the architecture project	Obtain formal project recognition, secure endorsement from corporate leadership, and gain support and commitment from relevant line management.	Statement of Architecture Work
Identify stakeholders and determine requirements	Identify stakeholders and requirements to allow for the architecture and related work to be communicated and aligned.	Stakeholder Catalog Communications Plan
Assess readiness for business transformation	Evaluate the organization's change readiness, guiding the architecture's scope and highlighting areas requiring attention.	Business Transformation Readiness Assessment
Elaborate on business goals and Architecture Principles	Validate and refine the business goals and Architecture Principles identified in the previous phase.	Validated business objectives, goals, and drivers Validated Architecture Principles
Create a high-level architecture vision	Develop a high-level description of the architecture solution that will meet business requirements and outline how it aligns with the overall business strategy.	Solution Concept Diagram
Define the Target Architecture value propositions and KPIs	Develop a business case and define performance metrics and measures.	Value propositions and KPIs

(*continued*)

Table 12-6. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Identify the business transformation risks and mitigation activities	Identify risks and assess their level and potential frequency.	Business transformation risks Mitigating activities
Develop Statement of Architecture Work	Develop Statement of Architecture Work and secure formal approval.	Statement of Architecture Work

Establish the Architecture Project: Enterprise Architecture serves as a critical business capability, with each iteration of the Architecture Development Method ideally managed as a project within the organization's existing project management framework. In certain scenarios, architecture projects may function independently, while in others, they may be integrated into broader initiatives. Regardless of the context, it is essential to plan and oversee architectural activities using the organization's established best practices.

To ensure project success, it is vital to obtain formal project recognition, secure endorsement from corporate leadership, and gain support and commitment from relevant line management. To agree on the scope of the work – and thus define the architecture project – it is necessary to prepare a Statement of Architecture Work. This document includes the architecture *project description and scope*, the *architecture vision statement*, and the architecture *project plan and schedule*.



Initiate the Project: Recognize the architecture endeavor as a formal project within your organization's project management framework. This approach ensures structured oversight and alignment with existing processes.

Determine Project Independence: Assess whether the architecture project will operate as a standalone initiative or integrate into broader organizational programs. This decision influences resource allocation and strategic alignment.

Plan Using Best Practices: Utilize your organization's established best practices to meticulously plan and manage the architectural activities. This strategy promotes consistency and quality in execution.

Secure Formal Recognition: Obtain official acknowledgment of the project to legitimize its objectives and facilitate organizational support.

Gain Leadership Endorsement: Seek explicit approval and backing from corporate leadership to ensure the project aligns with strategic goals and receives necessary resources.

Engage Line Management: Communicate with relevant line managers to garner their support and commitment, fostering collaboration across departments.

Define Project Scope: Prepare a comprehensive Statement of Architecture Work. This document should detail the project's description, scope, and vision statement and include a project plan with a clear schedule.

Identify Stakeholders and Determine Requirements: The last step of the Preliminary Phase was to identify the stakeholders and determine their requirements. This step is continued in the Architecture Vision

Phase. Knowing who the stakeholders are ensures that the architecture and related work can be communicated and aligned. A critical part of communicating with stakeholders is the Communications Plan. Refer to Chapter 14, Sections 14.3.1 through 14.3.3, for instructions on how to perform stakeholder analysis, create a stakeholder mapping, and set up a Communications Plan.

Knowing who the stakeholders are and what requirements and concerns they have is an important part of creating the Solution Concept Diagram. This diagram is based on the organization's goals and objectives and stakeholder requirements and constraints. An example of this diagram is provided in Section 12.4.2.



Identify Stakeholders: Determine who has an interest in the architecture project. This includes individuals or groups affected by or influencing the project's outcome.

Analyze Stakeholders: Understand each stakeholder's role, concerns, and influence. This helps in prioritizing their needs and expectations.

Document Requirements: Gather and record the specific needs and constraints of each stakeholder. This ensures that the architecture aligns with organizational goals and objectives.

Develop a Communications Plan: Create a strategy for keeping stakeholders informed and engaged throughout the project. Effective communication fosters alignment and support.

Create the Solution Concept Diagram: Using the collected information, develop a visual representation of the proposed solution. This diagram should reflect organizational goals, stakeholder requirements, and any constraints identified.

Assess Readiness for Business Transformation: It is essential to evaluate and quantify an organization's preparedness for change. A key objective is to perform a Business Transformation Readiness Assessment. This assessment hinges on determining and analyzing a series of readiness factors. These factors can include the following:

- **Vision:** Does the organization have clearly defined goals and objectives for the transformation?
- **Leadership:** Can the architecture work count on strong executive sponsorship and commitment?
- **Stakeholder Engagement:** Active involvement and support from key stakeholders is paramount.
- **Governance:** Effective decision-making and accountability structures are an absolute necessity.
- **Culture and Communications:** Does the organizational culture support change and open communication?

The readiness factors mentioned above are just a few examples. Each of the factors should be assessed to determine potential risks and required interventions before executing a transformation initiative.

The insights from the Business Transformation Readiness Assessment help shape the architecture's scope, pinpoint necessary activities within the project, and identify risk areas that need addressing. Because this is a specific technique, how to conduct a Business Transformation Readiness Assessment is described in more detail in Chapter 14, Section 14.7.



Understand the Purpose: Recognize that the goal is to assess how ready the organization is for transformation.
Identify Key Factors: Focus on important areas such as vision, leadership, stakeholder engagement, governance, culture, and communication.
Evaluate Each Factor: Look at each area to find potential risks and figure out what actions might be needed before starting the transformation.
Analyze the Findings: Use the information from the assessment to define the scope of the architecture, outline necessary project activities, and highlight risk areas that need attention.

Elaborate on Business Goals and Architecture Principles:
The business objectives, goals, drivers, as well as the Architecture Principles identified in the previous phase are validated and refined in the Architecture Vision Phase. If these have already been established elsewhere in the organization, make sure they are up to date and address any areas of ambiguity. If not, reconnect with the originators of the Statement of Architecture Work to collaboratively define these fundamental elements. Once clarified, ensure that senior management provides its official approval.

Next, outline the constraints that will affect the architecture. These include both organization-wide and architecture-specific constraints, such as timelines, schedules, and available resources. Organization-wide constraints may be influenced by the foundational Enterprise and Architecture Principles identified in the Preliminary Phase or further detailed in Phase A.



Review Existing Business Objectives and Goals: Examine the current business objectives, goals, and drivers identified in earlier phases. Ensure they are up to date and accurately reflect the organization's strategic direction.

Validate and Refine Architecture Principles: Assess the Architecture Principles established previously. Confirm their relevance and applicability to the current business context. Address any ambiguities or inconsistencies.

Engage Stakeholders for Input: Collaborate with key stakeholders, including those who initiated the Statement of Architecture Work. Gather insights and feedback to ensure a comprehensive understanding of business needs.

Define Fundamental Elements Collaboratively: Work together with stakeholders to clearly articulate business objectives and Architecture Principles. Ensure these elements are well-defined and aligned with organizational goals.

Obtain Senior Management Approval: Present the refined business objectives and Architecture Principles to senior management. Seek formal approval to ensure organizational alignment and support.

Identify Organizational Constraints: Outline constraints that may impact the architecture, such as timelines, schedules, and resource availability. Consider organization-wide constraints influenced by Enterprise and Architecture Principles from the Preliminary Phase.

Document and Communicate Constraints: Clearly document identified constraints. Communicate these constraints to all relevant stakeholders to ensure transparency and informed decision-making.

Create a High-Level Architecture Vision: Develop a high-level description of the architecture solution that will meet business requirements and outline how it aligns with the overall business strategy. It usually involves preparing a business case to justify the need for architecture work, demonstrating how it will *deliver value* or *solve business challenges*.

The architecture vision document outlines the business goals, capabilities, and high-level architectural approach. It often includes a conceptual solution description and an explanation of how the proposed architecture supports the business strategy. The high-level vision can be visualized using the aforementioned Solution Concept Diagram. Although the name of this artifact may suggest a specific and detailed solution, this diagram is far from that. It simply shows the relationships between the organization's goals and objectives on the one hand and the stakeholder requirements and constraints on the other.



Understand Business Goals and Requirements: Begin by identifying your organization's key business objectives and the specific requirements that the architecture needs to address. This foundational understanding ensures that the architecture vision aligns with the overall business strategy.

Develop a Business Case: Prepare a compelling business case to justify the need for architectural work. Highlight how the proposed architecture will deliver value, address business challenges, and support strategic goals.

Outline the Architecture Vision: Create a document that articulates the high-level description of the proposed architecture solution. This should include business goals, capabilities, and a high-level architectural approach. Describe the overarching strategy for the architecture, ensuring it aligns with business goals.

Include a Conceptual Solution Description: Provide a conceptual overview of the proposed solution, detailing how it addresses the identified business requirements and supports the organization's strategy.

Visualize with a Solution Concept Diagram: Utilize a Solution Concept Diagram to illustrate the relationships between organizational goals, stakeholder requirements, and constraints. This visual representation helps stakeholders easily grasp the alignment between business objectives and the proposed architecture.

Define the Target Architecture Value Propositions and KPIs: In this step, a business case is developed that describes the architecture(s) and the organizational changes it will require. To get stakeholders on board, it is important to create value propositions. These are presented to each stakeholder group for review and approval. Another action to take during this step is to define performance metrics and accompanying measures that allow the architecture work and its results to be monitored. They should ensure that the business requirements are being met. Finally, business risk is assessed. This means that a high-level assessment of potential risks is performed, and the risks are identified and recorded in the Statement of Architecture Work. This applies to all outputs from this particular step.



Develop a Business Case: Clearly explain the proposed architecture and the organizational changes it entails.

Create Value Propositions: Identify and articulate the benefits for each stakeholder group to gain their support.

Present for Review and Approval: Share these value propositions with stakeholders and seek their feedback and endorsement.

Define Performance Metrics: Establish clear metrics to monitor the architecture's effectiveness and ensure business requirements are met.

Assess Business Risks: Conduct a high-level evaluation of potential risks, documenting them in the Statement of Architecture Work.

Identify the Business Transformation Risks and Mitigation Activities: Introducing change to an organization almost always involves some risk. This type of risk is called business transformation risk. It is an absolute must to identify the business transformation risks and assess their initial level and their potential frequency. A high-level mitigation strategy can be assigned to each risk, although this can also be done at a later stage when the Consolidated Gaps, Solutions, and Dependencies Matrix (see Section 12.8.2) is created in Phase E (Opportunities and Solutions).

Using the Risk Management technique described in Chapter 14, Section 14.8, the documented risks from the Consolidated Gaps, Solutions, and Dependencies Matrix are classified, mitigated (if possible), and otherwise managed.



Spot the Risks: Begin by identifying potential risks associated with the business transformation. These could range from employee resistance to technological challenges. Understanding what might go wrong is the first step toward prevention.

Assess the Impact: Once the risks are identified, evaluate their potential impact and how often they might occur. This assessment helps in prioritizing which risks need immediate attention and which ones can be monitored over time.

Develop Mitigation Strategies: For each identified risk, craft a high-level strategy to mitigate its impact. This might involve training programs, adopting new technologies, or revising existing processes. Detailed planning can be integrated during the creation of the Consolidated Gaps, Solutions, and Dependencies Matrix in Phase E (Opportunities and Solutions).

Document Risks: Maintain a comprehensive record of all identified risks and their corresponding mitigation strategies. This documentation will serve as a valuable reference throughout the transformation process.

Implement and Monitor: Put the mitigation strategies into action and continuously monitor their effectiveness. Be prepared to adjust the approach as needed to address any unforeseen challenges.

Develop Statement of Architecture Work: At the end of the Architecture Vision Phase, the architecture team should aim to secure formal approval of the Statement of Architecture Work and commitment to proceed to the more detailed phases of the Architecture Development Method. This typically includes securing necessary funding, resources, and time allocations.

Gaining and maintaining stakeholder engagement is crucial. The vision must clearly communicate the benefits and value to stakeholders, especially senior management and business owners. Ensuring that the architecture vision aligns with the organization's business strategy is essential to get buy-in from stakeholders. At this stage, it's also important to identify risks early on that could impact the project later, such as technological challenges or resistance from stakeholders.



Communicate Benefits Clearly: Articulate the benefits and value of the architecture work in a manner that resonates with stakeholders. Clear communication fosters buy-in and commitment.

Identify and Assess Risks: Proactively identify potential risks, such as technological challenges or stakeholder resistance. Assess their impact and develop mitigation strategies to address them effectively.

Document Resource Requirements: Detail the necessary funding, resources, and time allocations required for the project. A comprehensive resource plan is essential for informed decision-making.

Draft the Statement of Architecture Work: Compile the information gathered into a formal document. This statement should encompass the project's scope, objectives, resource requirements, risk assessment, and alignment with business strategy.

Seek Formal Approval: Present the Statement of Architecture Work to the relevant approval authority within the organization. Address any concerns raised and obtain formal approval to proceed.

Maintain Stakeholder Engagement: Continuously engage stakeholders throughout the project to ensure ongoing support and promptly address emerging issues.

12.4.2 Architecture Vision Phase Artifacts

Table 12-7 lists the artifacts that can be created during the execution of the Architecture Vision Phase.

Table 12-7. *Artifacts created during Phase A (Architecture Vision)*

ADM Phase	Artifacts	Used Entities
Phase A: Architecture Vision	Stakeholder Catalog	– (or stakeholder, once added to the Enterprise Metamodel)
	Value Chain Diagram	Value stream
	Solution Concept Diagram	Objective, requirement, and constraint
	Business Model Diagram	Actor, role, stakeholder (see above), contract, function, process, business capability, business service, product
	Business Capability Map	Business capability
	Value Stream Map	Value stream

Stakeholder Catalog: Identifies the stakeholders for the architecture initiative, their influence on the initiative, and their key questions, issues, or concerns that need to be addressed by the architecture framework.

Understanding the stakeholders and their requirements allows the architect to focus efforts in areas that meet the stakeholders' needs.

Due to the potentially sensitive nature of stakeholder mapping information and the fact that the Architecture Vision Phase is intended to be conducted using informal modeling techniques, no specific metamodel entities are used to generate a Stakeholder Catalog.

As I mentioned in Chapter 5, the Enterprise Metamodel allows the architect to add and remove entities from the model if they do not add value to the organization.

This is particularly useful in the context of the Stakeholder Catalog. Let me explain.

The Enterprise Metamodel provides a variety of entities that help architecture initiatives visualize the answer to questions such as why the initiative is taking place (drivers, goals, and objectives) and what needs to be done to shape the change (e.g., adjusting processes, engaging actors, and creating capabilities).

It is noteworthy that the entities needed to indicate for whom the architecture initiative is being implemented and what concerns are being addressed do not appear as such in the metamodel.

I am referring, of course, to the stakeholder entity.

This entity is a critical part of any architecture approach. The TOGAF Standard itself identifies the need for Stakeholder Management by stating that the creation of a Stakeholder Catalog is highly desirable. The Communications Plan also pays close attention to formulating a way to keep stakeholders involved in the architecture work.

The TOGAF Standard cites the potentially sensitive nature of stakeholder mapping as a reason for not including the stakeholder entity in the Enterprise Metamodel. However, in day-to-day practice it is highly desirable to be able to relate goals and objectives to stakeholders. When stakeholders are defined with appropriate roles, there is generally no risk of misuse of potentially sensitive information.

It is therefore advisable to pay special attention to creating or including an entity that represents stakeholders during the customization of the Enterprise Metamodel.

Table 12-8 shows an example of a Stakeholder Catalog. The artifact lists the stakeholders themselves and their key concerns and classifications and shows which other architectural artifacts might pique their interest or address their concerns.

Table 12-8. *Stakeholder Catalog*

Stakeholder	Key Concerns	Classification	Catalogs, Matrices, Diagrams, and Maps
Maria Summers (CEO)	The high-level drivers, goals, and objectives of the organization and how these are translated into an effective process to advance the business	Keep satisfied	Strategy/Goal Matrix Goal/Objective Matrix
Chris West (CIO)	Alignment of IT strategy with business goals, optimizing IT costs, and supporting business agility through technology	Key player	Goal/Objective Matrix Objective/Initiative Matrix

(*continued*)

Table 12-8. (*continued*)

Stakeholder	Key Concerns	Classification	Catalogs, Matrices, Diagrams, and Maps
James Noble (CFO)	Budget efficiency and cost control in organizational projects, long-term financial sustainability of architecture decisions	Keep informed	Application Portfolio Catalog Technology Portfolio Catalog
Anna Carter (Manager HR)	Supporting workforce needs in alignment with organizational transformation, ensuring employee adoption of new processes and technologies	Minimal effort	Business Process Map Organization/Business Process Matrix

Value Chain Diagram: The Value Chain Diagram represents the high-level processes or activities that an organization performs to deliver value to customers. Unlike the more formal Organization Map developed in Phase B (Business Architecture), this diagram emphasizes presentational impact. This diagram helps to understand how different business functions contribute to the overall value proposition. Its primary purpose is to quickly engage and align stakeholders around a specific change initiative, ensuring that all participants understand the overarching functional and organizational context of the architecture engagement.

A Value Chain Diagram typically includes *primary activities* (analogous to the core business functions of an organization) and *supporting activities*. The latter are functions that enable the primary activities.

The Value Chain Diagram has a horizontal or vertical flowchart structure. In the diagram, boxes represent the primary and supporting activities. The corresponding colors of the vertical and horizontal activities indicate the relationship between the two. The diagram should be read from left to right, showing the sequence and interdependencies between activities.

In the Value Chain Diagram as shown in Figure 12-4, activities are divided into *primary* and *support activities*. Primary activities contribute directly to the delivery of – in this case – healthcare services. In the example, the four main primary activities are:

- **Patient Admission and Triage:** Assessing and registering patients, determining urgency and care requirements.
- **Diagnosis and Treatment:** Conducting medical tests, diagnosing conditions, and providing necessary treatments or procedures.
- **Patient Care and Monitoring:** Continuous care, nursing, medication administration, and post-treatment monitoring.
- **Discharge and Follow-Up:** Ensuring proper recovery plans, follow-up consultations, and post-hospitalization care.

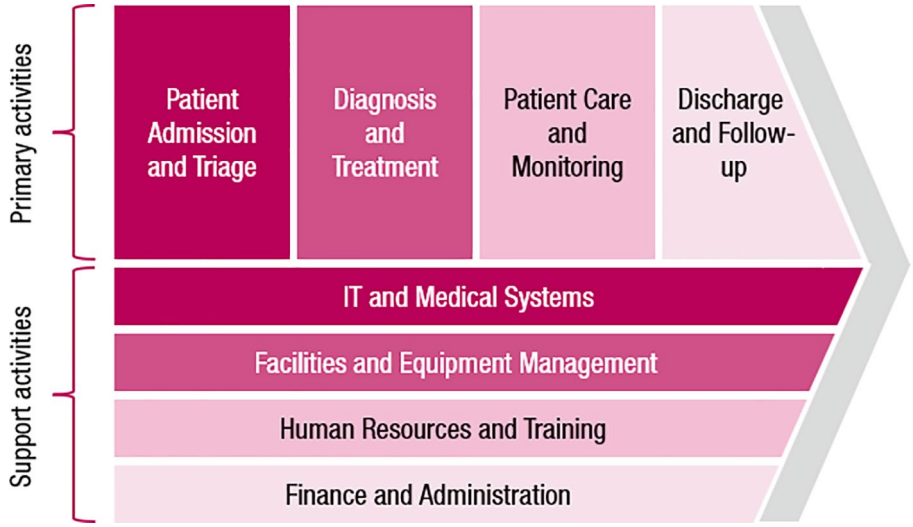


Figure 12-4. *Value Chain Diagram for a hospital*

Support activities enable primary activities to be performed efficiently. Four key support activities, shown in Figure 12-4, are:

- **IT and Medical Systems:** Managing electronic health records (EHR), hospital management software, and digital infrastructure
- **Human Resources and Training:** Recruiting, training, and managing healthcare professionals and support staff
- **Facilities and Equipment Management:** Maintaining hospital buildings, medical equipment, and supplies
- **Finance and Administration:** Budgeting, billing, insurance processing, and overall hospital governance

These activities ensure that the hospital delivers quality patient care efficiently while maintaining operational sustainability. As shown in the figure, the relationship between the primary and support activities is indicated by their matching colors:

- **Patient Admission and Triage ⇄ IT and Medical Systems**
  - Electronic health record (EHR) systems store patient data and streamline admissions.
  - IT systems manage triage workflows, ensuring proper patient prioritization.
- **Diagnosis and Treatment ⇄ Facilities and Equipment Management**
  - Medical equipment (MRI, X-ray, surgical tools) must be operational for accurate diagnosis and treatment.
  - Facility management ensures sterile, well-maintained environments for procedures.
- **Patient Care and Monitoring ⇄ Human Resources and Training**
  - Well-trained nurses, doctors, and support staff ensure effective patient care.
  - Ongoing medical training keeps staff updated on best practices and new treatments.
- **Discharge and Follow-Up ⇄ Finance and Administration**
  - Billing and insurance processing ensure proper financial clearance upon discharge.
  - Administrative teams handle scheduling for follow-ups and ensure patient records are updated.

Each support activity enables the smooth, efficient execution of its associated primary activity, ensuring the quality of the hospital's operations.

By analyzing this Value Chain Diagram, the hospital can identify areas where it can improve efficiency and reduce costs. For example, they may discover that they can improve their patient admission process by having their IT systems more efficiently manage and automate triage workflows. They may also discover that they can improve their operating rooms by investing in sterile, well-maintained environments. Analyzing a hospital's value chain provides a useful framework for identifying areas for improvement and optimizing performance. By optimizing its processes, the hospital can improve quality and create more value for its patients.

Solution Concept Diagram: This diagram offers a high-level overview of the proposed solution aimed at achieving the objectives of the architecture engagement. Unlike the more formal and detailed architecture diagrams developed in subsequent phases, this diagram serves as a preliminary *pencil sketch* of the anticipated solution at the start of the engagement.

This diagram may encapsulate key objectives, (stakeholder) requirements, and constraints for the engagement. It also may identify areas that require further investigation through formal architecture modeling. Its primary purpose is to quickly onboard and align stakeholders for a specific change initiative, ensuring that all participants comprehend the goals of the architecture engagement and how the proposed solution approach is expected to meet the organization's needs.

Business Model Diagram: A model describing the rationale for how an organization creates, delivers, and captures value. Another common name for this diagram is Business Model Canvas.

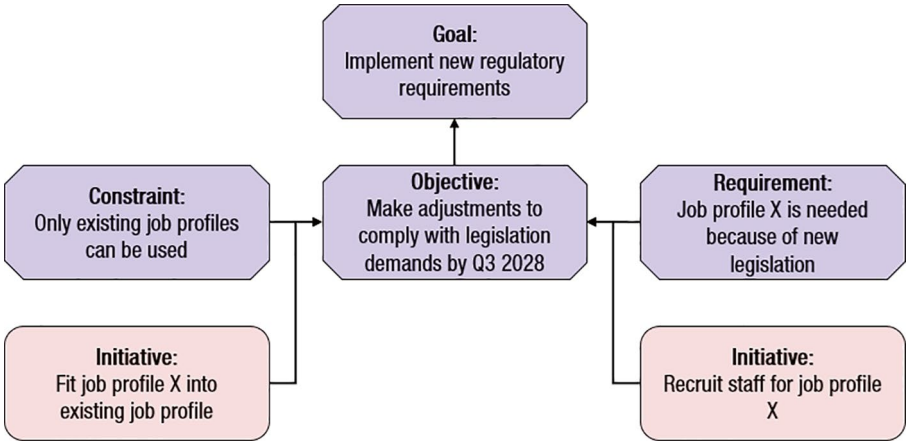


Figure 12-5. *Solution Concept Diagram*

The Business Model Diagram[1] shown in Figure 12-6 is an example of a diagram used in a hospital context [21].

[1] Figure 12-6 can be downloaded in a large format from https://eawheel.com/books/media/

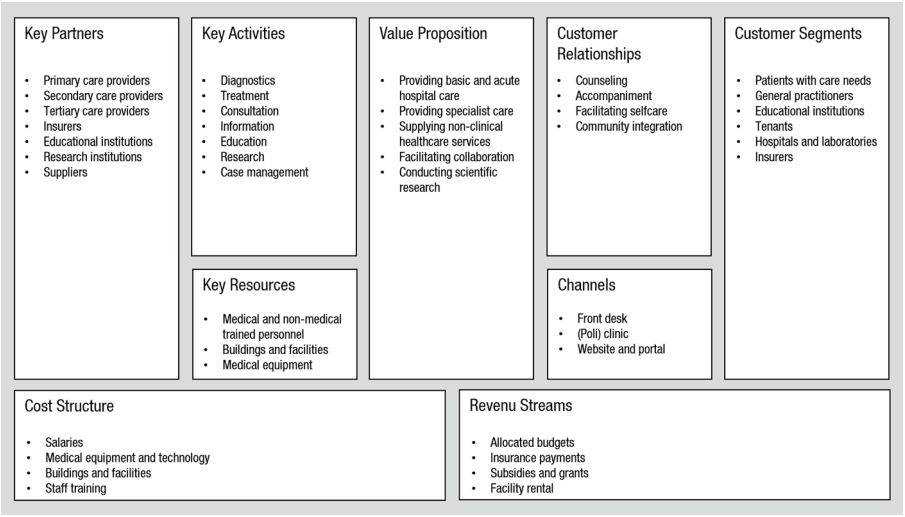


Figure 12-6. *Business Model Diagram of a hospital*

Note that the template of a Business Model Diagram or Canvas can contain the entities business collaboration, business interaction, business interface, resource, and value. These entities are not mentioned in the Foundation Enterprise Metamodel. It is advisable to consider them when tailoring the metamodel (as is described in Chapter 5, Section 5.2).

Business Capability Map: A diagram that shows the business capabilities that an organization needs to meet its purposes. During the Architecture Vision Phase, the diagram primarily shows capabilities related to achieving the organization's goals. It is not necessary to visualize and describe every capability the organization has at this stage.

The name of this artifact does not fully align with its description in the TOGAF Standard.

Please note that a *map* is a representation of architectural content in *textual format* (refer to Chapter 7, Section 7.4.2).

The explanation provided by the TOGAF Standard mentions that it should actually be a *diagram* ("A diagram that shows the business capabilities").

Based on the explanation, I went with showing an example of a Business Capability Diagram instead of a Business Capability Map.

The Business Capability Diagram shown in Figure 12-7 is a limited view of a Hospital Capability Diagram. This figure shows only a few capabilities for example purposes.

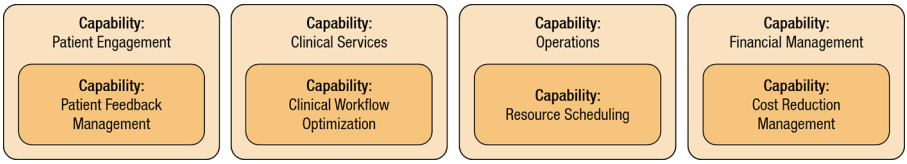


Figure 12-7. *Business Capability Diagram*

The Business Capability Diagram created during the Architecture Vision Phase is typically a high-level variant. Phase B (Business Architecture) of the Architecture Development Method updates the Business Capability Map artifact (see Section 12.5.3). During Phase B, more capability levels are added to the artifact.

Value Stream Map: A diagram representing an end-to-end collection of value-adding activities that create an overall result for a customer, stakeholder, or end user (see Figure 12-8).

Again, the name of the artifact should be Value Stream Diagram instead of Value Stream Map.

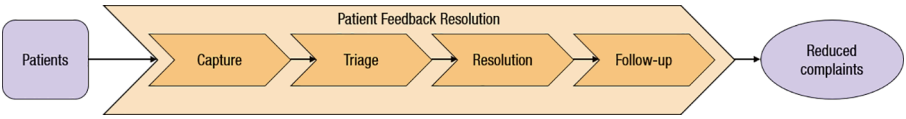


Figure 12-8. *Value Stream Diagram*

The Architecture Vision Phase sets the scope and specific objectives for the upcoming phases. It is key to ensuring that everyone involved in the project is aligned on what the architecture effort is aiming to achieve before moving forward into more detailed design and planning.

After the Architecture Vision Phase, the process moves to the Business Architecture Phase, where a more detailed view of the Business Architecture is developed, based on the high-level vision created in this phase.



Further Reading

More information about business models, including the Business Model Diagram or Canvas, can be found in the TOGAF Series Guide "Business Models", document number G18A.

12.5. Phase B: Business Architecture

Business Architecture serves as a comprehensive framework, offering multi-dimensional views of capabilities, end-to-end value delivery, information, and organizational structure. It intricately maps the relationships among these diverse business elements, including strategies, products, policies, initiatives, and stakeholders.

At its core, Business Architecture aligns these elements with overarching business goals, providing a crucial link to other domains within the organization. Understanding Business Architecture is essential – it lays the groundwork for all architectural endeavors across various domains. It should ideally be the first architecture-related activity undertaken unless already integrated into other organizational processes, such as organizational planning, strategic business planning, or business process re-engineering.

Practically speaking, a well-defined Business Architecture is invaluable. It not only showcases the business value of subsequent architectural initiatives to key stakeholders but also illustrates the return on investment for their support and engagement in future work.

The scope of work during Phase B is primarily guided by the architecture vision articulated in Phase A. Here, the business strategy outlines the goals, drivers, and success metrics, yet it doesn't specify the path to achieving them. That's where Business Architecture comes into play, providing a roadmap in Phase B to navigate toward those defined objectives.

12.5.1 Developing Business Architecture

The objectives of Phase B (Business Architecture) are twofold. The first one is to *craft the Target Business Architecture* to outline how the organization should function to meet business objectives. Strategic drivers defined in the Architecture Vision Phase are addressed, ensuring that they align with the Statement of Architecture Work and satisfy stakeholder expectations.

The second objective is to *identify potential components for the Architecture Roadmap*, pinpointing the gaps between Baseline and Target Business Architectures that must be bridged. The key to both objectives is to reuse existing material as much as possible.

As with all phases of the Architecture Development Method, all the reference materials, external to the organization and collected from previous phases, serve as input for the next phase. Phase B (Business Architecture) is no exception to this rule.

Looking internally, the non-architectural inputs such as business principles, objectives, goals, and drivers are further refined during this phase. The capabilities defined during the Preliminary Phase (which are needed to understand how to work with architecture) undergo an assessment during Phase B. Finally, a Communications Plan is drawn up to detail how to communicate with the various stakeholders and how to address their requirements and concerns.

As for the architectural inputs, the organizational model plays an important role in describing the scope of the impacted organization. Part of the model sheds a light on the current architecture maturity level of the organization in relation to working with architecture. Then there are the roles and responsibilities for the architecture team, which are contained and visualized using a RACI matrix (see Section 12.3.1). The scope of the impacted organization is further detailed with constraints on architecture work, budget requirements, and a governance and support strategy. This last part entails the establishment of an Architecture Governance Framework in order to manage the architecture activities. Table 12-9 lists the steps to perform Phase B (Business Architecture).

Table 12-9. *Steps to perform Phase B (Business Architecture)*

Step to Perform	Description of Step	Output Generated by Step
Select reference models, viewpoints, and tools	Choose relevant models and viewpoints from the Architecture Repository to match the business goals and stakeholder interests.	Views and viewpoints Required building blocks (catalogs, matrices, and diagrams) Architecture requirements
Develop Baseline and Target Business Architecture description	Describe the current and future Business Architecture to a degree that supports the architecture vision, building on the Architecture Repository's resources as needed.	Baseline Business Architecture description Target Business Architecture description
Perform gap analysis	Confirm internal consistency, identify any gaps between baseline and target, and validate that models align with business objectives.	Gap analysis results
Define candidate roadmap components	Using the baseline, target, and gap analysis, build an initial roadmap to guide future actions and prioritization across the remaining architecture phases.	Draft Architecture Roadmap Business Architecture Roadmap components

(*continued*)

Table 12-9. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Resolve impacts across the Architecture Landscape	Assess the Business Architecture's wider impacts across the organization and its alignment with existing or planned projects.	Validated building blocks and other artifacts
Conduct formal stakeholder review	Revisit the architecture's initial motivation and make any necessary refinements to ensure it's fit to support other architecture domains.	Statement of Architecture Work
Finalize the Business Architecture	Standardize building blocks, fully document them, and check alignment with business goals. Finalize documentation within the Architecture Repository.	Documented building blocks and requirements
Create the Architecture Definition Document	Complete all relevant sections in the Architecture Definition Document, adding models, reports, and graphics as needed to capture a clear architectural view.	Architecture Definition Document

Select Reference Models, Viewpoints, and Tools: Start by diving into the Architecture Repository to handpick Business Architecture resources, like reference models and patterns, that align with core business drivers and address stakeholder needs (their requirements). Next, zero in on the Business Architecture viewpoints that best communicate solutions (think operations, management, finance). Lastly, choose tools that fit the job for capturing and analyzing data with the selected viewpoints. Whether they

are simple documents or spreadsheets or more sophisticated modeling tools such as activity models and business process models, go with what's necessary to bring clarity and precision to the architecture.



Dive into the Architecture Repository: Begin by exploring the Architecture Repository to identify Business Architecture resources, such as reference models and patterns, that align with the organization's main business goals and meet stakeholder requirements.

Identify Suitable Viewpoints: Determine which Business Architecture viewpoints – such as those related to operations, management, or finance – will most effectively communicate the architectural solutions.

Select Appropriate Tools: Choose tools that are well suited for capturing and analyzing data within the selected viewpoints. These tools can range from simple documents and spreadsheets to more advanced modeling tools like activity models and business process models. Select the tools that will provide the necessary clarity and precision for the architectural work.

Develop Baseline and Target Business Architecture Description: Craft a baseline description of the current Business Architecture, covering as much as necessary to effectively build toward the Target Business Architecture. If an organization has existing architecture descriptions, they should be used as the basis for the baseline description. When developing the target description for the Business Architecture, focus on creating only what's necessary to support the architecture vision. The scope and level of detail to include will vary, depending on the importance of each business element in reaching the Target Architecture Vision. How much detail to include here depends on two things: first, how much of the existing structure will transition to the future architecture and, second, whether relevant architecture descriptions are already available. Utilize the Reference Library of the Architecture Repository to pinpoint the essential building blocks within the Business Architecture. If stakeholder

needs call for new models, use the initial reference models found in the Reference Library of the Architecture Repository as a baseline to shape these new elements. Also, consider exploring different Target Architecture options. These are best reviewed with stakeholders using the Architecture Alternatives and Trade-Offs approach (see Chapter 14, Section 14.9) to ensure the architecture aligns with business needs.



Start with the Baseline Description: Begin by documenting the current Business Architecture. Use existing descriptions as a starting point.

Create the Target Business Architecture: Focus on what's needed to support the architecture vision. Only include what's necessary to reach the target, based on the key business areas.

Consider the Scope and Detail: The amount of detail will depend on how much of the current architecture will carry over to the future and if any existing descriptions are available.

Use Reference Models: Use the Reference Library from the Architecture Repository to identify essential building blocks for the Business Architecture.

Explore Target Architecture Options: Look at different future architecture options and review them with stakeholders. Use the Architecture Alternatives and Trade-Offs method to ensure alignment with business needs.

Perform Gap Analysis: To ensure internal consistency and accuracy within the architecture models, dive into each view to identify and resolve potential conflicts. This phase is where models are rigorously validated against core principles, objectives, and constraints, making sure they are fit for purpose. Evaluate the models for completeness, confirming that they align seamlessly with all (stakeholder) requirements. Finally, apply gap analysis to pinpoint any gaps between the Baseline and Target Architecture, as detailed in Chapter 14, Section 14.5.

Any requirement, or change in requirement, that is outside of the scope defined in the Statement of Architecture Work must be submitted to the Requirements Repository for management through the governed requirements management process (see Section 12.12).



Check for Consistency: Go through all views in the architecture model and look for any conflicts or inconsistencies that need fixing.

Validate Against Principles: Make sure the models are in line with key principles, objectives, and constraints. This helps ensure they are effective and fit for purpose.

Verify Completeness: Ensure all stakeholder needs are met. Double-check that no requirement is missed.

Identify Gaps: Compare the Baseline Architecture to the Target Architecture. Spot any differences or gaps that need addressing.

Define Candidate Roadmap Components: After establishing the Baseline and Target Architectures and analyzing the gaps, creating an Architecture Roadmap becomes essential. The results of the gap analysis lead to the creation of work packages (or initiatives) related to the architecture to be developed. These initiatives are the components to be used in the Architecture Roadmap.

Think of the initial Architecture Roadmap as the raw draft. It's here to support a more refined roadmap that will evolve during Phase E (Opportunities and Solutions) (see Section 12.8). Add the Business Architecture components to the roadmap.



Create a Draft Architecture Roadmap: Based on the gaps, start putting together the Architecture Roadmap. Think of this as the first draft – it doesn't need to be perfect yet.

Break It Down Into Work Packages: From the gap analysis, develop work packages (or initiatives) that relate to the architecture that is being developed. These work packages are what will fill out the roadmap.

Keep Improving: The roadmap will evolve, especially in the Opportunities and Solutions Phase (Phase E). So think of the initial roadmap as something that will get better over time.

Add Business Architecture: Include the Business Architecture components in the roadmap. This is crucial for aligning with business goals.

Resolve Impacts Across the Architecture Landscape: With the Business Architecture in place, the next step is to assess any potential broader impacts or ripple effects. Now's the time to look across the Architecture Landscape and examine other architecture assets to answer key questions:

- Does this Business Architecture influence any established architectures already in place?
- Have any recent adjustments or updates been made that could affect this Business Architecture?
- Are there chances to reuse or extend elements from this Business Architecture to benefit other areas of the organization?
- Could this Business Architecture affect ongoing or upcoming projects?
- Could other projects – whether planned or currently in progress – have an impact on this Business Architecture?

Discuss the results of the questions posed with the relevant stakeholders and try to determine whether or not they should lead to adjustments to the architecture. If so, update the architecture accordingly to resolve any potential impacts.



Identify Influenced Architectures: Examine existing architectures to determine if the Business Architecture impacts any of them.

Review Recent Changes: Investigate any recent modifications or updates in the organization that might affect the Business Architecture.

Seek to Reuse Opportunities: Identify elements within the Business Architecture that could be reused or extended to benefit other areas of the organization.

Evaluate Project Interactions: Assess whether the Business Architecture could influence ongoing or upcoming projects, and vice versa.

Engage Stakeholders: Discuss the findings with relevant stakeholders to determine if any adjustments to the architecture are necessary.

Update the Architecture: Implement any agreed-upon changes to address potential impacts and ensure alignment across the Architecture Landscape.

Conduct Formal Stakeholder Review: To ensure the Business Architecture Phase effectively captures and models the essence of the business, it's essential to engage core business stakeholders: executives, process owners, and subject matter experts. Their perspectives ensure that the architecture truly reflects the organization's needs and goals. Gather these stakeholders to revisit the original objectives behind the architecture work and the Statement of Architecture Work. Together, evaluate if the proposed Business Architecture aligns with and supports the upcoming work in other architecture domains. Make refinements only where necessary to keep the architecture grounded and aligned with strategic goals.



Identify Key Stakeholders: Gather a diverse group of core business stakeholders, including executives, process owners, and subject matter experts. Their insights are invaluable in shaping an architecture that truly reflects the organization's needs and goals.

Revisit Original Objectives: Together with the stakeholders, review the initial objectives outlined in the Statement of Architecture Work. This ensures everyone is aligned and focused on the intended outcomes.

Evaluate Alignment: Assess whether the proposed Business Architecture aligns with and supports upcoming work in other architecture domains. This holistic evaluation helps maintain coherence across all areas.

Refine As Necessary: Make adjustments only where needed to keep the architecture grounded and aligned with strategic goals. This approach maintains focus and prevents unnecessary complexity.

Finalize the Business Architecture: Set clear standards for each building block, using reference models from the Architecture Repository's Reference Library wherever possible. Take the time to document each block in detail (see Chapter 7, Section 7.5.2.2, for instructions on how to do this).

Once that's squared away, do a final cross-check to ensure the architecture is aligned with overarching business goals. Document the *why* behind each decision; it's vital for linking back to organizational objectives. And when it comes to documenting requirements, ensure they connect directly to specific architecture elements. Try to complete all work products as much as possible.



Set Clear Standards for Each Building Block: Begin by defining clear standards for every component of the Business Architecture. Utilize reference models from the Architecture Repository's Reference Library wherever possible to ensure consistency and reliability.

Document Each Building Block in Detail: Take the time to thoroughly document each building block.

Ensure Alignment with Business Goals: Once the building blocks are well-documented, perform a final cross-check to ensure that the architecture aligns with the organization's overarching business goals. This alignment is vital for the architecture's effectiveness and relevance.

Document the Rationale Behind Each Decision: For every decision made during this process, document the reasoning behind it. This practice is essential for maintaining a clear link between architectural choices and organizational objectives, facilitating future reviews and adjustments.

Connect Requirements to Specific Architecture Elements: When documenting requirements, ensure that each one is directly linked to specific elements within the architecture. This connection enhances traceability and ensures that all requirements are adequately addressed within the architectural framework.

Complete All Work Products Thoroughly: Strive to complete all work products as comprehensively as possible. Thorough documentation and completion of these products will serve as a solid foundation for implementation and future architectural endeavors.

Create the Architecture Definition Document: To craft a *business-oriented* Architecture Definition Document, focus on the core goals and scope relevant to the architecture. Illustrate key aspects through visuals or outputs from modeling tools to highlight the architecture's strengths. At a minimum, the document should clearly present the Baseline and Target Business Architectures. When it's ready, share it with critical stakeholders

to gather feedback, ensuring refinements are made to align with organizational objectives. This iterative process is essential for achieving a comprehensive and goal-oriented architecture.



Identify Core Goals and Scope: Begin by clearly defining the main objectives and the scope of the architecture. This ensures that the Architecture Definition Document remains focused and relevant to the organization's needs.

Document Baseline and Target Architectures: Detail the Baseline and Target Business Architectures. This comparison highlights the necessary changes and developments required to achieve the goals.

Utilize Visual Aids: Incorporate diagrams, models, and other visual tools to effectively communicate key aspects of the architecture. Visual representations can simplify complex information and make the document more accessible to stakeholders.

Engage Stakeholders for Feedback: Share the Architecture Definition Document with key stakeholders to gather their insights and feedback. Their input is invaluable in ensuring that the architecture aligns with various perspectives and requirements within the organization.

Iterate and Refine: Use the feedback received to make necessary adjustments to the Architecture Definition Document. This iterative process helps in refining the architecture to better meet organizational objectives and address any concerns raised by stakeholders.

All the architecture work that needs to be done should be documented in what is called a *Statement of Architecture Work*. This document should be formally approved by the higher levels of management, in order to gain support and sponsorship for the architecture work to be done.

Since it is during this phase that actual architecture artifacts are going to be created, it is imperative to start using the Architecture Repository for documenting these artifacts. This leads to the initiation of the Enterprise

Continuum (see Chapter 7, Section 7.3.1), which is a conceptual framework and provides a lens through which to view the entire collection of architecture assets stored in the Architecture Repository. Among the assets contained in the repository are reusable building blocks, publicly available reference models, organization-specific reference models, and organization standards.

The architecture vision lays out a clear problem statement and defines the purpose of the Statement of Architecture Work – essentially, the *why* behind the architecture effort. This vision document can be enhanced with targeted views on specific architecture areas, business scenarios (if available), and an organized set of refined stakeholder requirements and concerns (which may be taken from the Solution Concept Diagram described earlier; see Figure 12-5).

12.5.2 Business Architecture Methods and Modeling Techniques

Several methods and techniques can be used to implement the steps described in the previous section. For instance, the following core Business Architecture methods are used to model the Business Architecture driven by the strategy scope from Phase A:

Applying Business Capabilities: The Business Capability Map developed in the Architecture Vision Phase acts as a high-level snapshot of the organization's core abilities, independent of its existing structure, business processes, information systems, and applications. Once outlined, these capabilities should be linked back to the specific organizational units, value streams, information systems, and strategic objectives in the architecture's scope. This mapping creates a more holistic view, enhancing the alignment and effectiveness of each area in reaching the broader goals.

Business Capability Heat Mapping: One widely used analysis technique is heat mapping, a method that provides a range of insights on core business capabilities. These insights might include maturity, effectiveness, performance, and even the value or cost each capability brings to the business. In a Business Capability Map, different colors represent specific attributes to give an at-a-glance view of each capability's status.

Take a maturity heat map, for instance. Here, a capability meeting the desired maturity level is highlighted in green, one level below target appears in yellow, and capabilities lagging by two or more levels show up in red. Colors can also reflect other conditions, for example, purple might indicate a capability that's not yet in place but highly desired or a capability with surplus funding and resources beyond what's necessary.

Remember the fictional healthcare institution described in Chapter 5 (Section 5.1.1)?

Figure 12-9 shows a portion of this institution's capabilities, using the Business Capability Heat Mapping technique. The capabilities that are slightly underperforming are shown in yellow. The capability in red is not functioning at all and requires immediate attention. The purple-colored capabilities do not yet exist and must be created (either developed or purchased).

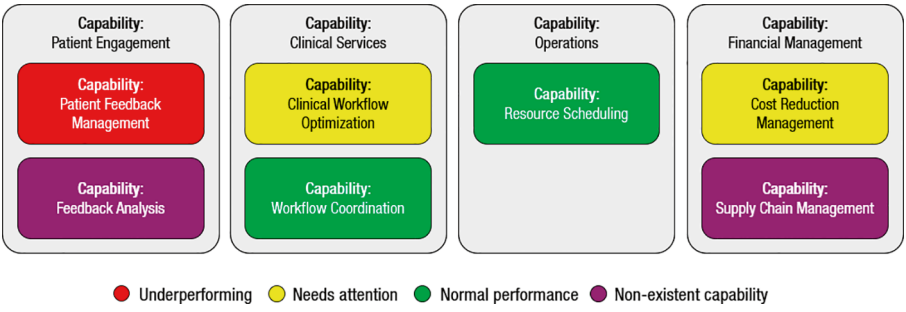


Figure 12-9. *Business Capability Heat Map*

> There is no scientific approach or calculation behind determining the maturity of a capability. Much comes down to stakeholder alignment and gut feeling. Business Capability Heat Mapping – and this is true of any type of heat mapping – requires a healthy dose of common sense.

For more information about the Business Capability Heat Mapping technique, see the *Further Reading* information box at the end of this section.

Applying Value Streams: Value streams provide valuable stakeholder context, helping them understand why certain business capabilities are critical for the organization. They set the stage by defining the *why* behind the need, while business capabilities focus on the *what* – the elements required for each phase of the value stream to deliver results.

Begin with an initial set of value stream models for the business, ideally documented in the Architecture Vision Phase. Whether working with a new or existing value stream, these models can be analyzed in the architecture's scope by heat mapping value stream stages or developing use cases that cover the entire value stream journey (for further details, see the "References" section at the end of this book).

The real power of this approach lies in mapping the relationship between each stage of a value stream and the corresponding business capabilities. This alignment allows for effective gap analysis – such as heat mapping – to pinpoint areas for capability development in line with the business value the value stream delivers to a specific stakeholder.

Applying the Organization Map: An Organization Map lays out the core organizational units, key partners, and crucial stakeholder groups that shape the enterprise ecosystem. Unlike a traditional organizational chart – which just shows who's reporting to whom – the Organization Map goes further, revealing the dynamic working relationships among these groups.

Think of it more as a network or web of connections rather than a rigid hierarchy. And while it is called a *map*, it's rather more of a *diagram*, since it is a graphical representation and not text-based.

This diagram plays a central role in Business Architecture because it sets the organizational stage for everything else in the architectural process. While capability mapping reveals *what* an organization does and value stream mapping shows *how* it delivers value to specific stakeholders, the organizational map identifies the business units or third parties that own or use these capabilities and participate in the value streams. The Organization Map shows *who* holds these capabilities and *who* is involved in these value streams, whether they're internal teams or external partners.

Together with other methods described earlier, the Organization Map clarifies which units need to be looped into the architecture effort, who to approach about specific requirements, and even how to gauge the impact of strategic decisions.

Applying Information Maps: Diving into the Business Architecture Phase starts with pinpointing the core elements that drive the business forward. Think products, customers, strategies, and so forth. With a clear view of these essentials, critical information can be mapped out in terms that resonate with the business. This mapping isn't just about listing things; it's about seeing how these elements interconnect, creating a foundational Information Map that sets the stage for the future.

The true power of this approach comes to life when matrices are built between key information concepts and business capabilities. These connections reveal the direct links between essential information on the one hand and the capabilities that turn it into real value on the other. This is fundamental to the essence of Business Architecture. These Information Maps and their ties to business capabilities aren't just one-off exercises. They'll serve as a valuable reference across future architecture phases, impacting everything from data insights to applications and infrastructure.

Applying Modeling Techniques: Beyond core techniques like capability maps, value streams, Organization Maps, and Information Maps, various modeling methods can be beneficial based on specific needs, for instance:

- **Activity Models:** Describe the organization's business activities. They detail internal data exchanges between activities and also cover interactions with external activities outside the model's scope. This versatility empowers organizations to tailor their approach, modeling only what is relevant and insightful, ensuring alignment with business goals.

The Object Management Group has developed the Business Process Modeling Notation (BPMN), a standard for business process modeling that includes a language with which to specify business processes, their tasks/steps, and the documents produced [20].

- **Use-Case Models:** Think of use-case models as a straightforward map of how a business process flows in an organization. They break down complex business processes into use cases (tasks or functions) and actors (the people or roles involved) in a way that's easy to follow. Each part of this model is visualized through use-case diagrams and detailed specs that make it simple to see who's doing what and when.
- **Logical Data Model:** The logical data model, or what some might call a class model, organizes information by defining key entities, their properties, and the acceptable values for each. Plus, it captures the connections between different entities, like using

is-a relationships for sharper analysis. For example, imagine *sedan* being a type of *car*, which is a type of *vehicle*. If there's a rule that applies to all vehicles, you can set it once, and it'll automatically apply across the board, whether you're talking about cars, trucks, or any other vehicle. Class models take this a step further by adding behaviors (methods) to each entity, giving them specific actions to perform. Typically, Application and Information Architects lean on the class diagram, while Business Architects focus on the logical data model.

Business Scenarios: This technique is complementary to the Business Architecture Phase of the Architecture Development Method. This technique evaluates, elaborates on, and/or modifies the assumptions behind an architecture project. It involves understanding and documenting the most important elements of a business scenario in successive iterations. Failing to examine all elements of a business scenario may result in an incomplete solution; however, care must be taken to avoid unnecessary repetition.

The Business Scenarios technique can be used in various phases of Enterprise Architecture development, primarily the Preliminary, Architecture Vision, and Business Architecture phases. However, it can also be used in other phases if necessary. See the *Further Reading* information box at the end of Section 12.5.3 for more information on applying the method.



Further Reading

For more information on the heat mapping technique, refer to the TOGAF Series Guide “Business Capabilities, Version 2”, document number G211.

A baseline example of Value Stream Heat Mapping can be found in the TOGAF Series Guide “Value Streams”, document number G178.

The BIZBOK Guide offers extensive information about heat mapping capabilities. See the chapter on capability mapping:

https://www.businessarchitectureguild.org/page/BIZBOK

For further guidance on using Organization Maps, see the TOGAF Series Guide “Organization Mapping”, document number G206.

Additional information on Information Maps is available in the TOGAF Series Guide “Information Mapping”, document number G190.

More information about Business Scenarios can be found in the TOGAF Series Guide “Business Scenarios”, document number G176.

12.5.3 Phase B (Business Architecture) Artifacts

Table 12-10 lists the artifacts that can be created during the execution of Phase B (Business Architecture).

Table 12-10. *Artifacts created during Phase B (Business Architecture)*

ADM Phase	Artifacts	Used Entities
Phase B: Business Architecture	Organization/Actor Catalog	Organization unit, actor, role, location (when Location Catalog is not maintained)
	Driver/Goal/Objective Catalog	Organization unit, driver, goal, objective, measure (may optionally be included)
	Role Catalog	Role
	Business Service/ Function Catalog	Organization unit, function, business service, application service (may optionally be included here)
	Location Catalog	Location, logical/physical application component, logical/physical technology component
	Process/Event/Control/ Product Catalog	Process, event, control, product
	Contract/Measure Catalog	Business service, application service (optionally), contract, measure
	Business Capabilities Catalog	Business capability
	Business Capability Map	Business capability, entities that flesh out the business capability
	Value Stream (Stages) Catalog	Business capability, value stream

(*continued*)

Table 12-10. (*continued*)

ADM Phase	Artifacts	Used Entities
	Value Stream Map	Business capability, value stream
	Value Stream/Capability Matrix	Business capability, value stream
	Strategy/Capability Matrix	Business capability, value stream, course of action
	Capability/Organization Matrix	Business capability, value stream, organization unit
	Organization Map	Organization unit, actor, role, stakeholder (if available)
	Business Glossary Catalog	–
	Business Interaction Matrix	Organization unit, function, business service
	Actor/Role Matrix	Actor, role
	Business Footprint Diagram	Organization unit, function, business service
	Business Service/ Information Diagram	Business service, business information
	Functional Decomposition Diagram	Capability
	Product Lifecycle Diagram	Product
	Goal/Objective/Business Service Diagram	Goal, objective, business service

(*continued*)

Table 12-10. (*continued*)

ADM Phase	Artifacts	Used Entities
	Business Use-Case Diagram	Actor, role, process, function (later on, it can include data, application, and technology details)
	Organization Decomposition Diagram	Actor, role, location
	Process Flow Diagram	Process, all mappings related to the process entity
	Business Event Diagram	Event, process
	Information Map	Business information

Organization/Actor Catalog: The purpose of this catalog is to capture a definitive listing of everyone that interacts with IT, including users and owners of IT systems. The Organization/Actor Catalog can be referenced when developing requirements in order to test for completeness.

For example, requirements for an application that services customers can be tested for completeness by verifying exactly which customer types or business roles need to be supported and whether there are any particular requirements or restrictions for user types. The Organization/Actor Catalog (see Table 12-11) can also contain business roles.

Table 12-11. *Organization/Actor Catalog*

Organization Unit	Actor	Business Role
Board of Directors	Maria Summers	Chief Executive Officer
Finance	James Noble	Chief Financial Officer
IT Department	Chris West	Chief Information Officer
Quality and Safety	Elena Petrova	Medical Director
	Kevin O'Brien	Patient Experience Manager
Clinical Operations	Susan Delgado	Chief Nursing Officer
	Benjamin Lee	Chief Medical Information Officer
	Hannah Duarte	Charge Nurse/Unit Manager

The Organization/Actor Catalog can be visualized using a diagram as well. Figure 12-10 shows the Organization/Actor Diagram in which the organization units *Quality and Safety* and *Clinical Operations* are visualized.

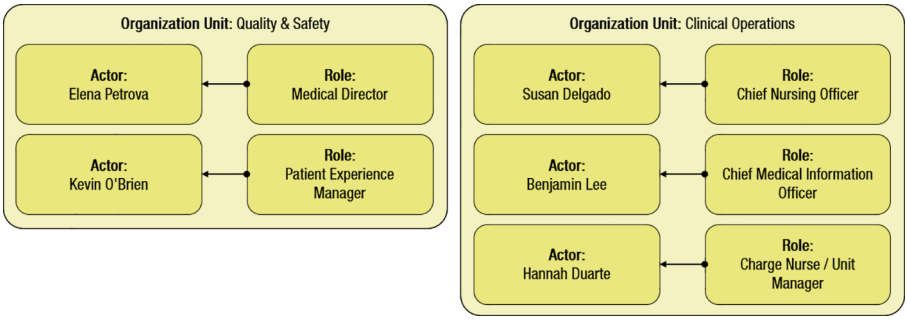


Figure 12-10. *Organization/Actor Diagram*

Driver/Goal/Objective Catalog: Provide a cross-organizational reference of how an organization meets its drivers in practical terms through goals, objectives, and (optionally) measures (see Table 12-12).

Publishing a definitive breakdown of drivers, goals, and objectives allows change initiatives within the organization to identify synergies across the organization (e.g., multiple organizations attempting to achieve similar objectives), which in turn allow stakeholders to be identified and related change initiatives to be aligned or consolidated.

Table 12-12. *Driver/Goal/Objective Catalog*

Driver	Goal	Objective
Patient-Centered Care	Improve patient satisfaction.	Achieve 90% rating in patient satisfaction surveys by Q4 2028.
	Enhance patient safety.	Decrease hospital-acquired infection rates by 15% by Q4 2029.
Operational Excellence	Optimize clinical workflows.	Reduce patient discharge processing time by 30 minutes by April 2029.
	Increase staff productivity.	Realize a 10% increase in key clinical throughput metrics by Q2 2030.
Financial Sustainability	Reduce operational costs.	Cut non-clinical supply expenses by 10% by December 2030.

Role Catalog: This artifact lists the different business roles within an organization. It can be used with other information, such as authorization levels or zones, to create a RACI matrix. The Role Catalog (see Table 12-13) is a valuable resource for identifying the impacts of organizational change management, defining job functions, and executing end-user training.

Table 12-13. *Role Catalog*

Role	Description
Medical Director	Oversees clinical risk, incident response, and safety programs.
Patient Experience Manager	Manages patient satisfaction and service recovery processes.
Chief Nursing Officer	Leads nursing operations, staffing, patient care quality.
Chief Medical Information Officer	Leads EHR adoption and clinical IT optimization.
Charge Nurse/Unit Manager	Coordinates shift-level operations, reports issues, escalates concerns.

Business Service/Function Catalog: The Business Service/Function Catalog provides a functional decomposition that can be filtered, reported on, and queried. It supplements graphical Functional Decomposition Diagrams.

The catalog (see Table 12-14) can be used to identify an organization's capabilities and understand the level at which governance is applied to its functions. This decomposition can identify new capabilities required to support business changes and determine the scope of change initiatives, applications, or technology components.

Table 12-14. *Business Service/Function Catalog*

Business Service	Description	Business Function	Capability
Patient Feedback Routing Service	Receives, categorizes, and routes patient feedback (complaints, compliments, safety concerns) to the appropriate business units for action.	Patient Experience Management	Patient Experience Management
Infection Risk Incident Escalation Service	Automatically escalates infection-related feedback or audit findings to the Infection Control Team for assessment and containment action.	Clinical Risk and Safety Oversight	Clinical Workflow Optimization Patient Feedback Management
Care Team Resource Scheduling Service	Manages automated scheduling of staff shifts, aligning personnel availability with patient load forecasts and unit needs.	Workforce Planning and Scheduling	Resource Scheduling
Cost Efficiency Analytics Service	Provides real-time and historical insights into departmental costs per treatment or service, enabling financial decision support.	Financial Performance Management	Cost Reduction Management

Since the Business Service/Function Catalog combines business services and functions, a matrix would have been a better choice. Cross-mapping between the two entities provides a clearer picture of their relationship. For this reason, a Business Service/Function Matrix is shown in Table 12-15.

Table 12-15. *Business Service/Function Matrix*

	Business Function			
Business Service ▼	Patient Experience Management	Clinical Risk and Safety Oversight	Workforce Planning and Scheduling	Financial Performance Management
Patient Feedback Routing Service	X			
Infection Risk Incident Escalation Service		X		
Care Team Resource Scheduling Service			X	
Cost Efficiency Analytics Service				X

Location Catalog: Documents all relevant physical or virtual locations where business operations, services, or architectural elements reside. This includes offices, clinical facilities, data centers, or cloud zones. It helps architects understand geographic dependencies, constraints (e.g., regulatory), and the distribution of capabilities, applications, or data. It supports planning for disaster recovery, latency, availability, and compliance. An example is shown in Table 12-16.

Table 12-16. *Location Catalog*

Location ID	Location Name	Type	Purpose	Applications Used
LOC-001	Central Hospital Campus	Physical Site	Inpatient care, surgery, admin HQ	EHR system (Epic), Staff Scheduling Tool
LOC-002	Data Center – Region A	Physical Site	Hosts EHR, feedback systems	Patient Feedback Repository, Infection Alert Engine
LOC-003	Telehealth Operations Hub	Virtual Location	Remote consultations and triage	Virtual Consultation Platform

The Location Catalog can also be visualized as a diagram. Figure 12-11 illustrates this.

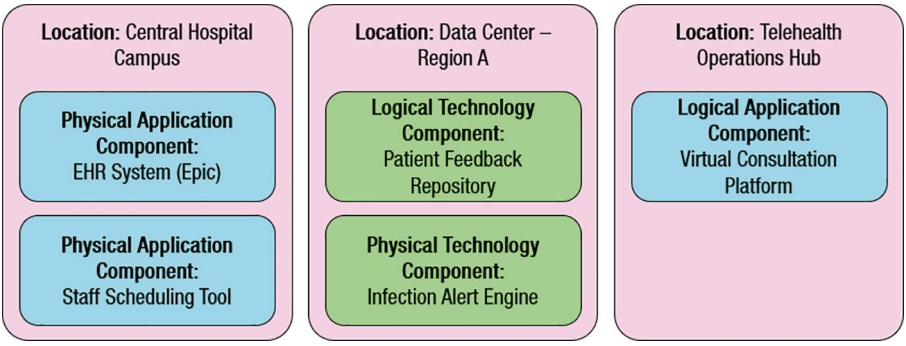


Figure 12-11. *Location Diagram*

Process/Event/Control/Product Catalog: Captures key elements of a business process, including the events that trigger it, the controls that constrain it, and the products or outcomes it delivers. This catalog (see Table 12-17) helps architects understand how business processes are governed and what inputs/outputs are involved. It supports regulatory compliance, Risk Management, and operational alignment by documenting relationships among business activities and their surrounding context.

Table 12-17. *Process/Event/Control/Product Catalog*

Process	Event	Control	Product
Patient Feedback Routing	Submission of patient survey	Feedback categorization rules	Routed feedback ticket
Infection Incident Escalation	Infection-related complaint logged	Escalation protocol (within 24 hours)	Infection control alert
Staff Schedule Generation	Weekly schedule trigger	Labor regulations, staffing ratios	Shift roster
Cost Analysis Reporting	End of quarter	Financial reporting standards	Cost performance report

Figure 12-12 shows a graphical representation of the first line from the Process/Event/Control/Product Catalog.

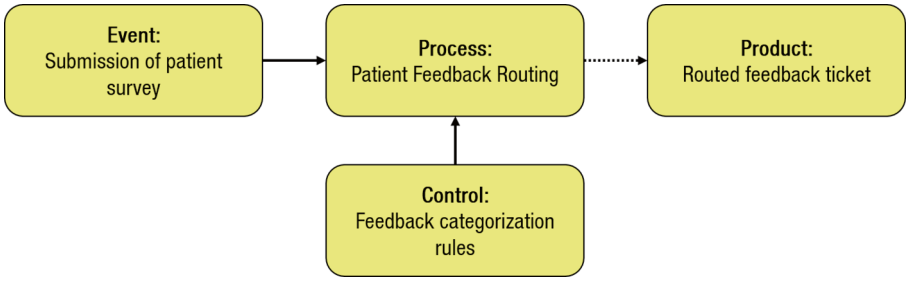


Figure 12-12. *Process/Event/Control/Product Diagram*

Contract/Measure Catalog: Documents the performance expectations, agreements, and service levels between business services and their consumers or providers. It helps align services with business goals by explicitly defining KPIs, SLAs, OLAs, and regulatory commitments. This catalog (see Table 12-18) supports architecture traceability by linking services to their value delivery and compliance metrics.

Table 12-18. *Contract/Measure Catalog*

Contract/ Measure Name	Applies to Service	Consumer	Provider	Measure Type	Target Value
Patient Feedback SLA	Patient Feedback Routing Service	Patient Experience Office	IT/ Experience Platform Vendor	SLA[2]	95% routed within 24 hours
Infection Escalation OLA	Infection Risk Incident Escalation	Quality and Safety Department	Clinical Risk System	OLA[3]	100% within 12 hours
Staffing Optimization KPI	Care Team Resource Scheduling Service	Nursing Administration	Scheduling System/HR	KPI	<5% manual overrides
Cost Efficiency Metric	Cost Efficiency Analytics Service	CFO/Finance	BI Platform/ Finance Analysts	KPI	Cost per case < baseline

Business Capabilities Catalog: A structured list of the core business capabilities an organization needs to fulfill its mission and strategic objectives. Each capability describes *what* the business does (not *how*)

[2] Service Level Agreement: a formal contract or agreement between a service provider and a client that outlines the expected level of service, performance metrics, and potential consequences if those standards aren't met.

[3] Operational Level Agreement: an internal agreement within a company that defines how different departments or teams will work together to deliver services and meet the terms of a Service Level Agreement (SLA).

and remains stable even when processes or systems change. The catalog (see Table 12-19) supports planning, impact analysis, and capability-based investment decisions. It typically includes capability name, description, maturity level, responsible organization, and related goals or services.

Table 12-19. *Business Capabilities Catalog*

Capability Name	Description	Maturity Level	Organization Unit	Related Goal
Patient Feedback Management	Ability to collect, route, and act on patient feedback to improve care and safety.	Ad hoc	Patient Experience Office	Improve patient satisfaction. Enhance patient safety.
Clinical Workflow Optimization	Ability to design, implement, and monitor efficient, standardized clinical paths.	Defined	Clinical Operations	Optimize clinical workflows.
Resource Scheduling	Ability to assign and manage personnel and assets based on demand forecasts.	Managed	Nursing Administration	Increase staff productivity.
Cost Reduction Management	Ability to monitor, analyze, and optimize costs without compromising quality.	Repeatable	Finance and Supply Chain	Reduce operational costs.

Business Capability Map: A map showing the business capabilities an organization needs to achieve its goals. Business capabilities can be broken down into sub-capabilities. The Business Capability Map can also be supplemented with entities that flesh out the capabilities. The Business Capabilities Catalog can be used to expand the Business Capability Map created in Phase A. Figure 12-13 provides a graphical representation of the map in the form of a Business Capability Diagram.

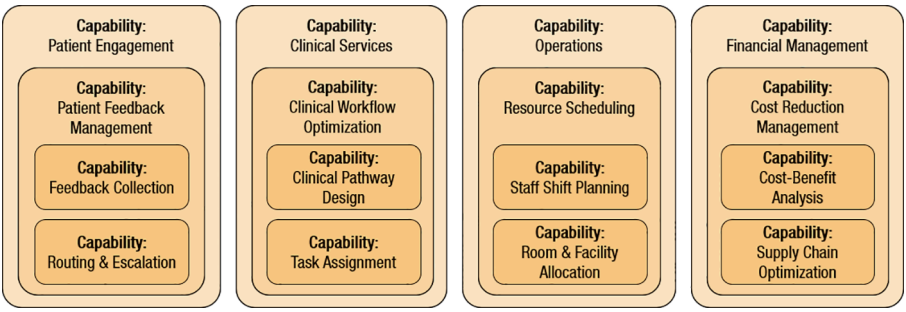


Figure 12-13. *Business Capability Diagram*

Value Stream (Stages) Catalog: A structured listing of the end-to-end collections of value-adding activities that describe *how an organization delivers value to its stakeholders*. Each entry outlines a value stream's name, description, triggering stakeholder, stages, and expected outcomes. This catalog (see Table 12-20) helps link business capabilities to customer-focused outcomes and supports strategic alignment.

Table 12-20. *Value Stream Catalog*

Value Stream Name	Description	Stages	Stakeholder	Outcome
Inpatient Care Delivery	Provides complete inpatient treatment from admission to discharge.	Admission, Diagnosis, Treatment, Discharge	Patients Clinicians	Safe, timely recovery Improved bed turnover
Patient Feedback Resolution	Captures and resolves patient feedback to improve quality.	Capture, Triage, Resolution, Follow-Up	Patients Quality Office	Reduced complaints Safety issue mitigation
Cost Performance Review	Analyzes and improves cost-effectiveness of clinical services.	Data Collection, Analysis, Action Plan	Finance Department Heads	Reduced cost per treatment Optimized spending

The Value Stream Catalog can be visualized by using a Value Stream Diagram for each of the listed value streams. Figure 12-14 illustrates the value stream for Patient Feedback Resolution.

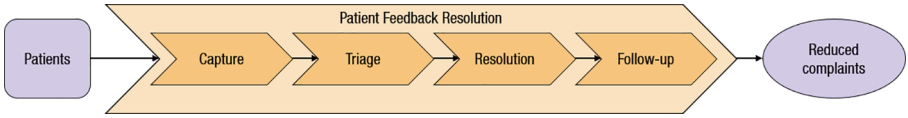


Figure 12-14. *Value Stream Diagram for Patient Feedback Resolution*

Value Stream Map: Illustrates an end-to-end collection of activities that add value and create an overall result for a customer, stakeholder, or end user.

The Value Stream (Stages) Catalog can serve as the basis for creating a Value Stream Map. The catalog can be supplemented with capabilities related to the stages of the value streams. The *Outcome* column in the Value Stream (Stages) Catalog can be replaced with a *Capabilities* column.

Value Stream/Capability Matrix: The purpose of this matrix is to link the stages of a value stream (what the organization delivers to stakeholders) to the business capabilities (what the organization needs to perform those stages). It's a key tool in capability-based planning and helps identify which capabilities support which parts of value delivery.

Table 12-21. *Value Stream/Capability Matrix*

Value Stream Name	Value Stream Stage	Capability
Patient Feedback Resolution	Capture	Patient Feedback Management, Digital Channel Integration
	Triage	Clinical Risk Assessment, Quality and Safety Monitoring
	Resolution	Incident Response Coordination, Clinical Workflow Optimization
	Follow-Up	Patient Communication Services, Service Quality Evaluation

The Value Stream/Capability Matrix, as shown in Table 12-21, can be visualized by using a Value Stream/Capability Diagram. Figure 12-15 illustrates the value stream for Patient Feedback Resolution. Using the *Business Capability Heat Mapping* technique (see Section 12.5.2) on this artifact reveals capability gaps, redundancies, and areas for improvement.

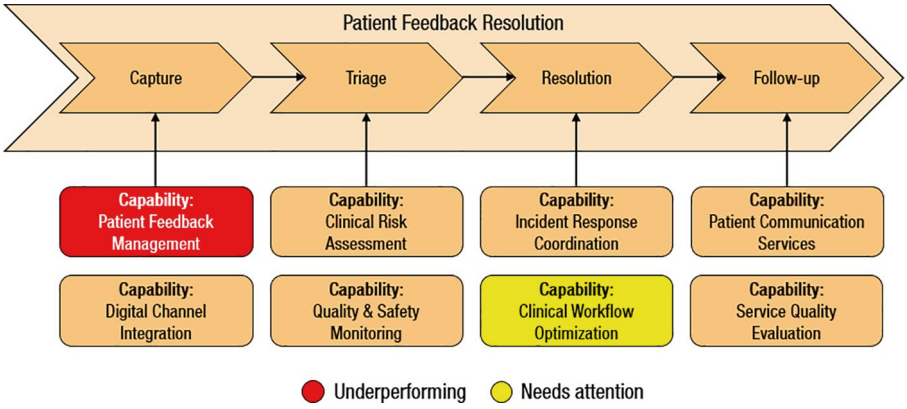


Figure 12-15. *Value Stream/Capability Diagram for Patient Feedback Resolution*

Strategy/Capability Matrix: Shows the capabilities required to support specific strategy statements. It aligns the organization's strategic goals with the business capabilities needed to realize them. It helps stakeholders see how strategic intents translate into organizational capabilities, ensuring traceability and focus in the architecture.

This matrix (see Table 12-22) supports capability-based planning, enabling prioritization of investments and architectural change by showing where critical capability development is required to meet strategic goals.

Table 12-22. *Strategy/Capability Matrix*

Strategic Goal	Capability	Priority
Improve patient satisfaction	Patient Engagement	High
Enhance patient safety	Patient Engagement	High
Optimize clinical workflows	Clinical Services	Medium
Increase staff productivity	Operations	Medium
Reduce operational costs	Financial Management	Low

Capability/Organization Matrix: Lists the organization elements that implement each capability. The matrix (see Table 12-23) maps business capabilities to the organizational units responsible for delivering or supporting them. It helps clarify ownership, accountability, and potential gaps or overlaps in capability delivery.

Table 12-23. *Capability/Organization Matrix*

	Organization Unit			
Capability ▼	Patient Experience Office	Quality and Safety	Clinical Operations	Finance
Patient Feedback Management	Primary	Contributor	Contributor	
Clinical Workflow Optimization			Primary	
Resource Scheduling			Contributor	
Cost Reduction Management				Primary

Organization Map: A map showing the relationships between the primary entities that make up the organization, its partners, and stakeholders. The map (see Table 12-24) lists the organization unit entities and their types, as well as their relationships to other entities on the map.

Table 12-24. *Organization Map*

Organization Unit	Type	Related to
Patient Experience Office	Business unit	Healthcare Institution
Quality and Safety Department	Business unit	Healthcare Institution Ministry of Health
Clinical Operations	Business unit	Healthcare institution Medical Equipment Supplier
Finance	Business unit	Healthcare Institution Insurance Partner Training Institute Medical Equipment Supplier
Nursing Administration	Business unit	Healthcare Institution Training Institute
Ministry of Health	Partner	Quality and Safety Department
Medical Equipment Supplier	Supplier	Clinical Operations Finance
Training Institute	Supplier	Nursing Administration Finance
Insurance Partner	Partner	Finance

The Organization Map can also be visualized using an Organization Diagram, as illustrated in Figure 12-16.

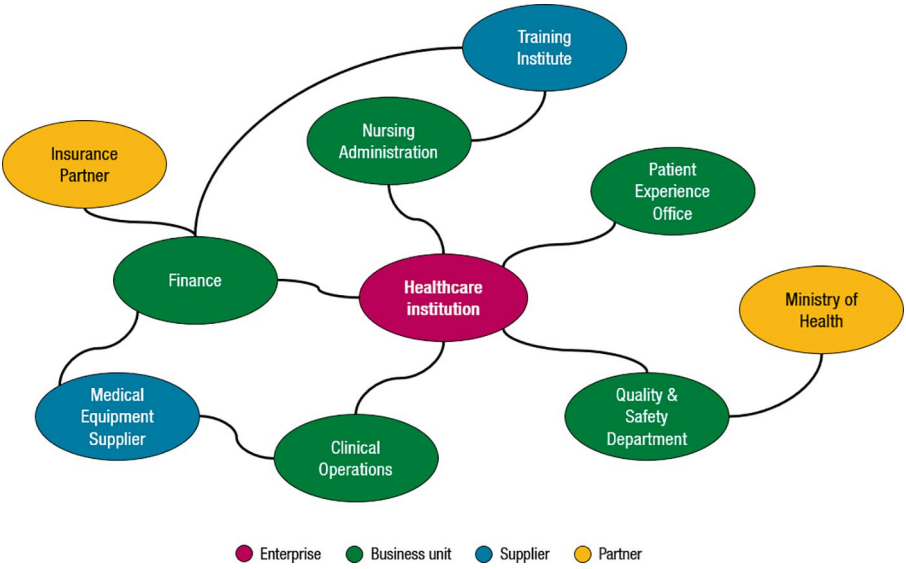


Figure 12-16. *Organization Diagram*

Business Glossary Catalog: A structured list of business terms and their unambiguous definitions that are used throughout the organization. This ensures semantic consistency across business processes, applications, and data models. The glossary (see Table 12-25) is essential for bridging the communication gap between business and IT stakeholders by standardizing the terminology used in requirements, reports, services, and interfaces. This artifact evolves throughout the Architecture Development Method.

Table 12-25. *Business Glossary Catalog*

Term	Definition	Synonyms	Usage Context	Capability
Feedback Case	A documented instance of patient feedback, complaint, or compliment.	Feedback Ticket	Patient Experience Portal, Quality Review	Patient Experience Management
Clinical Pathway	A standardized, evidence-based care plan for specific conditions.	Care Protocol	Workflow Optimization, EHR Task Flows	Clinical Services
Cost per Patient Day	Total operating cost divided by the number of patient days in a time period.	Per Diem Cost	Financial KPIs, Departmental Cost Reviews	Financial Management

Business Interaction Matrix: This matrix models the interactions between business roles and functions. It helps identify collaboration points, dependencies, and responsibility relationships within an organization. The matrix (see Table 12-26) is particularly useful for clarifying who does what and how organizational units engage with business capabilities.

Table 12-26. *Business Interaction Matrix*

	Business Function			
Business Role ▼	Patient Feedback Routing	Infection Risk Escalation	Resource Scheduling	Cost Efficiency Analysis
Patient Experience Manager	Performs	Not Involved	Not Involved	Not Involved
Infection Control Officer	Receives/Acts On	Performs	Not Involved	Not Involved
Nursing Administration	Not Involved	Informed	Performs	Supports
Financial Analyst	Not Involved	Not Involved	Not Involved	Performs

Actor/Role Matrix: Shows which actors perform which roles to support the definition of security and skill requirements (see Table 12-27).

Personally, I think this is one of the least common artifacts you create as an architect.

The reason is that the same information can easily be requested from the HR department.

It is questionable whether there is added value in keeping track of actors and roles in the form of an Actor/Role Matrix in the Architecture Repository. This creates an additional source of information that belongs with the HR department.

Understanding actor-to-role relationships is key to defining training needs, user security settings, and organizational change management.

Table 12-27. *Actor/Role Matrix*

Actor	Role
Maria Summers	Chief Executive Officer
James Noble	Chief Financial Officer
Chris West	Chief Information Officer
Elena Petrova	Medical Director
Kevin O'Brien	Patient Experience Manager
Susan Delgado	Chief Nursing Officer
Benjamin Lee	Chief Medical Information Officer
Hannah Duarte	Charge Nurse/Unit Manager

Business Footprint Diagram: This diagram illustrates the connections among business goals, organizational units, business functions, and services. It also maps these functions to the technical components that deliver the necessary capabilities.

The diagram provides clear traceability between technical components and the business goals they satisfy, demonstrating ownership of the identified services.

A Business Footprint Diagram only shows the key facts that link organizational unit functions to delivery services. They are used as communication platforms for senior-level (CxO) stakeholders. Figure 12-17 only shows the key entities and has removed the relations between them for better readability.

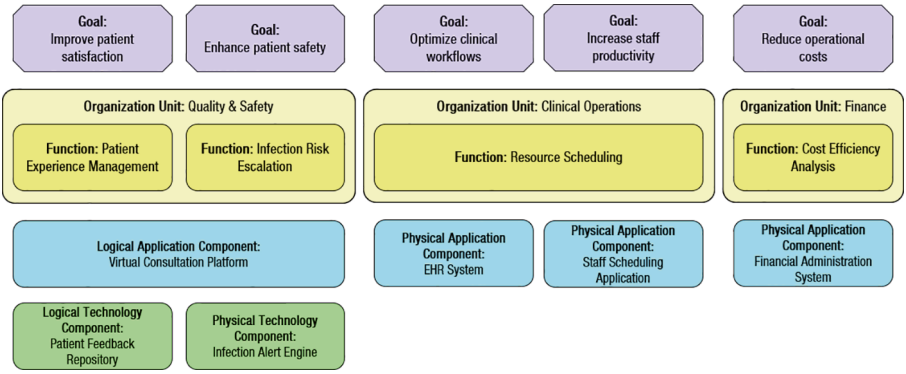


Figure 12-17. *Business Footprint Diagram*

Business Service/Information Diagram: Illustrates how business services consume or produce business and data objects (information). It shows the relationship between services (what the business offers) and the information they rely on or generate, helping ensure that business processes, applications, and data are aligned. This diagram (see Figure 12-18) is valuable during Phase B (Business Architecture) and Phase C (Data Architecture) to clarify which services handle specific information types and to identify potential redundancies, gaps, or integration needs.

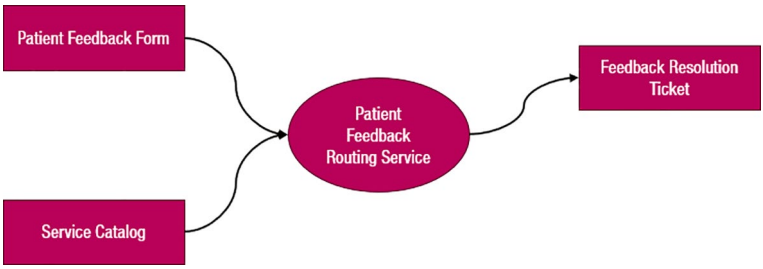


Figure 12-18. *Business Service/Information Diagram*

The Business Service/Information Diagram can also be visualized in tabular format using a Business Service/Information Catalog (see Table 12-28).

Table 12-28. *Business Service/Information Catalog*

Business Service	Consumes	Produces
Patient Feedback Routing Service	Patient Feedback Form, Service Catalog	Feedback Resolution Ticket
Infection Risk Incident Escalation	Infection Alert Report	Infection Incident Case File
Care Team Resource Scheduling Service	Staff Availability Data, Patient Census	Staff Schedule, Shift Alerts
Cost Efficiency Analytics Service	Cost per Service Data, Treatment Outcomes	Departmental Cost Report

Functional Decomposition Diagram: The purpose of the Functional Decomposition Diagram (See Figure 12-19) is to break down high-level business capabilities or functions into their constituent sub-functions. It helps architects and stakeholders understand *what* an organization does, supporting capability-based planning and aligning architecture with strategic objectives. This diagram provides a single-page overview of organizational functionality relevant to the architecture scope. It shows top-level business capabilities or functions and the logical grouping of sub-functions beneath them. A Functional Decomposition Diagram is very similar to a multi-level Business Capabilities or Function Catalog.

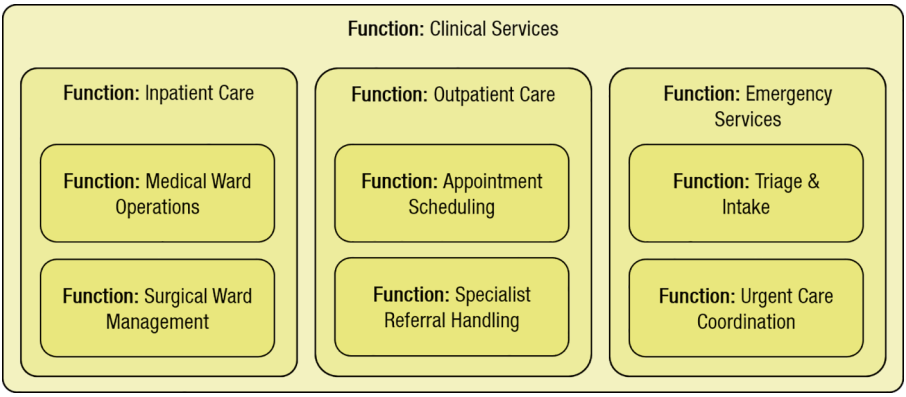


Figure 12-19. *Functional Decomposition Diagram (using business functions)*

Product Lifecycle Diagram: This diagram illustrates the stages a product or service goes through from inception to retirement. It provides a structured view of how products relate to business capabilities and processes. This allows architects to assess dependencies, transitions, and impacts. The diagram usually maps products to lifecycle phases, such as *concept*, *development*, *operation*, and *retirement*. This diagram is useful for managing portfolios, understanding transitions, and planning change initiatives. Each product is tracked individually, often with time-bound markers, enabling strategic alignment with organizational goals.

Figure 12-20 shows that the Patient Feedback Application will be operational until the end of Q2 2028. After that, it will be retired. Its successor, the Patient Feedback Application v2, is under development and will become operational at the start of Q3 2028. The Procurement Application will retire in Q2 2030. It will be replaced by the Supply Chain Management Application. Development of the replacement application will begin in Q3 2029.

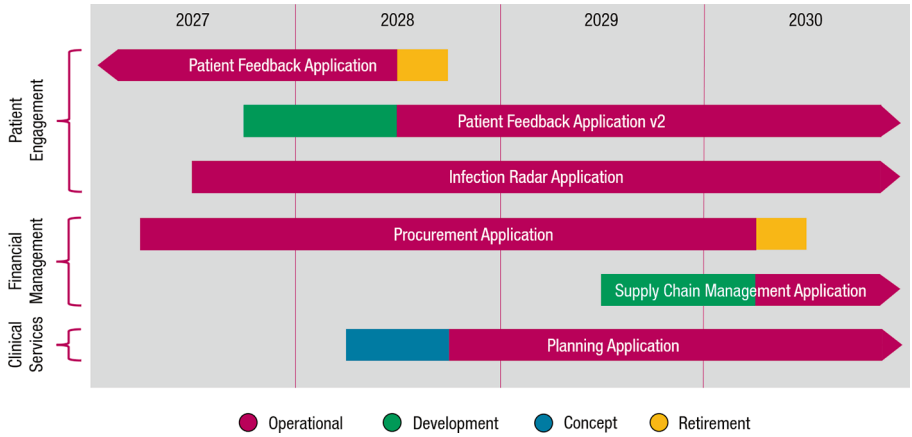


Figure 12-20. *Product Lifecycle Diagram*

As with most diagrams, the Product Lifecycle Diagram can also be visualized as a catalog. This is illustrated in Table 12-29.

Table 12-29. *Product Lifecycle Catalog*

Product Name	Lifecycle Stage	Timeframe	Related Capability
Patient Feedback Application	Operational	Q1 2025 to Q2 2028	Patient Engagement
Infection Radar Application	Operational	Q2 2027 to Q4 2035	Patient Engagement
Procurement Application	Operational	Q1 2027 to Q4 2029	Financial Management
Patient Feedback Application v2	Development	Q3 2027 to Q2 2028	Patient Engagement

(*continued*)

Table 12-29. (*continued*)

Product Name	Lifecycle Stage	Timeframe	Related Capability
Supply Chain Management Application	Development	Q2 2029 to Q1 2030	Financial Management
Planning Application	Concept	Q1 2028 to Q3 2028	Clinical Services
Patient Feedback Application	Retirement	Q3 2028	Patient Engagement
Procurement Application	Retirement	Q2 2030	Financial Management

Goal/Objective/Business Service Diagram: Visually shows how business goals are supported by objectives and how those objectives are realized through specific business services. This helps trace strategic intent through measurable targets to operational functionality. This Goal/Objective/Business Service Diagram (See Figure 12-21) is a derivative of the Business Footprint Diagram (Figure 12-17).

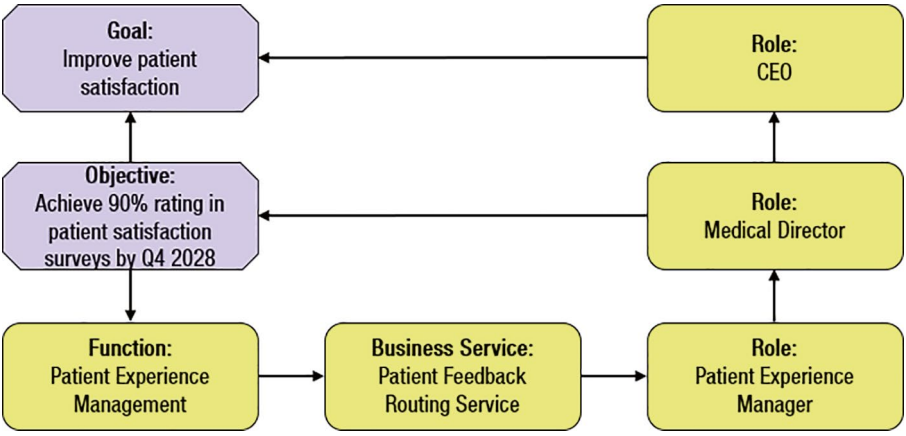


Figure 12-21. *Goal/Objective/Business Service Diagram*

Business Use-Case Diagram: Represents the interactions between business actors (roles) and business services or processes. It clarifies system boundaries and stakeholder interests by highlighting who uses what. Business Use-Case Diagrams are useful for identifying requirements and stakeholder concerns. It can also be used to map roles to business services, functions, or processes. As the architecture evolves, use cases can progress from the business level to include data, application, and technology details.

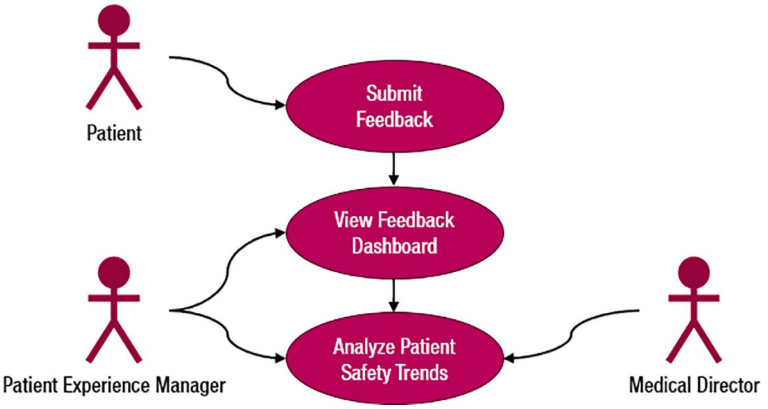


Figure 12-22. *Business Use-Case Diagram*

Organization Decomposition Diagram: Describes the links between actor, roles, and location within an organization tree.

An Organization Decomposition Diagram should provide a chain of command of owners and decision-makers in the organization. Although it is not the intent of the Organization Decomposition Diagram to link goal to organization, it should be possible to intuitively link the goals to the stakeholders from the Organization Decomposition Diagram.

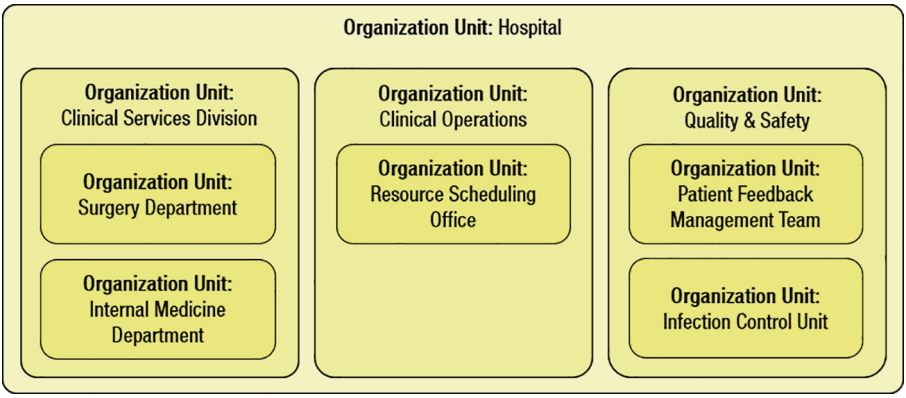


Figure 12-23. *Organization Decomposition Diagram*

Process Flow Diagram: The purpose of a Process Flow Diagram is to depict all mappings related to the *process* metamodel entity.

Process Flow Diagrams illustrate the sequential flow of control between activities and may use swimlane techniques to depict the ownership and execution of process steps. For instance, the application supporting a process step can be displayed as a swimlane.

In addition to showing a sequence of activities, these diagrams can detail the controls that apply to a process, the events that trigger or result from the completion of a process, and the products generated from process execution.

Process Flow Diagrams are useful for elaborating on the architecture with subject matter experts because they allow experts to describe *how the job is done* for a particular function.

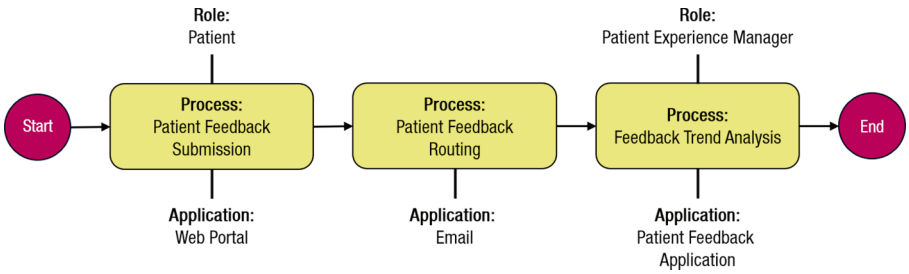


Figure 12-24. *Process Flow Diagram*

Business Event Diagram: Shows how business events trigger business processes, functions, or services and their resulting outcomes. A business event is an occurrence, either internal or external, that requires a response from the organization. This diagram (See Figure 12-25) helps model an organization's reactive behavior, which is essential for aligning processes with unpredictable or time-sensitive stimuli. Events may originate from users, systems, or environmental changes and often lead to process initiation, decision-making, or notifications.

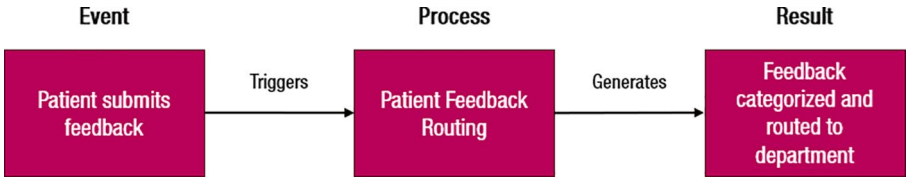


Figure 12-25. *Business Event Diagram*

Information Map: This map shows a collection of information concepts and their relationships to one another. It provides a structured view of an organization's key information concepts and how they relate to business capabilities, value streams, and processes. Information concepts reflect the business vocabulary. The Information Map (see Table 12-30) helps identify what information is critical to the organization, where it flows, and how it connects to decision-making and operational

performance. The Information Map organizes information into categories and shows relationships between concepts to support traceability, governance, and integration.

Table 12-30. *Information Map*

Information Concept	Information Concept Category	Definition	Related Information Concepts
Patient Feedback	Transactional	Input submitted by patients about their experience, including complaints, compliments, and safety concerns.	Patient ID, Feedback Category, Department
Feedback Category	Reference	Standard classification of feedback types such as hygiene, wait time, or staff behavior.	Patient Feedback
Scheduled Staff Roster	Transactional	A daily or weekly assignment list showing which staff are scheduled for shifts.	Staff Member, Shift Time, Unit
Staff Availability	Reference	Availability status of individual staff based on contracts, leave, or training.	Scheduled Staff Roster

The *transactional* category of an information concept refers to information generated by business activities or events. This type of information is dynamic, time-sensitive, and event-driven. It is frequently updated or changed and tied to a specific process or workflow.

The *reference* category assigned to information concepts refers to stable, reusable, and governed information. It supports categorization, validation, and consistent interpretation across processes. This type of information changes slowly, if at all, over time. It can be used for lookup, classification, or validation.

Phase B (Business Architecture) provides critical inputs to other phases of the Architecture Development Method. The insights gained about the business help guide decisions in subsequent phases:

- Business Architecture informs how data and applications should be structured to support business processes during Phase C (Information Systems Architectures).
- The Technology Architecture (established in Phase D (Technology Architecture)) will be aligned with the business needs defined in Phase B.
- Projects and solutions are identified in Phase E (Opportunities and Solutions), based on the business requirements from the Business Architecture Phase.

It is critical to devote sufficient time and attention to Phase B because it is the foundation for all subsequent architectural work.

12.6. Phase C: Information Systems Architectures

The purpose of the Information Systems Architectures Phase is to develop a blueprint of the information systems that support the Business Architecture. It identifies the key information systems required to implement the business capabilities and defines the Data and Application Architectures. The architects assess the existing systems and determine the gaps that need to be addressed. Key deliverables from this phase include data models and application portfolios.

Phase C is typically divided into two distinct yet interconnected architectures:

- **Data Architecture:** Focuses on the organization's data assets, data management practices, and governance of data. It describes how data is stored, managed, and utilized across the organization.
- **Application Architecture:** Focuses on the individual applications needed by the organization and how they interact with each other and with users. This includes the design of application components and their relationships, ensuring they align with the business goals.

The TOGAF Standard refers to the BIZBOK Guide for the development of Business Architecture. The BIZBOK Guide addresses the concept of information and plays an important role in understanding the relationship between business processes and functions and the use of information enabled by applications. The information concept thus assumes a role between Business Architecture and Data Architecture.

Unfortunately, this concept is not yet sufficiently reflected in the Fundamental Content of the TOGAF Standard.

The framework explains the concept of information mapping in the Series Guide "Information Mapping" [22] and bases the creation of an Information Map on the content as described in the BIZBOK Guide [5].

The two architectures that make up Phase C (Information Systems Architectures) are detailed in the following sections.

12.6.1 Developing Data Architecture

The objectives of Phase C (Data Architecture) are consistent with Phase B (Business Architecture). The first objective is to *craft the Target Data Architecture* to outline how the organization should function to meet business objectives. For example, identifying the data entities and their relationships and defining data storage, handling, and integration approaches.

The second objective is to *identify potential Data Architecture components for the Architecture Roadmap*, pinpointing the gaps between Baseline and Target Data Architectures that must be bridged. The key to both objectives is to reuse existing material as much as possible.

Table 12-31. *Steps to perform Phase C (Data Architecture)*

Step to Perform	Description of Step	Output Generated by Step
Select reference models, viewpoints, and tools	Choose relevant models and viewpoints from the Architecture Repository to match the business goals and stakeholder interests.	Views and viewpoints Required building blocks (catalogs, matrices, and diagrams) Architecture requirements
Develop Baseline and Target Data Architecture description	Describe the current and future Data Architecture to a degree that supports the architecture vision, building on the Architecture Repository's resources as needed.	Baseline Data Architecture description Target Data Architecture description

(*continued*)

Table 12-31. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Perform gap analysis	Confirm internal consistency, identify any gaps between baseline and target, and validate that models align with business objectives.	Gap analysis results
Define candidate roadmap components	Using the baseline, target, and gap analysis, build an initial roadmap to guide future actions and prioritization across the remaining architecture phases.	Data Architecture Roadmap components
Resolve impacts across the Architecture Landscape	Assess the Data Architecture's wider impacts across the organization and its alignment with existing or planned projects.	Validated building blocks and other artifacts
Conduct formal stakeholder review	Revisit the architecture's initial motivation and make any necessary refinements to ensure it's fit to support other architecture domains.	Updated Statement of Architecture Work
Finalize the Data Architecture	Standardize building blocks, fully document them, and check alignment with business goals. Finalize documentation within the Architecture Repository.	Documented building blocks and requirements
Update the Architecture Definition Document	Complete all relevant sections in the Architecture Definition Document, adding models, reports, and graphics as needed to capture a clear architectural view.	Updated Architecture Definition Document

The steps defined in Table 12-31 are the same as those in the previous phase, Phase B (Business Architecture). Because phases B, C, and D all deal with creating specific parts of the Enterprise Architecture (Business Architecture, Data Architecture, and Application Architecture, respectively), the steps don't really differ between the three phases. The minor differences between the phases are described below. For a detailed description of all the steps, refer to Table 12-9 in Section 12.5.1.

Select Reference Models, Viewpoints, and Tools: Refer to the Architecture Repository to select Data Architecture resources, like reference models and patterns, that align with core business drivers and address stakeholder needs (requirements). Zero in on the Data Architecture viewpoints.



Access the Architecture Repository: Begin by exploring the organization's Architecture Repository to identify available resources pertinent to Data Architecture.

Identify Relevant Reference Models and Patterns: Within the repository, locate Data Architecture reference models and patterns that resonate with the organization's business objectives and stakeholder expectations.

Select Appropriate Viewpoints: Choose Data Architecture viewpoints that effectively address and represent the concerns of stakeholders, ensuring comprehensive coverage of their requirements.

Evaluate Tools and Techniques: Assess the tools and techniques available in the repository to determine which ones best support the selected models and viewpoints, facilitating efficient architecture development.

Align with Business Drivers: Ensure that the selected models, viewpoints, and tools are in harmony with the organization's core business drivers, promoting coherence between Data Architecture and business strategy.

Document the Selection Process: Maintain clear documentation of the choices made and the rationale behind them, providing transparency and a reference for future architectural endeavors.

Develop Baseline and Target Data Architecture Description: Develop a baseline description of the current Data Architecture, covering as much as necessary to effectively build toward the Target Data Architecture. Use existing architecture descriptions if available, since they should be used as the basis for the baseline description. When developing the target description for the Data Architecture, focus on creating only what's necessary to support the architecture vision. Use the Architecture Repository Reference Library to locate the essential building blocks within the Data Architecture. As suggested in the previous phase, if stakeholder needs require new models, use the initial reference models found in the Reference Library of the Architecture Repository as a baseline to shape these new elements.

Consider exploring different Target Architecture options and review them with stakeholders using the Architecture Alternatives and Trade-Offs approach (see Chapter 14, Section 14.9) to ensure the architecture aligns with business needs.



Understand the Current Data Architecture: Take stock of all existing data assets, systems, and processes. Document how data flows within the organization and identify any existing issues or gaps.

Gather Existing Documentation: Locate any current architecture descriptions or data flow diagrams. Use these documents as a foundation to build the Baseline Data Architecture.

Define the Future Data Architecture Needs: Clarify the organization's goals and how data supports these objectives. Determine what changes or improvements are necessary in the Data Architecture to meet future demands.

Use Available Resources: Consult the Architecture Repository Reference Library to find standard building blocks and best practices. Incorporate these elements to ensure consistency and efficiency in the Target Data Architecture.

Develop the Target Data Architecture: Design the Target Data Architecture that aligns with the organization's vision. Focus on creating components that are essential to support business objectives without overcomplicating the design.

Explore Different Options: Consider various architectural approaches and solutions. Evaluate the pros and cons of each option in the context of the organization's needs.

Engage Stakeholders for Feedback: Present the proposed Target Data Architecture options to key stakeholders. Discuss potential trade-offs and gather input to ensure the architecture aligns with business requirements.

Refine and Finalize the Data Architecture: Incorporate feedback from stakeholders to adjust the design. Ensure that the final architecture is practical, scalable, and aligned with the organization's strategic goals.

Perform Gap Analysis: This phase is where models are rigorously validated against core principles, objectives, and constraints, making sure they are fit for purpose. To ensure internal consistency and accuracy within the architecture models, dive into each view to identify and resolve potential conflicts. Evaluate the models for completeness, confirming that they align seamlessly with all stakeholder requirements. Finally, apply gap analysis to pinpoint any gaps between the Baseline and Target Architecture, as detailed in Chapter 14, Section 14.5.



Validate Models Against Core Principles: Begin by thoroughly reviewing each model to ensure they adhere to the organization's fundamental principles, objectives, and constraints. This step is crucial to confirm that the models are appropriate and aligned with the overarching goals.

Ensure Internal Consistency: Next, examine each view within the architecture models to identify any inconsistencies or conflicts. Addressing these issues is essential for maintaining the accuracy and reliability of the models.

Confirm Completeness: After resolving internal conflicts, assess the models to verify that they comprehensively meet all stakeholder requirements. This ensures that no critical elements are overlooked.

Conduct Gap Analysis: Finally, perform a gap analysis to detect any discrepancies between the Baseline and Target Architectures. This involves identifying areas where the current architecture falls short of meeting the target state.

Define Candidate Roadmap Components: After establishing the Baseline and Target Architectures and analyzing the gaps, it is essential to create components for the Architecture Roadmap. The results of the gap analysis lead to the creation of work packages (or initiatives) related to the architecture to be developed. These initiatives are the components to be used in the Architecture Roadmap. In this step, the Data Architecture

components are added to the draft roadmap created in the previous phase. A more refined roadmap is developed during Phase E (Opportunities and Solutions) (see Section 12.8).



Review the Gap Analysis Results: Begin by thoroughly examining the outcomes of the gap analysis. This will help in understanding the specific areas where the current architecture falls short in meeting the Target Architecture.

Identify Necessary Initiatives: For each identified gap, determine the initiatives or work packages required to address and close these gaps. These initiatives should be clearly defined and directly related to achieving the Target Architecture.

Add to the Draft Architecture Roadmap: Compile all the identified initiatives related to Data Architecture into the Draft Architecture Roadmap. This roadmap should outline the sequence and priority of each work package, providing a clear path from the Baseline to the Target Architecture.

Prepare for Further Refinement in Phase E: Recognize that this draft is not final. The Architecture Roadmap will undergo further refinement during Phase E (Opportunities and Solutions). This phase focuses on identifying and evaluating potential solutions, ensuring that the initiatives are feasible and aligned with business objectives.

Resolve Impacts Across the Architecture Landscape: Look across the Architecture Landscape and examine other architecture assets to answer key questions, such as the following:

- Does the Data Architecture influence any established architectures, and vice versa?
- Are there chances to reuse or extend elements from this Data Architecture to benefit other areas of the organization?

- Could this Data Architecture impact current or upcoming projects, and vice versa?

Discuss the results of the questions posed with the relevant stakeholders and try to determine whether or not they should lead to adjustments to the architecture. If so, update the architecture accordingly to resolve any potential impacts.



Review Existing Architectures: Examine current architectural assets to identify interactions between the Data Architecture and other established architectures.

Engage Stakeholders: Discuss the findings with relevant stakeholders to determine if adjustments to the architecture are necessary.

Update the Data Architecture: If changes are needed, revise the architecture to address potential impacts and ensure alignment across the organization.

Conduct Formal Stakeholder Review: It's essential to engage key stakeholders to ensure the Data Architecture Phase effectively captures and models the essence of the organization. The stakeholder perspectives ensure that the architecture truly reflects the organization's needs and goals. Gather these stakeholders to revisit the original objectives behind the architecture work and the Statement of Architecture Work. Evaluate if the proposed Data Architecture aligns with and supports the upcoming work in other architecture domains. Make refinements where necessary, based on stakeholder feedback.



Assemble Key Stakeholders: Identify and bring together individuals who have a vested interest in the architecture's success.

Revisit Original Objectives: Collaboratively review the initial goals set for the architecture work to ensure alignment.

Examine the Statement of Architecture Work: Assess this foundational document to confirm that the Data Architecture aligns with its directives.

Evaluate Alignment with Other Domains: Determine if the proposed Data Architecture supports and integrates well with forthcoming efforts in other architectural areas.

Collect Stakeholder Feedback: Encourage stakeholders to provide insights and perspectives on the current architectural approach.

Implement Necessary Refinements: Based on the feedback received, make adjustments to the Data Architecture to better meet organizational objectives.

Finalize the Data Architecture: Set clear standards for each building block, using reference models from the Architecture Repository's Reference Library wherever possible. Meticulously document each block (see Chapter 7, Section 7.5.2.2, for instructions on how to do this).

Afterward, do a final cross-check to ensure the architecture is aligned with overarching business goals. Document the *why* behind each decision; it's vital for linking back to organizational objectives.



Set Clear Standards for Each Building Block: Begin by defining clear standards for every component of the Data Architecture. Utilize reference models available in the Architecture Repository's Reference Library to ensure consistency and industry alignment.

Document Each Building Block Thoroughly: For each component, create detailed documentation.

Cross-Check Alignment with Business Goals: After documenting, perform a comprehensive review to ensure that the Data Architecture aligns with the organization's overarching business goals. This alignment is vital for the architecture's effectiveness and relevance.

Document the Rationale Behind Each Decision: For every decision made during this process, document the reasoning behind it. This practice is essential for maintaining a clear connection between your Data Architecture and the organization's objectives, facilitating future reviews and adjustments.

Update the Architecture Definition Document: Update the Architecture Definition Document by presenting the Baseline and Target Data Architectures. Share the document with the key stakeholders to gather feedback, ensuring refinements are made to keep the architecture aligned with organizational objectives. This iterative process is essential for achieving a comprehensive and goal-oriented architecture.



Share with Key Stakeholders: Distribute the updated Architecture Definition Document to essential stakeholders. Their insights are invaluable for ensuring the architecture meets organizational needs.

Gather and Incorporate Feedback: Collect feedback from stakeholders and refine the document accordingly. This iterative process helps in addressing concerns and improving the architecture.

Align with Organizational Objectives: Ensure that the updated architecture aligns with our organization's goals and strategies. This alignment is key to delivering value.

Similar to Phase B (Business Architecture), all the architecture work that needs to be done should be documented in what is called a *Statement of Architecture Work*. This document should be formally approved by the higher levels of management, in order to gain support and sponsorship for the architecture work to be done.

The Data Architecture Phase is key in ensuring the data infrastructure of an organization is designed to meet business goals efficiently, is secure, and is scalable for future needs.

12.6.2 Phase C (Data Architecture) Artifacts

Table 12-32 lists the artifacts that can be created during the execution of Phase C (Data Architecture).

Table 12-32. *Artifacts created during Phase C (Data Architecture)*

ADM Phase	Artifacts	Used Entities
Phase C: Data Architecture	Data Entity/Data Component Catalog	Data entity, logical/physical data component
	Data Entity/Business Function Matrix	Data entity, function (optionally, data entity relationship to owning organization unit)
	Application/Data Matrix	Logical application component, data entity
	Conceptual Data Diagram	Data entity
	Data Dissemination Diagram	Data entity, business service, logical/physical application component
	Data Security Diagram	Role, organization unit, logical/physical application component
	Data Migration Diagram	Data entity, location
	Data Lifecycle Diagram	Data entity, process

Data Entity/Data Component Catalog: This catalog identifies and maintains a list of the data used across the organization, relating data entities to data components showing how data entities are structured. Its purpose is to identify all data used in the organization and to encourage effective data use. This artifact (see Table 12-33) is based on the Information Map created previously (see Section 12.5.3). The Information Map defined information concepts. According to the TOGAF Standard, this entity is comparable to the logical data component entity. Mapping these concepts to logical or physical data components results in the Data Entity/Data Component Catalog.

Table 12-33. *Data Entity/Data Component Catalog*

Data Entity	Data Component	Business Service
Patient Feedback Record	Patient Feedback	Patient Feedback Routing Service
	Feedback Category	
Staff Schedule	Scheduled Staff Roster	Care Team Resource Scheduling Service
	Staff Availability	

The Data Entity/Data Component Catalog, when visualized graphically, resembles the illustration in Figure 12-26 (which uses only the *Patient Feedback Record* data entity).



Figure 12-26. *Data Entity/Data Component Diagram*

Data Entity/Business Function Matrix: The purpose of the Data Entity/Business Function Matrix is to depict the relationship between data entities and business functions within the organization. Business functions are supported and realized by business processes. The mapping of the data entity to the business function entity enables the organization to gain insight into the assigned ownership of data entities. The matrix illustrates where data is being used in the organization.

In my experience, it is much more common for an organization to want insight into what data is being used by what processes. In practice, mapping data entities to business functions will be less common than mapping data entities to processes.

In most cases, people in an organization are very familiar with the processes they perform on a daily basis. The mapping of data entities to processes will therefore provide a more helpful picture than the mapping to business functions. The latter entity generally paints a much more general picture than the process entity.

In this sense, the creation of a Data Entity/Business Process Matrix offers more advantages than the matrix proposed in the TOGAF Standard.

A Data Entity/Business Function Matrix (or Data Entity/Business Process Matrix) supports a gap analysis to determine whether any data entities are missing and need to be created. For illustrative purposes, Table 12-34 lists both business functions and processes.

Table 12-34. *Data Entity/Business Function and Process Matrix*

	Business Function		**Business Process**	
Data Entity ▼	Patient Experience Management	Workforce Planning and Scheduling	Patient Feedback Routing	Staff Schedule Generation
Patient Feedback Record	X		X	
Staff Schedule		X		X

Application/Data Matrix: Shows the relationship between applications (application components) and the data entities they create, use, update, or delete. It helps identify data duplication, integration needs, and application ownership over specific data. This matrix supports decisions about data centralization, application rationalization, and system interfaces. The matrix (see Table 12-35) typically lists data entities (rows) and applications (columns), with indicators like C (Create), R (Read), U (Update), and D (Delete) to show interactions.

Table 12-35. *Application/Data Matrix*

	Application Component			
Data Entity ▼	Patient Feedback Application	Infection Radar Application	Planning Application	Supply Chain Management Application
Patient Feedback Record	C, R, U, D			
Infection Incident Report	R	C, R, U		
Staff Schedule			R	
Department Cost				C, R, U

Conceptual Data Diagram: The primary purpose of the Conceptual Data Diagram (see Figure 12-27) is to illustrate the relationships between critical data entities within an organization. It is developed to address the concerns of business stakeholders.

It shows logical views of the relationships between critical data entities within the organization. It is largely comparable to the *Information Map* mentioned in Section 12.5.3.

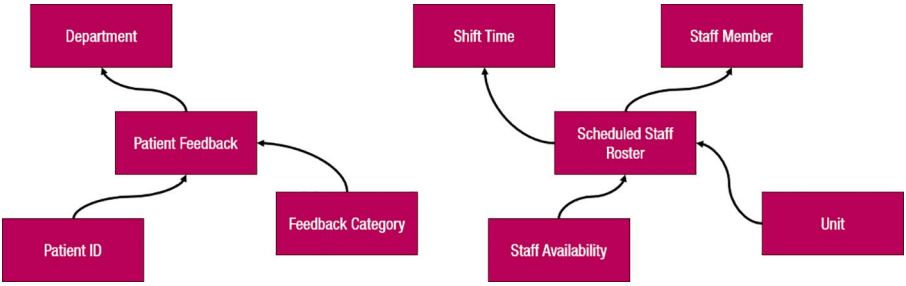


Figure 12-27. *Conceptual Data Diagram*

Data Dissemination Diagram: Shows the relationship between data entities, business services, and application components. The diagram (see Figure 12-28) shows how the *logical data components* are to be *physically realized by application components*. This allows effective sizing to be carried out and refined. Moreover, by assigning business value to data, an indication of the business criticality of application components can be gained.

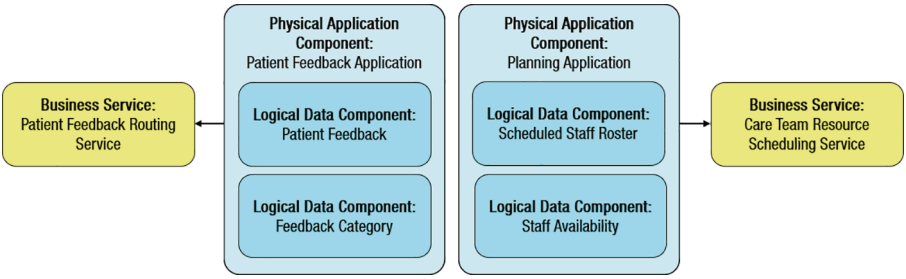


Figure 12-28. *Data Dissemination Diagram*

Data Security Diagram: Illustrates which data is accessed by which roles, organizational units, and applications. This information can be presented as a diagram or a catalog (see Table 12-36). The catalog displayed here maps data entities to security classifications and illustrates the access roles and protection mechanisms applied to each. Additional entities, such as business functions, business services, and locations, may

be added as well. The catalog specifies which roles or systems can read, write, or update specific data sets, thereby helping to ensure compliance with privacy, confidentiality, and regulatory requirements.

Table 12-36. *Data Security Catalog*

Data Entity	Security Classification	Access Roles	Protection Mechanism
Patient Feedback Form	Internal Use	Patient Experience Manager	Secure web access, pseudonymization
Infection Incident Log	Confidential	Quality and Safety	RBAC, immutable audit trail, limited access via VPN
Staff Credential Database	Restricted	HR Admin, Compliance Officer	Encrypted, multi-factor access

Data Migration Catalog: This catalog shows how data is extracted from baseline or source location(s) and loaded into target location(s). It may show where data is transformed and/or cleansed on the way. In Phase C (Data Architecture) of the Architecture Development Method, the catalog (see Table 12-37) is likely to be at an overview level. Later, it may be elaborated to show which source data items map to which target data items. For illustration purposes, a Data Migration Diagram is also shown (Figure 12-29), visualizing the first row of the catalog.

Table 12-37. *Data Migration Catalog*

Source System	Target System	Data Entities Migrated	Transformation Activities
Patient Feedback Application	Patient Feedback Application v2	Patient feedback, feedback categories	Reformat text fields, reclassify codes
Manual Excel Reports	Quality Management System	Infection-related incident logs	Normalize dates, map legacy IDs
Archived Survey Data (CSV)	Analytics Data Lake	Patient satisfaction scores, free text	Clean nulls, tokenize free text
Intranet Forms Repository	Service Recovery Workflow Tool	Feedback escalation requests	Convert form fields to structured JSON

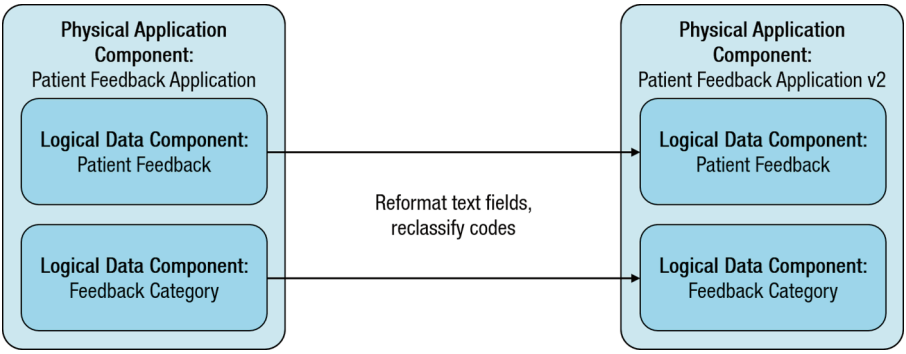


Figure 12-29. *Data Migration Diagram showing the first row of the Data Migration Catalog*

Data Lifecycle Diagram: The Data Lifecycle Diagram is an essential part of managing business data throughout its lifecycle from conception until disposal within the constraints of the business process. The diagram (see Figure 12-30) shows how a data entity progresses through various stages in its lifecycle. Typically, this includes the stages of creation, modification, use, storage, archival, and deletion.

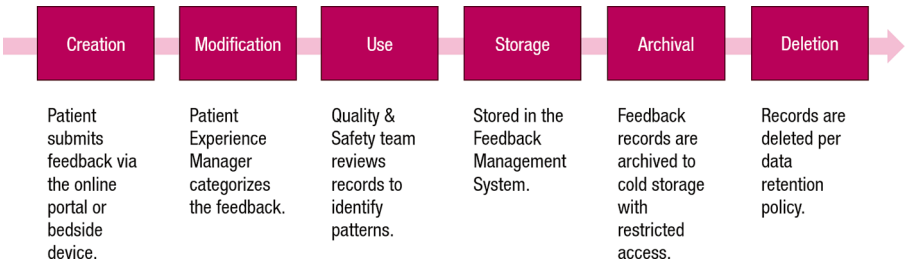


Figure 12-30. *Data Lifecycle Diagram*

Phase B (Business Architecture) provided the business context and requirements for the data described in this phase. The Data Architecture Phase influences the infrastructure and technology choices, especially for databases, storage, and data management platforms. It works in parallel with the Application Architecture to ensure data is effectively used by software and applications.

12.6.3 Developing Application Architecture

The objectives of Phase C (Application Architecture) are consistent with Phase B (Business Architecture). The first objective is to *craft the Target Application Architecture* to outline how the organization should function to meet business objectives.

The second objective is to *identify potential Application Architecture components for the Architecture Roadmap*, pinpointing the gaps between Baseline and Target Application Architectures that must be bridged. The key to both of these goals is the same as Phase B: reuse existing material as much as possible.

Table 12-38. *Steps to perform Phase C (Application Architecture)*

Step to Perform	Description of Step	Output Generated by Step
Select reference models, viewpoints, and tools	Choose relevant models and viewpoints from the Architecture Repository to match the business goals and stakeholder interests.	Views and viewpoints Required building blocks (catalogs, matrices, and diagrams) Architecture requirements
Develop Baseline and Target Application Architecture description	Describe the current and future Application Architecture to a degree that supports the architecture vision, building on the Architecture Repository's resources as needed.	Baseline Application Architecture description Target Application Architecture description
Perform gap analysis	Confirm internal consistency, identify any gaps between baseline and target, and validate that models align with business objectives.	Gap analysis results
Define candidate roadmap components	Using the baseline, target, and gap analysis, build an initial roadmap to guide future actions and prioritization across the remaining architecture phases.	Application Architecture Roadmap components

(*continued*)

Table 12-38. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Resolve impacts across the Architecture Landscape	Assess the Application Architecture's wider impacts across the organization and its alignment with existing or planned projects.	Validated building blocks and other artifacts
Conduct formal stakeholder review	Revisit the architecture's initial motivation and make any necessary refinements to ensure it's fit to support other architecture domains.	Updated Statement of Architecture Work
Finalize the Data Architecture	Standardize building blocks, fully document them, and check alignment with business goals. Finalize documentation within the Architecture Repository.	Documented building blocks and requirements
Update the Architecture Definition Document	Complete all relevant sections in the Architecture Definition Document, adding models, reports, and graphics as needed to capture a clear architectural view.	Updated Architecture Definition Document

The steps defined in Table 12-38 are the same as those in Phase B (Business Architecture). Because phases B, C, and D all deal with creating specific parts of the Enterprise Architecture (Business Architecture, Data Architecture, and Application Architecture, respectively), the steps don't really differ between the three phases. The minor differences between the phases are described below. For a detailed description of all the steps, refer to Table 12-9 in Section 12.5.1.

Select Reference Models, Viewpoints, and Tools: Refer to the Architecture Repository to select Application Architecture resources, like reference models and patterns, that align with core business drivers and address stakeholder requirements and constraints. Zero in on the Application Architecture viewpoints.



Check the Architecture Repository: Start by looking into the Architecture Repository. This is where all the organization's architecture resources are stored.

Find Application Architecture Resources: Within the repository, search for materials related to Application Architecture. These might include reference models and patterns that have been used before.

Match Resources to Business Goals: Review these materials to see which ones align with the main business objectives. Choose the ones that best support what the business aims to achieve.

Consider Stakeholder Needs: Think about the requirements and limitations of stakeholders. Ensure the selected resources address their concerns and fit within any constraints they have.

Select Appropriate Viewpoints: Identify the Application Architecture viewpoints that are most relevant. These perspectives will help in understanding and communicating the architecture effectively.

Develop Baseline and Target Application Architecture Description: Develop a baseline description of the current Application Architecture, covering as much as necessary to effectively build toward the Target Application Architecture:

- Use existing architecture descriptions if available.
- Focus on creating only what's necessary to support the architecture vision.

- Use the Architecture Repository Reference Library to locate the essential building blocks within the Application Architecture.
- Explore different Target Architecture options and review them with stakeholders using the Architecture Alternatives and Trade-Offs approach.



Develop the Baseline Application Architecture: Begin by documenting the current state of the organization's Application Architecture. If existing architecture descriptions are available, utilize them to save time and ensure consistency. Focus on capturing only the details necessary to support the architecture vision, avoiding unnecessary complexity.

Identify Essential Building Blocks: Leverage the Architecture Repository Reference Library to locate the fundamental components within the Application Architecture. This resource will help to identify existing building blocks that can be reused or adapted for the Target Architecture.

Explore Target Architecture Options: Consider various options for the Target Application Architecture. Engage stakeholders in reviewing these alternatives using the Architecture Alternatives and Trade-Offs approach. This collaborative process ensures that the selected architecture aligns with business objectives and stakeholder expectations.

Perform Gap Analysis: This step validates and refines architecture models to ensure accuracy, alignment, and completeness. The models are rigorously tested against principles, objectives, and constraints to confirm they are suitable for purpose. Each architecture view is examined for consistency and conflicts, while completeness is verified against stakeholder requirements. A gap analysis is then performed to identify differences between the Baseline and Target Architecture, as outlined in Chapter 14, Section 14.5.



Validate models against core principles: Begin by thoroughly reviewing each model to ensure they adhere to the organization's fundamental principles, objectives, and constraints. This step is crucial to confirm that the models are appropriate and aligned with the overarching goals.

Ensure internal consistency: Next, examine each view within the architecture models to identify any inconsistencies or conflicts. Addressing these issues is essential for maintaining the accuracy and reliability of the models.

Confirm completeness: After resolving internal conflicts, assess the models to verify that they comprehensively meet all stakeholder requirements. This ensures that no critical elements are overlooked.

Conduct gap analysis: Finally, perform a gap analysis to detect any discrepancies between the Baseline and Target Architectures. This involves identifying areas where the current architecture falls short of meeting the target state.

Define Candidate Roadmap Components: After establishing the Baseline and Target Architectures and analyzing the gaps, it is essential to create components for the Architecture Roadmap. The results of the gap analysis lead to the creation of work packages (or initiatives) related to the architecture to be developed. These initiatives are the components to be used in the Architecture Roadmap. In this step, the Application Architecture components are added to the draft roadmap created in Phase B (Business Architecture).



Identify the Baseline and Target Architectures: Start by understanding where the architecture currently stands (Baseline) and where it needs to go (Target). This will provide a foundation for everything.

Analyze the Gaps: Figure out the differences between the Baseline and Target. What's missing? What needs improvement?

Define Initiatives: From the gap analysis, come up with initiatives that will help bridge the gaps. These are the action points for developing the architecture.

Add Application Architecture Components: Now add the Application Architecture elements to the Architecture Roadmap.

Resolve Impacts Across the Architecture Landscape: Look across the Architecture Landscape and examine other architecture assets to answer key questions, such as the following:

- Does the Application Architecture influence any established architectures, and vice versa?
- Are there chances to reuse or extend elements from this Application Architecture to benefit other areas of the organization?
- Could this Application Architecture impact current or upcoming projects, and vice versa?

Discuss the results of the questions posed with the relevant stakeholders and try to determine whether or not they should lead to adjustments to the architecture. If so, update the architecture accordingly to resolve any potential impacts.



Look at the Whole Architecture Landscape: Check if the Application Architecture impacts other parts of the business and how other architectures might influence it.

Ask Key Questions: Does this Application Architecture affect other existing architectures, or vice versa? Can we reuse parts of this Application Architecture elsewhere? How could this Application Architecture affect ongoing or future projects?

Discuss These Findings with the Relevant People: Make sure to talk through the answers with stakeholders and decide if changes are needed in the architecture.

Adjust the Architecture If Necessary: If there are any issues or improvements identified, update the architecture to address them.

Conduct Formal Stakeholder Review: It's essential to engage key stakeholders to ensure the Application Architecture Phase effectively captures and models the essence of the organization. The stakeholder perspectives ensure that the architecture truly reflects the organization's needs and goals. Gather these stakeholders to revisit the original objectives behind the architecture work and the Statement of Architecture Work. Evaluate if the proposed Application Architecture aligns with and supports the upcoming work in other architecture domains. Make refinements where necessary, based on stakeholder feedback.



Gather the Right People: Start by bringing together the key stakeholders. These are the people whose input is critical for making sure the architecture reflects the organization's real needs.

Check the Original Goals: Review the initial objectives and the Statement of Architecture Work. This will help ensure the work aligns with what was originally planned.

Align with Other Domains: Evaluate whether the proposed Application Architecture fits with and supports the upcoming work in other areas of architecture.

Refine Based on Feedback: Make changes based on the feedback from stakeholders to ensure the architecture is on track and effective.

Finalize the Application Architecture: Set clear standards for each building block, using reference models from the Architecture Repository's Reference Library wherever possible. Document each block in detail (see Chapter 7, Section 7.5.2.2).

Do a final cross-check to ensure the architecture is aligned with overarching business goals. Document the *why* behind each decision; it's vital for linking back to organizational objectives.



Set Standards for Each Building Block: Start by defining clear standards for every component. Use reference models from the Architecture Repository's Reference Library whenever possible. This provides a solid foundation to build upon.

Document Each Building Block: Record detailed information about each building block.

Align with Business Goals: Do a final check to make sure the architecture supports the overall business objectives.

Explain the Decisions: Write down the reasoning behind each architectural decision. This is crucial for connecting each decision back to the organization's goals.

Update the Architecture Definition Document: Update the Architecture Definition Document by presenting the Baseline and Target Application Architectures. Share the document with the key stakeholders to gather feedback, ensuring refinements are made to keep the architecture aligned with organizational objectives. This iterative process is essential for achieving a comprehensive and goal-oriented architecture.



Update the Architecture Definition Document: Add details about both the Baseline and Target Application Architectures.

Share with Key Stakeholders: Distribute the updated document to those who matter most in the decision-making process.

Gather Feedback: Collect input from stakeholders to ensure the document aligns with the organization's goals.

Refine the Application Architecture: Use the feedback to make necessary adjustments, ensuring continuous alignment with business objectives.

Iterate the Process: This feedback loop is key to refining the architecture and keeping it relevant as the business evolves.

Similar to the previous phase, all the architecture work that needs to be done should be documented in the *Statement of Architecture Work*.

Phase C (Application Architecture) focuses on defining a structured, cohesive, and efficient application landscape that supports business processes and objectives while considering technical, strategic, and operational requirements. The Application Architecture Phase ensures that the applications are interoperable and can exchange data effectively.

12.6.4 Phase C (Application Architecture) Artifacts

Table 12-39 lists the artifacts that can be created during the execution of Phase C (Application Architecture).

Table 12-39. *Artifacts created during Phase C (Application Architecture)*

ADM Phase	Artifacts	Used Entities
Phase C: Application Architecture	Application Portfolio Catalog	Application service, logical/physical application component
	Interface Catalog	Logical/physical application component
	Application/Organization Matrix	Logical application component, organization unit, actor, business service
	Role/Application Matrix	Logical application component, role
	Application/Function Matrix	Logical application component, function, business service
	Application Interaction Matrix	Application service, logical/physical application component
	Application Communication Diagram	Logical/physical application component, data entity
	Application and User Location Diagram	Logical/physical application component, location
	Application Use-Case Diagram	Actor, role, logical application component
	Enterprise Manageability Diagram	Logical/physical application component, logical technology component

(*continued*)

Table 12-39. (*continued*)

ADM Phase	Artifacts	Used Entities
	Process/Application Realization Diagram	Logical application component, process
	Software Engineering Diagram	Physical application component (composed of packages, modules, services, and operations)
	Application Migration Diagram	Logical/physical application component, staging areas (plateaus), logical technology component
	Software Distribution Diagram	Physical application component, physical technology component, location

Application Portfolio Catalog: A structured list of all the software applications used within the organization. It includes information on the purpose, lifecycle status, and how the applications support business capabilities and processes. This catalog (see Table 12-40) helps assess redundancy, alignment, interoperability, and risk in the application landscape. It supports gap analysis, rationalization, and planning in later phases.

Table 12-40. *Application Portfolio Catalog*

Application Name	Business Function Supported	Purpose	Lifecycle Status	Interfaces
Patient Feedback Application	Patient Experience Management	Collects and routes patient feedback	Production	EHR, CRM
Planning Application	Workforce Planning and Scheduling	Staff rostering	Development	HR System, Alert System
Workflow Management Application	Workforce Planning and Scheduling	Shift optimization	Production	HR System, Alert System
Financial Administration System	Financial Performance Management	Financial reporting	Production	ERP, BI Tool
Supply Chain Management Application	Financial Performance Management	Cost analytics and financial reporting	Development	ERP, BI Tool

Creating an Application Portfolio Catalog provides a comprehensive overview of all the applications used within an organization. It also reveals where applications with similar functionality are being used. For example, consider situations in which application components are used to run reports, take screenshots, or store and process customer data. It's more common than one might think for an organization to have several application components in use that provide virtually the same functionality.

Without a complete view of all application components, problems and challenges are likely to arise. For instance, different departments may request various applications when the desired functionality is often already available within the organization. Since the availability of the requested functionality cannot be readily verified, requests for new application components are often approved too quickly. This eventually leads to an overpopulation of the application landscape and the need for an organization to rationalize its applications, a process that can take several years.

In the absence of an Application Portfolio Catalog, a second problem that can arise is that costs are incurred that are not fully visible. For instance, an organization may discover that it is spending a lot of money on software licenses, yet the breakdown of those costs is unclear. The lack of an Application Portfolio Catalog hinders the ability to get a clear picture of license costs. Often, it is only after creating an Application Portfolio Catalog that it becomes apparent that license costs are driven by duplicate application functionality. Removing application components with duplicate (or sometimes triple or quadruple!) functionality from the application landscape can significantly reduce license costs.

The Application Portfolio Catalog provides an overview of the organization's applications. To take full advantage of the catalog, additional information should be added to it.

For example, consider adding confidentiality, integrity, and availability (CIA) information to the artifact. This provides a valuable overview of the information used by applications and the additional security measures required for compliance. When visualized as a diagram, the catalog provides a clear, comprehensible overview of the application landscape and its security-related points of attention.

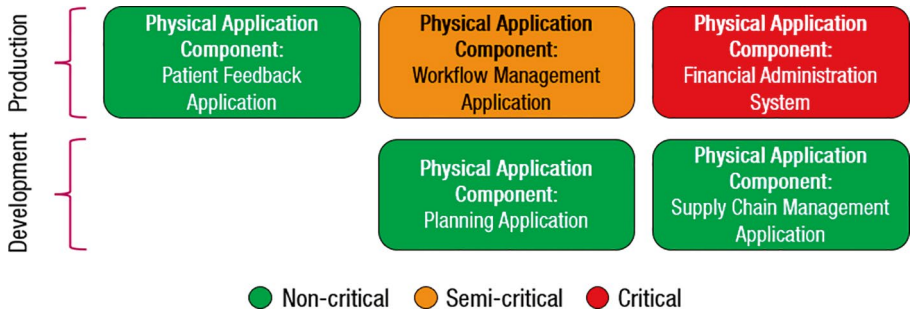


Figure 12-31. *Application Portfolio Diagram with CIA score*

Of course, the Application Portfolio Catalog can be enriched with various information components. For example, consider adding the location property to the catalog to indicate whether applications are on-premises or in the cloud. Another option is to add the capabilities automated by the applications to the list. This will show where work needs to be done to eliminate manual activities. Displaying that information through color views transforms the catalog into a powerful tool.

Interface Catalog: The purpose of the Interface Catalog is to scope and document the interfaces between applications to enable the overall dependencies between applications to be scoped as early as possible.

Applications will create, read, update, and delete data within other applications; this will be achieved by some kind of interface, whether via a batch file that is loaded periodically, a direct connection to another application's database, or some form of API or web service.

The mapping of one application component entity to another is an important step as it enables understanding the degree of interaction between applications. It allows an organization to understand the number and types of interfaces between applications and the degree of duplication of interfaces between applications.

The Interface Catalog displayed in Table 12-41 only shows four columns. This is done to improve readability. A complete catalog contains additional columns, including *data exchanged*, *protocol*, *frequency*, and *security*.

Table 12-41. *Interface Catalog*

Interface ID	Source System	Target System	Interface Type
IF-001	Patient Feedback Application	Quality Management Dashboard	REST API
IF-002	Patient Feedback Application	Nursing Admin Portal	REST API
IF-003	Nursing Admin Portal	Patient Feedback Application	REST API
IF-004	Workflow Management Application	Nursing Admin Portal	REST API
IF-005	Financial Administration System	Cost Analytics Dashboard	SQL View

Application/Organization Matrix: This matrix shows the relationship between applications and organizational units within the organization. Organizational units perform business operations. Some of these operations and services will be supported by applications. Mapping the application component to organization units provides insight into which applications are used by which organization units. This allows an organization to understand the application support requirements of the business services and processes carried out by a particular organizational unit. An example is shown in Table 12-42.

Table 12-42. *Application/Organization Matrix*

Application Component ▼	Organization Unit: Patient Experience Office	Quality and Safety	Clinical Operations	Finance
Patient Feedback Application	X	X	X	
Workflow Management Application			X	
Financial Administration System				X

Role/Application Matrix: The Role/Application Matrix illustrates the relationship between applications and the business roles that utilize them within an organization.

People within an organization interact with applications. During these interactions, they assume a specific role to perform a task. The matrix assigns application usage to specific roles within the organization. This matrix (see Table 12-43) is essential to any transition to role-based computing. Consequently, it is primarily used in the context of security.

Table 12-43. *Role/Application Matrix*

Role ▼	Application Component: Patient Feedback Application	Workflow Management Application	Financial Administration System
Patient Experience Manager	X		
Nursing Administration		X	
Financial Analyst			X

Application/Function Matrix: Demonstrates the relationship between applications and business functions within the organization. Business functions are performed by organizational units. Some of the business functions and services will be supported by applications. The Application/Function Matrix (see Table 12-44) demonstrates which applications are used by which business functions.

Table 12-44. *Application/Function Matrix*

	Business Function		
Application Component ▼	Patient Feedback Routing	Resource Scheduling	Cost Efficiency Analysis
Patient Feedback Application	X		
Workflow Management Application		X	
Financial Administration System			X

Application Interaction Matrix: The purpose of the Application Interaction Matrix is to illustrate the communication relationships between applications. The matrix (see Table 12-45) shows the equivalent of the Interface Catalog in the form of a map of application interactions. It is a two-dimensional table with logical or physical application components in both the rows and columns. The application service entity is displayed in the intersecting cells.

Table 12-45. *Application Interaction Matrix*

	Application Component			
Application Component ▼	Patient Feedback Application	Quality Management Dashboard	Workflow Management Application	Nursing Admin Portal
Patient Feedback Application		Push safety alerts		Submit complaints
Quality Management Dashboard	Read feedback data		View escalation	
Workflow Management Application		Escalate issue		Show nurse tasks
Nursing Admin Portal	Submit patient input	Report hygiene issues	View shift plan	

Application Communication Diagram: Illustrates how different applications interact by showing the flow of data between them across interfaces. It details which applications exchange information and which protocols or mechanisms are used. Optionally, it also details which business services or processes those exchanges support.

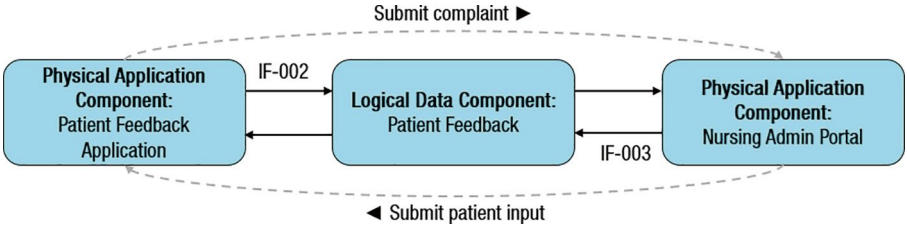


Figure 12-32. *Application Communication Diagram*

Application and User Location Diagram: Shows where applications are used in relation to user groups and locations. It helps visualize the distribution of application services and their geographic dependencies. This diagram clearly depicts business locations where business users typically interact with applications. It can also be used to illustrate the location where the application infrastructure is hosted.

In practice, this diagram will rarely be used. Adding the application location to the Application Portfolio Catalog eliminates the need for an Application and User Location Diagram.

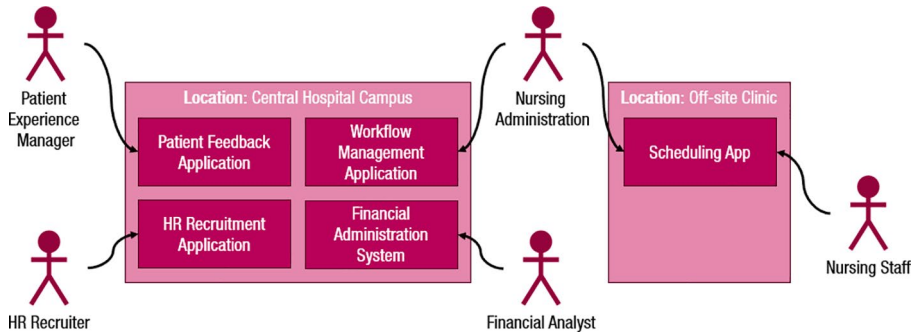


Figure 12-33. *Application and User Location Diagram*

Application Use-Case Diagram: Illustrates how business users (actors) interact with application components through specific application services to support business processes or capabilities. It describes who uses what, how, and for what purpose at the application layer. The Application and User Location Diagram can be expanded to include the application services used by users (actors) and applications.

Enterprise Manageability Diagram: This diagram is essentially a filter of the Application Communication Diagram. It illustrates how the organization's applications will be monitored, controlled, and supported

to ensure reliable operation in alignment with business needs. The diagram identifies management services, such as monitoring, alerting, and logging. It defines the relationships between system elements and their manageability controls and supports the operational governance of the architecture. Analysis can reveal duplication, gaps, and opportunities in an organization's IT service management operation.

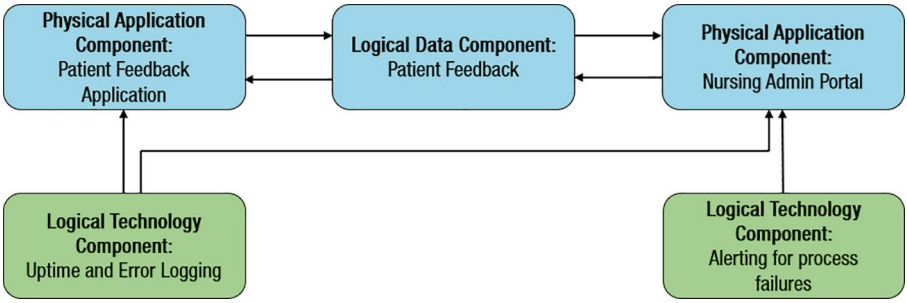


Figure 12-34. *Enterprise Manageability Diagram*

Process/Application Realization Diagram: The Process/Application Realization Diagram shows how a business process is supported by one or more application components. The diagram maps the execution of a process to the applications that enable or automate it, highlighting integration points and data flows.

The Process/Application Realization Diagram enhances the Application Communication Diagram by adding any sequencing constraints. It can also pinpoint process efficiency improvements that could reduce interaction traffic between applications.

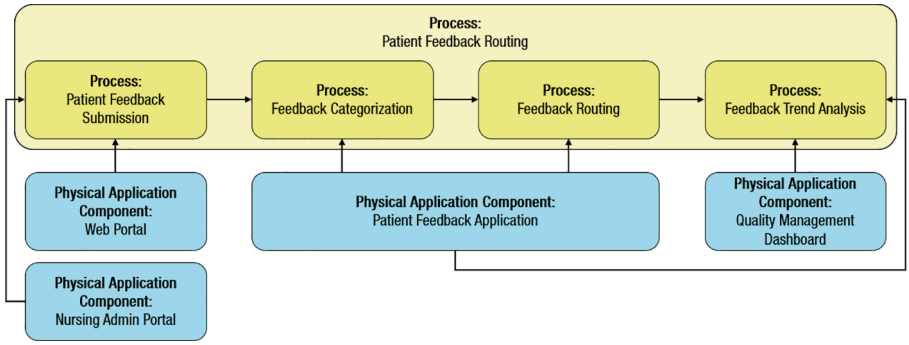


Figure 12-35. *Process/Application Realization Diagram*

Software Engineering Diagram: From a development perspective, it breaks applications into packages, modules, services, and operations. It enables more detailed impact analysis when planning migration stages and analyzing opportunities and solutions. It is ideal for application development and management teams when working with complex environments.

Since none of the three core architecture roles (see Chapter 10, Section 10.2) creates this artifact, an example is omitted here.

Application Migration Diagram: Identifies application migration from baseline to target application components. It enables a more accurate estimation of migration costs by showing precisely which applications and interfaces need to be mapped between migration stages.

It would identify temporary applications, staging areas, and the infrastructure required to support migrations (e.g., parallel run environments and others).

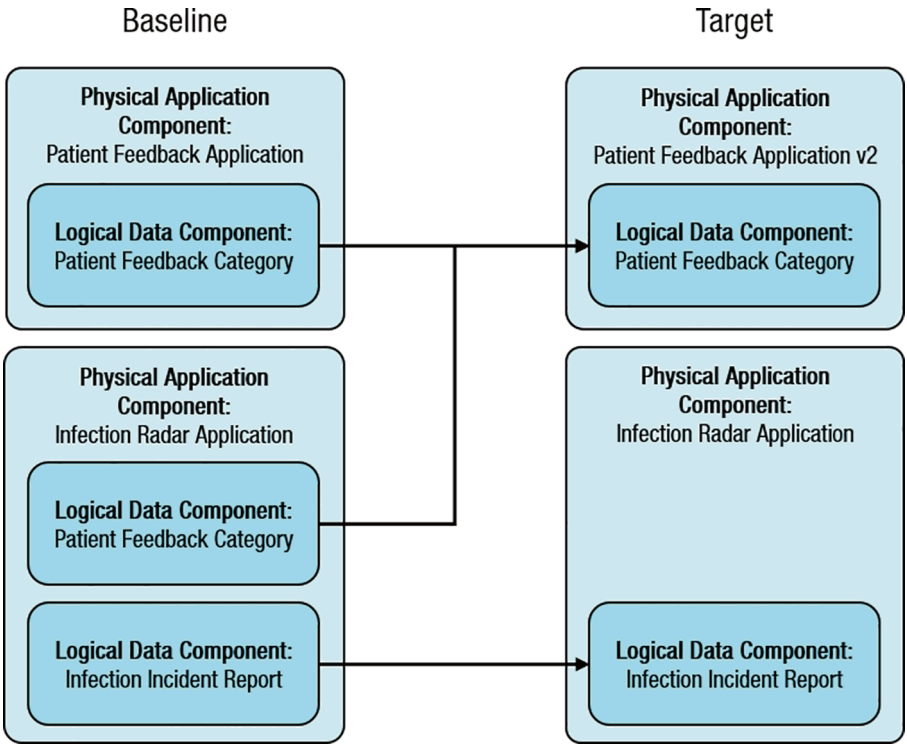


Figure 12-36. *Application Migration Diagram*

Software Distribution Diagram: This diagram illustrates the distribution of physical applications across physical technology and its location. It is useful for system upgrades or application consolidation projects.

This diagram largely follows the Application and User Location Diagram. A similar result can be achieved by adding application locations to the Application Portfolio Catalog. This eliminates the need for a Software Distribution Diagram.

Phase C (Information Systems Architectures) builds on the understanding of the business goals and objectives established in Phase B (Business Architecture).

Once the information systems architecture is defined, it serves as an input to the next phase (Phase D (Technology Architecture)), which focuses on the underlying technology infrastructure required to support the applications and data.

Any gaps identified in Phase C will drive planning and decisions (see Section 12.8) about how to implement or update systems in later phases.

In summary, the Information Systems Architectures Phase is essential for defining the structure of data and applications that enable the organization's Business Architecture. It helps bridge the gap between the business strategy and the technology infrastructure by specifying the systems that manage data and processes critical to business functions.

12.7. Phase D: Technology Architecture

Phase D (Technology Architecture) focuses on defining the technology infrastructure necessary to support the Business, Data, and Application Architectures defined in the previous phases of the Architecture Development Method. This phase, which may follow the Application Architecture (Phase C), is geared toward determining the most suitable technology components, platforms, and services needed to enable efficient operation of the organization's applications. In Phase D, architects identify hardware, software, and network infrastructure requirements and determine technology standards to ensure consistency, scalability, and security. The goal is to create a well-defined Technology Architecture that aligns with the broader organizational goals while ensuring interoperability and future readiness.

12.7.1 Developing Technology Architecture

The objectives of Phase D (Technology Architecture) – as well as those of Phase C (Data and Application Architectures) – are consistent with Phase B (Business Architecture). The first objective is to *craft the Target Technology Architecture* to outline how the organization should function to meet business objectives.

This seems like a logical first step, but in nine out of ten organizations, IT is viewed as a separate entity with its own planning and strategy. The idea behind the TOGAF Standard is to link the architecture work that takes place within each architecture domain to the business goals.

The Technology Architecture defines the IT infrastructure, systems, and services required to run the organization's applications and meet the business needs.

The second objective is to *identify potential Technology Architecture components for the Architecture Roadmap*, locating the gaps between Baseline and Target Technology Architectures that need to be addressed. It is important to reuse existing material as much as possible.

Table 12-46. *Steps to perform Phase D (Technology Architecture)*

Step to Perform	Description of Step	Output Generated by Step
Select reference models, viewpoints, and tools	Choose relevant models and viewpoints from the Architecture Repository to match the business goals and stakeholder interests.	Views and viewpoints Required building blocks (catalogs, matrices, and diagrams) Architecture requirements
Develop Baseline and Target Technology Architecture description	Describe the current and future Technology Architecture to a degree that supports the architecture vision, building on the Architecture Repository's resources as needed.	Baseline Technology Architecture description Target Technology Architecture description
Perform gap analysis	Confirm internal consistency, identify any gaps between baseline and target, and validate that models align with business objectives.	Gap analysis results
Define candidate roadmap components	Using the baseline, target, and gap analysis, build an initial roadmap to guide future actions and prioritization across the remaining architecture phases.	Technology Architecture Roadmap components

(*continued*)

Table 12-46. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Resolve impacts across the Architecture Landscape	Assess the Technology Architecture's wider impacts across the organization and its alignment with existing or planned projects.	Validated building blocks and other artifacts
Conduct formal stakeholder review	Revisit the architecture's initial motivation and make any necessary refinements to ensure it's fit to support other architecture domains.	Updated Statement of Architecture Work
Finalize the Technology Architecture	Standardize building blocks, fully document them, and check alignment with business goals. Finalize documentation within the Architecture Repository.	Documented building blocks and requirements
Update the Architecture Definition Document	Complete all relevant sections in the Architecture Definition Document, adding models, reports, and graphics as needed to capture a clear architectural view.	Updated Architecture Definition Document

The steps defined in Table 12-46 are the same as those in Phase B (Business Architecture). Phases B, C, and D all deal with creating Baseline and Target Architectures. Therefore, the steps don't really differ between the three phases. The minor differences between the phases are described below. For a detailed description of all the steps, refer to Table 12-9 in Section 12.5.1.

Select Reference Models, Viewpoints, and Tools: Refer to the Architecture Repository to select Technology Architecture resources, like reference models and patterns, that align with core business drivers and address stakeholder requirements and constraints. Focus on the Technology Architecture viewpoints.



Explore the Architecture Repository: Begin by examining the Architecture Repository. This is the go-to resource for Technology Architecture materials, including reference models and patterns.

Align with Business Drivers: Identify the core business drivers of the organization. Understanding these will help choose resources that support and enhance these key objectives.

Address Stakeholder Needs: Consider the requirements and constraints of the stakeholders. Selecting resources that meet their expectations ensures the architecture will be well-received and effective.

Focus on Technology Architecture Viewpoints: Pay special attention to viewpoints specific to Technology Architecture. These perspectives will help to understand and address the technical aspects of the architecture.

Develop Baseline and Target Technology Architecture Description: Develop a baseline description of the current Technology Architecture, covering as much as necessary to effectively build toward the Target Technology Architecture:

- Use existing architecture descriptions if available.
- Focus on creating only what's necessary to support the architecture vision.
- Use the Architecture Repository Reference Library to locate the essential building blocks within the Technology Architecture.

- Explore different Target Architecture options and review them with stakeholders using the Architecture Alternatives and Trade-Offs approach.



Understand the Baseline Technology Architecture: Gather existing documentation and resources that detail the organization's current technology infrastructure. Focus on collecting only the information necessary to support the architecture vision, avoiding unnecessary details.

Identify Essential Building Blocks: Utilize the Architecture Repository Reference Library to locate and understand the fundamental components within the current Technology Architecture.

Explore Potential Target Architectures: Investigate various options for the organization's Target Technology Architecture. Consider different technologies, structures, and strategies that could be implemented.

Evaluate and Compare Options with Stakeholders: Engage with key stakeholders to review the potential Target Architectures. Use the Architecture Alternatives and Trade-Offs approach to assess the advantages and disadvantages of each option.

Select the Most Suitable Target Architecture: Based on stakeholder feedback and thorough evaluation, choose the Target Technology Architecture that best aligns with the organization's goals and needs.

Document the Baseline and Target Architectures: Clearly document both the Baseline and Target Technology Architectures. Ensure that this documentation is accessible to all relevant parties within the organization.

Perform Gap Analysis: The step focuses on ensuring that architecture models are accurate, consistent, and aligned with objectives and stakeholder needs. Each view is reviewed to resolve conflicts,

completeness is verified, and a gap analysis is carried out to highlight differences between the Baseline and Target Architecture, as outlined in Chapter 14, Section 14.5.



Validate models against core principles: Begin by thoroughly reviewing each model to ensure they adhere to the organization's fundamental principles, objectives, and constraints. This step is crucial to confirm that the models are appropriate and aligned with the overarching goals.

Ensure internal consistency: Next, examine each view within the architecture models to identify any inconsistencies or conflicts. Addressing these issues is essential for maintaining the accuracy and reliability of the models.

Confirm completeness: After resolving internal conflicts, assess the models to verify that they comprehensively meet all stakeholder requirements. This ensures that no critical elements are overlooked.

Conduct gap analysis: Finally, perform a gap analysis to detect any discrepancies between the Baseline and Target Architectures. This involves identifying areas where the current architecture falls short of meeting the target state.

Define Candidate Roadmap Components: After establishing the Baseline and Target Architectures and analyzing the gaps, create components for the Architecture Roadmap. Use the results of the gap analysis to create work packages (initiatives) related to the architecture to be developed. These initiatives are to be used in the Architecture Roadmap. The Technology Architecture components are added to the draft roadmap created in Phase B (Business Architecture).



Review Baseline and Target Architectures: Begin by thoroughly understanding both the Baseline and Target Architectures. This comprehension sets the foundation for identifying necessary changes.

Conduct a Gap Analysis: Examine the differences between the Baseline and Target Architectures. This analysis will highlight the gaps that need to be addressed to achieve the desired Target Architecture.

Identify Required Work Packages: Based on the gap analysis, determine specific initiatives or projects – referred to as work packages – that will bridge these gaps. These work packages represent the actionable steps needed for development.

Develop the Architecture Roadmap: Organize and sequence the identified work packages into a coherent Architecture Roadmap. This roadmap serves as a strategic plan, outlining the path from the Baseline Architecture to the Target Architecture.

Incorporate Technology Components: Integrate relevant Technology Architecture components into the Draft Architecture Roadmap. This integration ensures that technological considerations are aligned with business objectives and are part of the overall transition strategy.

Resolve Impacts Across the Architecture Landscape: Look across the technology landscape and examine other architecture assets to answer key questions, such as the following:

- Does the Technology Architecture influence any established architectures, and vice versa?
- Are there chances to reuse or extend elements from this Technology Architecture to benefit other areas of the organization?
- Could this Technology Architecture impact current or upcoming projects, and vice versa?

Discuss the results with the relevant stakeholders and determine whether or not they should lead to adjustments to the architecture. If so, update the architecture accordingly.



Review the Technology Landscape: Begin by examining the current Technology Architecture. Identify its components, integrations, and dependencies within the organization.

Identify Interdependencies: Determine if the Technology Architecture affects existing architectures or if established architectures influence the Technology Architecture.

Spot Reuse Opportunities: Look for elements within the Technology Architecture that can be reused or extended to benefit other areas of the organization.

Assess Project Impacts: Evaluate whether the Technology Architecture impacts current or upcoming projects, and vice versa.

Engage Stakeholders: Discuss the findings with relevant stakeholders to understand their perspectives and gather additional insights.

Decide on Adjustments: Collaborate with stakeholders to determine if the identified impacts warrant changes to the architecture.

Update the Technology Architecture: If adjustments are necessary, revise the architecture to reflect the agreed-upon changes, ensuring alignment with organizational objectives.

Conduct Formal Stakeholder Review: It's essential to engage key stakeholders to ensure the Technology Architecture Phase effectively captures and models the essence of the organization. The stakeholder perspectives ensure that the architecture truly reflects the organization's needs and goals. Gather these stakeholders to revisit the original objectives behind the architecture work and the Statement of Architecture Work. Evaluate if the proposed Technology Architecture aligns with and supports the upcoming work in other architecture domains. Make refinements where necessary, based on stakeholder feedback.



Schedule a Review Meeting: Coordinate a time and place that accommodates all identified stakeholders. Ensure the meeting is set up well in advance to allow for maximum participation.

Revisit Original Objectives: At the beginning of the meeting, remind everyone of the initial goals behind the architecture project and review the Statement of Architecture Work. This sets a clear context for the discussion.

Present the Proposed Technology Architecture: Share the current state of the Technology Architecture with the stakeholders. Use diagrams, documents, and presentations to clearly convey the information.

Collect Feedback: Encourage stakeholders to share their thoughts on the architecture. Focus on whether it aligns with the organization's objectives and supports future work in other architectural areas.

Analyze Feedback: After the meeting, review all input received. Identify common themes, concerns, and suggestions that emerged during the discussion.

Refine the Technology Architecture: Make necessary adjustments to the Technology Architecture based on the feedback. Ensure that these changes enhance alignment with organizational goals and support other architectural domains.

Document Changes: Clearly record all modifications made to the architecture. This documentation will be valuable for future reference and for maintaining transparency.

Communicate Updates: Share the revised Technology Architecture with all stakeholders. Highlight the changes made and explain how stakeholder feedback was incorporated.

Seek Final Approval: Request confirmation from stakeholders that the revised architecture meets the organization's needs and is ready for implementation.

Finalize the Technology Architecture: Set clear standards for each building block, using reference models from the Architecture Repository's Reference Library wherever possible. Document each block in detail (see Chapter 7, Section 7.5.2.2).

Do a final cross-check to ensure the architecture is aligned with overarching business goals. Document the *why* behind each decision; it's vital for linking back to organizational objectives.



Establish Clear Standards for Each Component: Define explicit standards for every building block within the Technology Architecture. Utilize reference models from the Architecture Repository's Reference Library to ensure consistency and leverage established best practices.

Document Each Component Thoroughly: For every building block, provide detailed documentation. This should include its purpose, functionality, relationships, and any other pertinent information.

Ensure Alignment with Business Goals: Conduct a comprehensive review to verify that the Technology Architecture supports and advances the overarching business objectives. This alignment is crucial for the architecture's effectiveness and relevance.

Record the Rationale Behind Decisions: Document the reasoning for each architectural decision made. This practice is vital for maintaining traceability and demonstrating how each choice contributes to achieving organizational goals.

Update the Architecture Definition Document: Update the Architecture Definition Document by presenting the Baseline and Target Technology Architectures. Share the document with the key stakeholders to gather feedback, ensuring refinements are made to keep the architecture aligned with organizational objectives. This iterative process is essential for achieving a comprehensive and goal-oriented architecture.



Share with Key Stakeholders: Distribute the updated document to relevant stakeholders, including business leaders, IT teams, and other pertinent parties, to gather their insights and feedback.

Incorporate Feedback for Alignment: Revise the document based on the feedback received to ensure the architecture remains in sync with organizational objectives and addresses any concerns raised.

Iterate for Comprehensive Technology Architecture: Repeat the process as necessary, refining the architecture to achieve a thorough and goal-oriented framework that supports the organization's mission and vision.

Similar to the previous phase, all the architecture work that needs to be done should be documented in the *Statement of Architecture Work*.

Building on the information gathered in the previous phases, the Technology Architecture Phase focuses on defining the infrastructure and technology required to support the information systems. Architects consider factors such as hardware, software, networking, security, and integration requirements.

In summary, the Technology Architecture Phase is where the organization defines the technical building blocks and infrastructure necessary to support its business, application, and data needs. It serves as a bridge between the conceptual designs created in earlier phases and the actual implementation of technology solutions.

12.7.2 Phase D (Technology Architecture) Artifacts

Table 12-47 lists the artifacts that can be created during the execution of Phase D (Technology Architecture).

Table 12-47. *Artifacts created during Phase D (Technology Architecture)*

ADM Phase	Artifacts	Used Entities
Phase D: Technology Architecture	Technology Standards Catalog	Technology service, logical/physical technology component
	Technology Portfolio Catalog	Technology service, logical/physical technology component
	Application/Technology Matrix	Logical/physical application component, application service, logical/physical technology component
	Environments and Locations Diagram	Logical/physical technology component, logical/physical application component, location, actor
	Platform Decomposition Diagram	Logical/physical technology component
	Processing Diagram	Logical/physical technology component, logical/physical application component, actor, location
	Networked Computing/ Hardware Diagram	Logical/physical application component, logical/physical technology component, function
	Network and Communications Diagram	Logical/physical technology component

Technology Standards Catalog: The catalog outlines the approved technologies, products, and standards that support infrastructure, platforms, and networks used for delivering solutions. It helps ensure interoperability, maintainability, and compliance across systems. The catalog (see Table 12-48) provides a snapshot of the organization's standard technologies that can be or are deployed.

Table 12-48. *Technology Standards Catalog*

Standard Name	Technology Type	Version	Status	Comments
Red Hat Enterprise Linux	OS/Server Platform	10	Approved	Standard for clinical server deployments
PostgreSQL	Database Management System	17.5	Approved	Preferred for analytics and cost reporting modules
HL7 FHIR	Interoperability API	R4 R6	Approved Under Review	Required for patient data exchange with EHR systems

Technology Portfolio Catalog: This catalog identifies and maintains a list of all technology in use across the organization. This includes hardware, infrastructure software, and application software. An agreed-upon technology portfolio supports the lifecycle management of technology products and versions, as well as the definition of technology standards.

The Technology Portfolio Catalog (see Table 12-49) establishes the foundation for the remaining matrices and diagrams. It is typically the starting point of the Technology Architecture Phase. Technology registries and repositories (such as a CMDB application) provide input for this catalog from baseline and target perspectives.

Table 12-49. *Technology Portfolio Catalog*

Technology Component	Technology Type	Version	Lifecycle Status
Epic EHR System	Application Platform	2025	Active
ServiceNow Feedback Portal	Web Platform	Zurich Q4 2025	Active
Power BI Reporting	Analytics Tool	2.143	Active

The Technology Portfolio Catalog can be expanded to include additional columns, such as the organizational unit using the technology and the specific owner (business role) of the technology component.

Application/Technology Matrix: Shows the relationships between applications and the technology platforms (infrastructure and middleware) that support them. It helps identify technical dependencies, opportunities for reuse, and potential risks due to outdated or unsupported technologies. It should be aligned with and complement one or more Platform Decomposition Diagrams. An example of an Application/Technology Matrix is shown in Table 12-50.

Table 12-50. *Application/Technology Matrix*

	Technology Component				
Application Component ▼	Epic EHR	Web Platform	Azure Cloud	PostgreSQL	Power BI
Patient Feedback Application		X	X	X	
Workflow Management Application	X				X
Planning Application		X		X	
Supply Chain Management Application			X		X

Environments and Locations Diagram: This diagram depicts the *physical* and *virtual* locations in which application and technology components operate. It helps architects understand deployment constraints, operational risks, disaster recovery requirements, and regional dependencies. An example is shown in Table 12-51.

Table 12-51. *Environments and Locations Diagram*

Location/Environment	Type	Purpose
Central Hospital Campus	Physical Site	Inpatient care, surgery, admin HQ
Outpatient Clinic	Physical Site	Ambulatory care and diagnostics
Data Center – Region A	Physical Site	Hosts EHR
Telehealth Operations Hub	Virtual Environment	Remote consultations and triage
Azure Cloud Environment	Virtual Environment	Hosts patient feedback platform and analytics dashboards

The Environments and Locations Diagram is essentially a filter of the Location Catalog that was created during Phase B (see Table 12-16 in Section 12.5.3).

Platform Decomposition Diagram: This artifact illustrates the structure of technology platforms and their interrelationships. The diagram (see Figure 12-37) provides an overview of the organization's technology platform and covers all aspects of the infrastructure platform. It typically includes servers, operating systems, networks, and databases *grouped into physical and logical platforms*. The diagram can be expanded to map the technology platform to the appropriate application components.

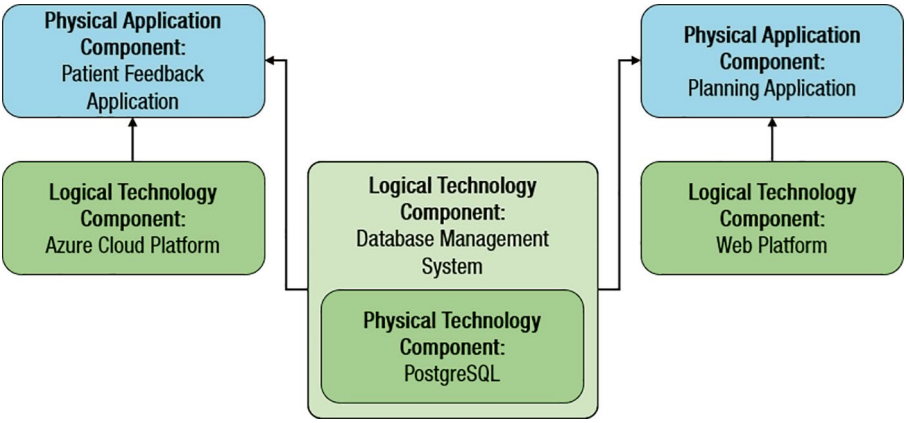


Figure 12-37. *Platform Decomposition Diagram*

Processing Diagram: The diagram illustrates how components are organized and deployed on a technology platform. These groups are called deployment units and typically represent business capabilities, services, applications, or technology components. The diagram helps identify which components should be packaged together and how the deployment units connect and interact with each other.

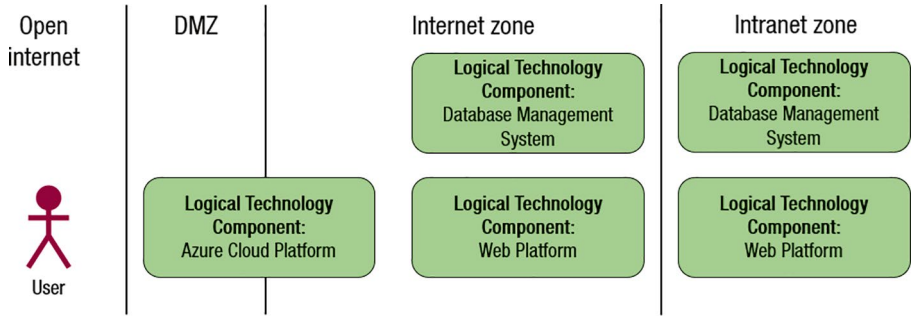


Figure 12-38. *Processing Diagram*

Networked Computing/Hardware Diagram: Documenting the mapping between logical applications and the technology components that support them is essential. This diagram illustrates the logical view of application components within a distributed network computing environment. It is useful for understanding where each application is deployed. The scope of the diagram can be defined to cover a specific application, business function, or the entire organization. If developed at the organizational level, the network computing landscape can be depicted in an application-agnostic way.

The Networked Computing/Hardware Diagram is a more detailed version of the Processing Diagram. The logical technology components shown in the Processing Diagram (see Figure 12-38) have been replaced by logical application components (as shown in Figure 12-39).

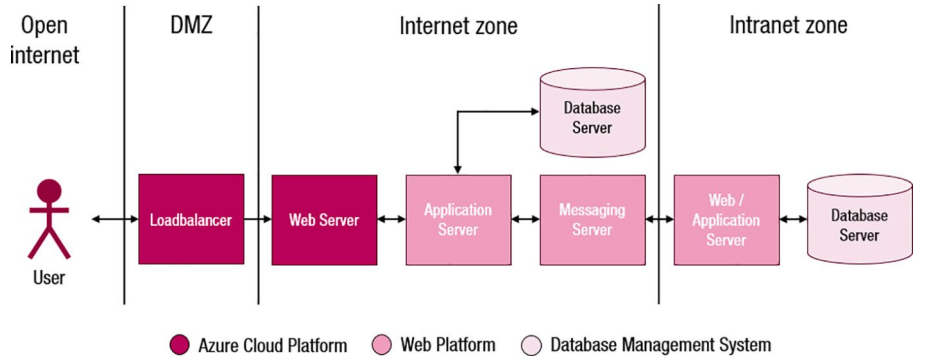


Figure 12-39. *Networked Computing/Hardware Diagram*

Network and Communications Diagram: Shows how technology components such as servers, clients, databases, and devices are connected across networks, emphasizing infrastructure, boundaries, and communication paths, not data formats. An example is shown in Figure 12-40.

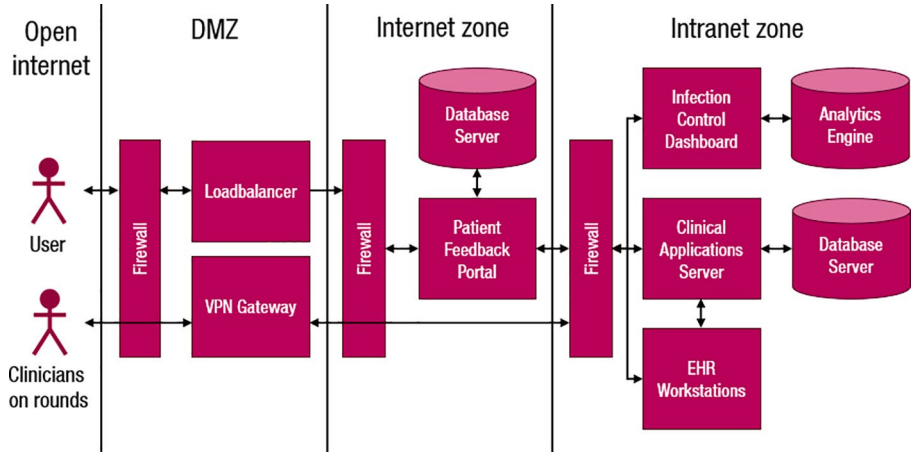


Figure 12-40. *Network and Communications Diagram*

Since the preceding phase, Phase C (Information Systems Architectures) (consisting of Data and Application Architectures), defines the applications that need technology support, the Technology Architecture must align with these architectures. Furthermore, the Technology Architecture serves as a foundation for Phase E (Opportunities and Solutions), where the focus is on identifying and prioritizing the implementation of architectural solutions.

12.8. Phase E: Opportunities and Solutions

The fifth alphabetized phase of the Architecture Development Method focuses on identifying and prioritizing the potential solutions that can deliver the capabilities required to achieve the Target Architecture. Phase E (Opportunities and Solutions) is primarily intended to identify projects or programs (in the form of work packages or initiatives) that will ultimately implement the Target Architecture. It is used to determine the most effective ways to realize the architecture vision and deliver business value.

Phase E focuses on evaluating potential implementation approaches – such as buy, build, or reuse – and assessing dependencies, opportunities, and constraints to create a feasible Architecture Roadmap (detailed in 12.8.3) for implementation.

12.8.1 Defining Opportunities and Solutions

Phase E (Opportunities and Solutions) has three key objectives. The first is to *identify the work packages* required to realize the Target Architecture. The second objective is to *determine whether an incremental approach is required* to achieve the first objective. The third and final key objective is to *define the building blocks* needed to complete the Target Architecture.

Input from the previous phases of the Architecture Development Method is required to identify the work packages (or initiatives) that will lead to the implementation of the Target Architecture. Phase E creates a first complete version of the Architecture Roadmap based on the gap analysis and candidate Architecture Roadmap components from phases B, C, and D.

During the execution of Phase E, the architect determines whether an incremental approach is required to reach the Target Architecture. If so, one or more Transition Architectures must be identified that will deliver ongoing business value.

The third objective of Phase E is to define the overall Solution Building Blocks (see Chapter 7, Section 7.5.2.5) to complete the Target Architecture based on the Architecture Building Blocks (see Chapter 7, Section 7.5.2.4). During this phase, the Architecture Building Blocks are further detailed so that clearly specified Solution Building Blocks can be defined. Table 12-52 lists the steps to perform Phase E (Opportunities and Solutions).

Table 12-52. *Steps to perform Phase E (Opportunities and Solutions)*

Step to Perform	Description of Step	Output Generated by Step
Determine/confirm key corporate change attributes	Align the architecture with an organization's business culture. This involves creating an Implementation Factor Catalog and assessing the transition capabilities of organizational units.	Implementation Factor Catalog
Determine business constraints for implementation	Identify business drivers that could compromise the Target Architecture. If there are any constraints, try to resolve them.	
Review and consolidate gap analysis results from phases B–D	Consolidate and integrate gap analysis results from the Business, Information Systems, and Technology Architectures.	Consolidated Gaps, Solutions, and Dependencies Matrix
Consolidate and reconcile interoperability requirements	Identify and address constraints that may hinder interoperability.	
Refine and validate dependencies	Analyze dependencies by grouping related activities.	Work Package Portfolio Map
Confirm readiness and risk for business transformation	Review the Business Transformation Readiness Assessment to inform the Architecture Roadmap and implementation strategy.	Updated Consolidated Gaps, Solutions, and Dependencies Matrix

(*continued*)

Table 12-52. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Formulate Implementation and Migration Strategy	Develop an Implementation and Migration Strategy by selecting a strategic approach and applying methodologies to effectively implement the Target Architecture while mitigating identified risks.	Implementation and Migration Strategy
Identify and group major work packages	Assess and address gaps in business capabilities in order to enable the organization to effectively plan and implement necessary changes aligned with strategic objectives.	Work Package Portfolio Map
Identify Transition Architecture(s)	Identify Transition Architecture(s) to provide structured, value-driven steps toward achieving the Target Architecture, tailored to the organization's implementation strategy and capacity for change.	Required building blocks (catalogs, matrices, and diagrams) Architecture Definition Increments Table Transition Architecture(s)
Create Architecture Roadmap and Implementation and Migration Plan	Create a structured approach to transition from Baseline to Target Architectures through a detailed Architecture Roadmap and Implementation and Migration Plan.	Architecture Roadmap Implementation and Migration Plan

Determine/Confirm Key Corporate Change Attributes: Implementing Enterprise Architecture effectively requires aligning it with the organization's unique business culture. A crucial step in this process is developing an *Implementation Factor Catalog*. This catalog serves as a repository for decisions related to architecture implementation and migration. Additionally, it's essential to assess the transition capabilities of the involved organizational units, considering their culture and abilities, as well as evaluating the organization's overall culture and skill sets.

Documenting the outcomes of these assessments in the Implementation Factor Catalog is vital. For organizations with a well-established Enterprise Architecture practice, this step may be straightforward. However, establishing the catalog is necessary to maintain a record of decisions made, ensuring a structured and adaptable approach to architecture implementation.

The Implementation Factor Catalog is an invaluable tool for documenting elements that influence the architecture's Implementation and Migration Plan. This catalog should encompass a comprehensive list of factors, detailed descriptions, and the resulting actions or constraints to consider during planning.

Typically, these factors include risks, issues, and assumptions. It is also advisable to include dependencies, actions, and impacts. By meticulously documenting these elements, organizations can ensure a more structured and effective approach to implementing and migrating their architecture.

One could argue that the Implementation Factor Catalog and the Consolidated Gaps, Solutions, and Dependencies Matrix are artifacts because they are a catalog and a matrix.

Although catalogs, diagrams, maps, and matrices are generally referred to as artifacts, they must contain architectural entities to be called an artifact (see Chapter 7, Section 7.4).

Since both the Implementation Factor Catalog and the Consolidated Gaps, Solutions, and Dependencies Matrix do not contain entities, they are not actual artifacts.



Evaluate Company-Wide Culture and Skills: Don't forget to look at the organization as a whole. You'll need to assess the overall culture and the skills available in the company to support the architecture implementation.

Create an Implementation Factor Catalog: Develop an Implementation Factor Catalog. This is where the elements that influence the architecture's Implementation and Migration Plan (like culture and available skills) are documented. It acts as a roadmap for the whole process. Assess the Implementation Factor Catalog.

Document Everything: Record the results of these assessments. Keeping detailed notes is critical. It ensures that things stay on track and can be easily referred back to previous decisions.

Stay Adaptable: Understand that if the organization is new to Enterprise Architecture, creating this catalog might feel like a big step. But it's necessary to keep everything organized and adaptable for future changes.

Determine Business Constraints for Implementation: It is important to identify any business drivers that could compromise the implementation of the Target Architecture. If there are any constraints, try to resolve them by discussing them with the stakeholders. Work out the best possible solution by acting as a mediator, utilizing the *Architecture Alternatives and Trade-Offs technique*, as outlined in Chapter 14, Section 14.9.



Identify Potential Constraints: Start by understanding the business drivers that might limit how to implement the Target Architecture. Think about anything that could cause challenges.

Engage Stakeholders: Have conversations with key stakeholders to see if any constraints can be resolved. Clear communication is key here.

Collaborate on Solutions: Act as a mediator, bringing stakeholders together to find the best solutions. Use techniques like the Architecture Alternatives and Trade-Offs method to guide the decision-making.

Review and Consolidate Gap Analysis Results from Phases B–D: A crucial aspect of this process is the consolidation and integration of gap analysis results from the Business, Information Systems, and Technology Architectures. By assessing these gaps, organizations can identify potential solutions and understand interdependencies, facilitating the creation of a *Consolidated Gaps, Solutions, and Dependencies Matrix* (see Section 12.8.2). This matrix is instrumental in pinpointing Solution Building Blocks that can address multiple gaps and their corresponding Architecture Building Blocks.

The development of Solution Building Blocks is informed by candidate roadmap components identified in phases B, C, and D.

To effectively consolidate the gap analysis results from phases B, C, and D, it's advisable to compile them into a single, comprehensive list. This list should include the *identified gaps* alongside *potential solutions* and their *dependencies*. A recommended approach for determining these dependencies involves utilizing a set of views, such as the Business Interaction Matrix, the Data Entity/Business Function Matrix, and the Application/Function Matrix. These tools help in relating elements across different architecture domains, ensuring a holistic perspective.

Once the Consolidated Gaps, Solutions, and Dependencies Matrix is documented, the next step is to rationalize it by reorganizing the gap list and grouping similar items together. During this process, it's essential to refer to the Implementation Factor Catalog and review the implementation factors. Any additional factors identified should be incorporated into the Implementation Factor Catalog, ensuring a comprehensive and up-to-date reference.



Collect Gap Results: Gather the gap analysis findings from the Business, Information Systems, and Technology Architectures (phases B, C, and D).
Combine and Organize Gaps: Create a single, easy-to-read list that includes the gaps, possible solutions, and any dependencies between them.

Create a Gap–Solution Matrix: Use a Consolidated Gaps, Solutions, and Dependencies Matrix to show the gaps, solutions, and dependencies clearly.

Identify Building Blocks: Use the matrix to figure out which Solution Building Blocks can solve multiple gaps.

Check Dependencies: Use specific views (such as the Business Interaction Matrix and the Data Entity/Business Function Matrix) to map out dependencies. This will provide a complete view of how everything connects.

Build Architecture Roadmap: Use the solutions identified to create an Architecture Roadmap for addressing gaps, making sure to align everything with the broader architecture.

Consolidate and Reconcile Interoperability Requirements: Review the architecture vision, Target Architectures, Implementation Factor Catalog, and the Consolidated Gaps, Solutions, and Dependencies Matrix, to identify and address constraints that may hinder interoperability. This proactive approach minimizes conflicts arising from reused Solution

Building Blocks, commercial off-the-shelf products, and third-party service providers. When conflicts are identified, two primary strategies can be employed: either developing a building block that transforms or translates between conflicting components or modifying the specifications of the conflicting building blocks themselves. This ensures a cohesive and efficient architectural framework that aligns with organizational objectives.



Understand the Requirements: Start by looking over key documents like the architecture vision, Target Architectures, the Implementation Factor Catalog, and the Consolidated Gaps, Solutions, and Dependencies Matrix. These will help identify any potential issues or gaps that might make things hard to integrate smoothly.

Identify Problems: Check for anything that could cause problems between different systems, such as reused solutions, third-party tools, or products.

Find Solutions: If issues are found, do one of two things:

Create a solution to convert or link conflicting systems.

Change the systems or their specs to make them work better together.
Ensure Alignment: This process ensures everything works well together and supports the company's overall goals.

Refine and Validate Dependencies: To refine initial dependencies and identify constraints on Implementation and Migration Plans, it's essential to consider key dependencies. Analyzing them helps determine the implementation sequence and the necessary coordination. Grouping related activities forms a foundation for establishing projects, and examining these projects can reveal logical increments of deliverables. This approach also aids in scheduling the delivery of these increments. Documenting the assessment of these dependencies should be part of the Architecture Roadmap and any required Transition Architectures. Addressing dependencies is fundamental to effective migration planning.

The *Work Package Portfolio Map*, in conjunction with the *Architecture Roadmap* (both artifacts are detailed in Section 12.8.3), provides a cohesive method for mapping interdependencies between work packages. In particular, the Work Package Portfolio Map provides tools for grouping work packages. The Architecture Roadmap visualizes these different groups of work packages and also shows their interrelationships and dependencies in relation to the organizational goals. Refer to Figure 12-47 for an example of an Architecture Roadmap.



Refine Dependencies: Start by figuring out which tasks depend on each other. This will help to understand the order to work on things and where to coordinate.

Group Related Tasks: Group tasks that go together. This makes it easier to manage the project and plan delivery in manageable chunks.

Schedule and Plan: With the groups of tasks, plan when each group of work will be done. This helps avoid confusion and ensures timely delivery.

Document Dependencies: Write down the findings about these dependencies. This should be included in the Architecture Roadmap to help guide the migration process.

Use the Work Package Portfolio Map: This tool helps to see how activities are connected. It shows the relationship between different activities and aligns them with the organization's goals.

Confirm Readiness and Risk for Business Transformation: The *Business Transformation Readiness Assessment* plays a pivotal role in determining whether and to what extent an organization is able to adapt in accordance with the changes required as a result of architecture work. During Phase A (Architecture Vision), the Business Transformation Readiness Assessment is initiated. By reviewing its findings during Phase E (Opportunities and Solutions), organizations can gauge their preparedness

for transformation initiatives. This assessment directly influences the Architecture Roadmap and the Implementation and Migration Strategy, ensuring they are grounded in the organization's current state and readiness levels.

Identifying, classifying, and mitigating risks associated with the transformation effort is crucial. Documenting these risks in the Consolidated Gaps, Solutions, and Dependencies Matrix provides a structured approach to address potential challenges, facilitating a smoother transition toward the Target Architecture.



Assess Transformation Readiness: Start by evaluating whether the organization is ready for change, especially concerning the architecture work. This is done through a Business Transformation Readiness Assessment, which kicks off in Phase A.

Review Findings: Revisit the findings from the readiness assessment to gauge how prepared the organization is for transformation initiatives.

Link to the Architecture Roadmap: Use the assessment results to shape the Architecture Roadmap and Implementation and Migration Strategy, ensuring that these plans reflect the current state and readiness.

Identify Risks: Spot any risks that could arise during the transformation. Document these risks in the Consolidated Gaps, Solutions, and Dependencies Matrix.

Mitigate Risks: Address the identified risks systematically, making the transition smoother and more aligned with the Target Architecture.

Formulate Implementation and Migration Strategy: Implementing a Target Architecture requires a comprehensive Implementation and Migration Strategy to guide the process and structure any necessary Transition Architectures. The initial step involves determining a strategic approach to implement solutions or capitalize on opportunities. Three fundamental approaches include:

- **Greenfield:** Initiating a completely new implementation
- **Revolutionary:** Undertaking a radical change, such as a complete system overhaul
- **Evolutionary:** Adopting a strategy of gradual convergence, like parallel operations or phased introductions of new capabilities

Subsequently, it's essential to establish a strategic direction that addresses and mitigates risks identified in the Consolidated Gaps, Solutions, and Dependencies Matrix. Common implementation methodologies encompass:

- **Quick Wins:** Achieving immediate benefits through targeted initiatives
- **Achievable Targets:** Setting realistic goals that are attainable within the current constraints
- **Value Chain Method:** Enhancing the entire value chain to improve overall performance

These methodologies, along with identified dependencies, form the foundation for creating work packages. This process culminates in the agreement on the Implementation and Migration Strategy for the organization.



Decide on the Approach: Before anything, figure out how to start the implementation of the Target Architecture. There are three basic options: Greenfield, Revolutionary, and Evolutionary.

Plan the Direction: Next, set clear goals that focus on fixing the issues found in the assessment. Think about quick wins, achievable targets, and the entire value chain.

Build the Action Plan: Now, organize the plan by creating small, manageable work packages. These tasks will guide the strategy and ultimately lead to successful implementation.

Identify and Group Major Work Packages: To effectively address identified gaps in business capabilities, key stakeholders, planners, and Enterprise Architects should assess these deficiencies as outlined in the architecture vision and Target Architecture. Utilizing tools such as the Consolidated Gaps, Solutions, and Dependencies Matrix, along with the Implementation Factor Catalog, enables logical grouping of activities into work packages. This approach ensures that each gap is addressed with appropriate solutions, whether through new development, existing products, or purchasable solutions. Additionally, classifying current systems as *mainstream*, *contain*, or *replace* helps in planning their future roles within the organization. By decomposing work packages into capability increments and analyzing them concerning business transformation issues and strategic implementation approaches, organizations can effectively group these packages into portfolios and projects, considering dependencies and strategic goals. A key artifact that supports this method is the Work Package Portfolio Map (see Section 12.8.3).



Group Activities Into Work Packages: Use tools like the Consolidated Gaps, Solutions, and Dependencies Matrix, along with the Implementation Factor Catalog, to organize the activities into clear groups. This helps ensure that each gap is addressed with the right solution – whether building new systems, improving existing ones, or buying ready-made solutions.

Classify Current Systems: Evaluate the current systems. Decide if they need to be kept, improved, or replaced. This helps plan how they will fit into the future of the organization.

Break Work Packages Into Smaller Pieces: Split the work into manageable tasks that align with business goals and transformation efforts. This makes it easier to organize into portfolios and projects, taking into account dependencies and strategic priorities.

Use a Visual Tool: Use a visual map, like a Work Package Portfolio Map, to clearly show how everything fits together and how each piece will be completed.

Identify Transition Architectures: Implementing a Target Architecture often necessitates an incremental approach, introducing one or more Transition Architectures. These Transition Architectures serve as clear *milestones* on the roadmap toward the Target Architecture, each delivering measurable business value. It's important to note that the intervals between successive Transition Architectures need not be uniform.

The development of Transition Architectures should be grounded in the preferred implementation strategy; the Consolidated Gaps, Solutions, and Dependencies Matrix; the project and portfolio listings; and the organization's capacity for change. Creating an *Architecture Definition Increments Table* (see Section 12.8.2) supports these activities. Identifying challenging activities is crucial; unless compelling reasons dictate otherwise, it's advisable to schedule these after implementing activities that more readily provide missing capabilities.



Identify Transition Architectures: When moving toward a Target Architecture, it's smart to take small steps. This means introducing one or more Transition Architectures along the way. These steps act as milestones, each delivering tangible benefits. Record them in the Architecture Definition Increments Table.

Map Out the Road to Success: Make sure the Transition Architectures align with the overall strategy. Use tools like the Consolidated Gaps, Solutions, and Dependencies Matrix and the Architecture Definition Increments Table. Take into account the project portfolio and the organization's ability to adapt.

Focus On the Tough Stuff: Identify the hardest tasks first. But don't try to tackle them all at once! It's better to start with easier tasks that build up the necessary capabilities and schedule the tougher activities for later.

Create Architecture Roadmap and Implementation and Migration Plan: By consolidating work packages and Transition Architectures into a Draft Architecture Roadmap, organizations can outline a clear timeline for progression. This roadmap not only informs the Implementation and Migration Plan but also frames the migration planning in Phase F of the Architecture Development Method (see Section 12.9). It's essential that each identified Transition Architecture and corresponding work packages have a well-defined set of outcomes, demonstrating how their selection and timing contribute to realizing the Target Architecture.

The Draft Architecture Roadmap should mirror the level of detail found in the Architecture Definition Document developed during phases B, C, and D. If significant additional detail is required before implementation, it may indicate a transition to a different level of architecture.

The Implementation and Migration Plan (see Table 12-53) must detail the activities necessary to achieve the Architecture Roadmap, serving as the foundation for migration planning in Phase F. The plan's detail should align with that of the Architecture Roadmap, sufficiently identifying the projects and resources required to realize the roadmap.

Table 12-53. *Implementation and Migration Plan*

Phase	Timeframe	Work Packages	Key Milestones	Dependencies
1	Q2 2028 to Q4 2028	Implement new capability for Patient Engagement.	Implemented new capability	–
		Implement new capability for Financial Management.	Implemented new capability	–
2	Q3 2028 to Q2 2030	Improve capability for Clinical Services.	Improved performance of capability	Staff training complete
		Improve capability for Financial Management.	Improved performance of capability	–
3	Q4 2028 to Q4 2030	Implement and improve processes for Patient Engagement.	Improved and new process	Implemented new capability for Patient Engagement
		Improve processes for Financial Management.	Improved process	Improved capability for Financial Management

When developing the Implementation and Migration Plan, consider various approaches, such as a data-driven sequence where applications that create data are implemented first, followed by those that process the data. A thorough understanding of the dependencies and lifecycle of existing Solution Building Blocks (see Chapter 7, Section 7.5.2.5) is crucial for an effective plan.

Finally, update the architecture vision and Architecture Definition Document with any additional relevant outcomes from this phase to ensure all documentation reflects the current state and planned progression.



Draft the Architecture Roadmap: Start by consolidating the work packages and Transition Architectures into the Draft Architecture Roadmap. This gives a clear timeline for moving forward.

Link to Implementation and Migration: This roadmap will guide the Implementation and Migration Plan. It's key in Phase F.

Set Clear Outcomes: Each work package and Transition Architecture should have clear outcomes. These outcomes help show how each piece fits into the bigger picture of the Target Architecture.

Mirror Architecture Documents: The Draft Architecture Roadmap should reflect the detail level found in the Architecture Definition Document (from phases B, C, and D). If more detail is required, this may indicate a need for a new level of architecture planning.

The Opportunities and Solutions Phase is a bridge between defining the architecture and planning how to implement it, focusing on identifying and evaluating solution options and setting the stage for successful migration and implementation.

12.8.2 Phase E (Opportunities and Solutions) Artifacts

The Opportunities and Solutions Phase evaluates and selects the most appropriate solutions to address the gaps identified in the previous phases. Architects identify potential architecture options, perform a risk assessment, and recommend appropriate solutions to stakeholders. They consider both

internal development and external sourcing options. The output of this phase is a set of architecture specifications and implementation plans.

Table 12-54 lists the artifacts that can be created during the execution of Phase E (Opportunities and Solutions).

Table 12-54. *Artifacts created during Phase E (Opportunities and Solutions)*

ADM Phase	Artifacts	Used Entities
Phase E: Opportunities and Solutions	Project Context Diagram	Work package, organization unit, function, business service, process, logical application component, data entity, logical technology component
	Benefits Diagram	Stakeholder, once added to the metamodel, business capability, business service, product
	Implementation Factor Catalog	Risk
	Consolidated Gaps, Solutions, and Dependencies Matrix	Gap, work package
	Architecture Definition Increments Table	–
	Work Package Portfolio Map	Work package, goal, objective
	Architecture Roadmap	Work package, goal, objective

Project Context Diagram: This diagram shows the scope of one or more work packages (initiatives) that will be implemented as part of a broader transformation roadmap, depending on the size of the project. The diagram (see Figure 12-41) links the work packages to the organizational units, functions, services, processes, applications, data, and technology that will be affected by the project. It is a valuable tool for project portfolio management and project mobilization.

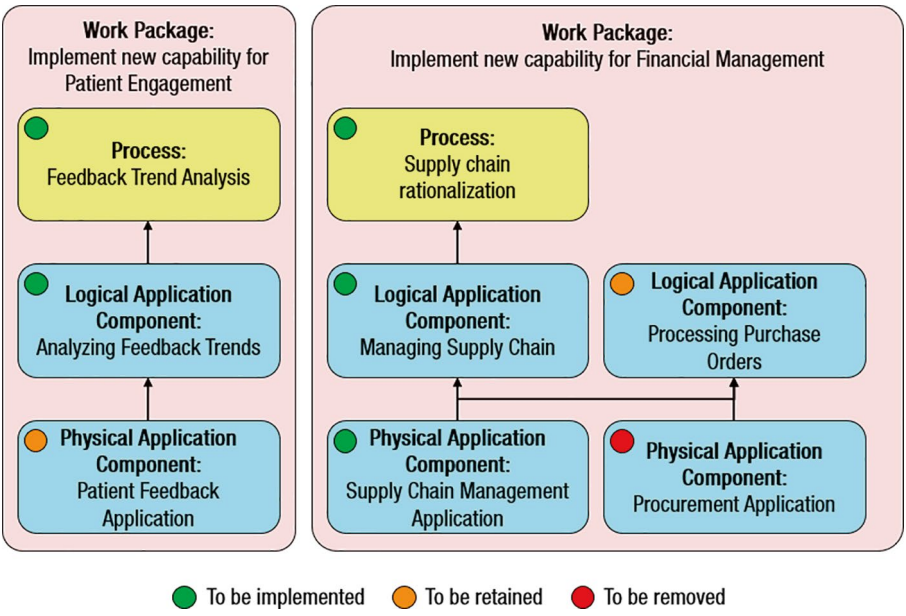


Figure 12-41. *Project Context Diagram*

Benefits Diagram: Shows opportunities identified in an architecture definition, classified according to their relative size, benefit, and complexity. This diagram, or catalog (see Table 12-55), can be used by stakeholders to make selection, prioritization, and sequencing decisions on identified opportunities.

Table 12-55. *Benefits Catalog*

Goal	Objective	Capability	Benefit
Improve patient satisfaction	Achieve 90% rating in patient satisfaction surveys by Q4 2028.	Patient Engagement Patient Feedback Management	Higher satisfaction ratings and service trust
Optimize clinical workflows	Reduce patient discharge processing time by 30 minutes by April 2029.	Clinical Services Clinical Workflow Optimization	Faster care delivery and improved satisfaction
Reduce operational costs	Cut non-clinical supply expenses by 10% by December 2030.	Financial Management Cost Reduction Management	Improved resource allocation and cost savings

Implementation Factor Catalog: This artifact is used to document elements that influence the architecture Implementation and Migration Plan. This catalog should include a comprehensive list of factors, their descriptions, and the deductions (i.e., conclusions) that indicate the corresponding actions or constraints to be considered when creating the migration plan. By systematically documenting these factors, a more robust and adaptable Implementation and Migration Plan can be developed, ensuring alignment with organizational objectives and external conditions.

The Implementation Factor Catalog ensures that each gap between the Baseline and Target Architectures is addressed with appropriate solutions. The catalog is used to effectively address identified gaps (during phases B, C, and D) by evaluating the deficiencies as outlined in the architecture vision and Target Architecture. The use of techniques such as the Implementation Factor Catalog helps with creating logical groups of activities and combine them into work packages.

In the *Deduction column* of Table 12-56, the contours of the work packages begin to take shape. For example, the inference that there is a need to *train personnel to use the new application* could result in a work package that includes the activity of *providing training options for personnel*. Another inference that the *new application will automate most application services* is likely to result in the creation of a work package containing the activity to *automate application services*.

Table 12-56. *Implementation Factor Catalog*

Factor Category	Factor	Description	Deduction
Effect (impact)	Change in applications	Replace HR Recruitment Application A with Application B to streamline storage of candidate information.	Need for personnel training to use new application. New application has major personnel savings and should be given priority.
Dependency	Consolidation of application services	Multiple information streams will be consolidated.	New application will automate most application services.
Action	Introduction of new HR Recruitment Application		

The Implementation Factor Catalog may also contain constraints that could hinder interoperability. To address these constraints, the same approach of creating work packages containing activities to address these constraints can be used.

Begin creating an Implementation Factor Catalog by listing elements that could affect the implementation of the architecture. Common factors include risks, assumptions, dependencies, and actions and effects. Create a table with the following columns:

- **Factor Category:** The type of factor (e.g., risk, assumption)
- **Factor:** A brief title or name for the factor
- **Description:** A concise explanation of the factor
- **Deduction:** Steps needed to address or manage the factor

Next, enter the type of factor and provide a short, descriptive name. Offer a succinct explanation, keeping it clear and to the point. List the necessary steps to address the factor, using bullet points for clarity.

To illustrate the steps outlined in the approach above, the first factor listed in Table 12-56 mentions a change in applications. The category assigned to the factor is Effect (impact). The description expands on the briefly stated factor by adding that HR Recruitment Application A will be replaced by HR Recruitment Application B to streamline the storage of candidate information. The deduction leads to the assumption that staff will need to be trained to use the new application and that the implementation of the application will lead to significant staff savings and should therefore be given priority. The bulleted lists are omitted in Table 12-56 to save space.

Using an Implementation Factor Catalog forces careful consideration of factors that could potentially affect the Implementation and Migration Plan.

Consolidated Gaps, Solutions, and Dependencies Matrix: This matrix is used to group the gaps that were identified during the gap analysis that took place during phases B, C, and D of the Architecture Development Method. The technique of using this matrix allows architects to assess potential solutions and their dependencies to the discovered gaps. The matrix is particularly useful for creating work packages. These sets of grouped activities will eventually be used to form the items on the Architecture Roadmap that will be created during Phase E (Opportunities and Solutions) and finalized during Phase F (Migration Planning).

During phases B, C, and D of the Architecture Development Method, the gap analysis results are consolidated and documented in the Consolidated Gaps, Solutions, and Dependencies Matrix. This matrix is instrumental in pinpointing Solution Building Blocks that can address multiple gaps and their corresponding Architecture Building Blocks (see Chapter 12, Section 12.8.1).

A Consolidated Gaps, Solutions, and Dependencies Matrix can be created by meticulously recording and consolidating the gaps. For each identified gap, propose a possible solution. Identify Solution Building Blocks that can effectively bridge these gaps. Link each Solution Building Block to the relevant Architecture Building Block(s) that it will modify or enhance. Note any interdependencies among the gaps, solutions, and architecture components. This will ensure a clear understanding of how changes may impact various parts of the architecture. Cluster similar gaps and solutions to streamline the implementation process. This grouping helps identify patterns and potential shared solutions.

Refer to the Implementation Factor Catalog to understand the factors that influence implementation, such as organizational constraints, resources, and timelines.

As new factors emerge during the rationalization process, add them to the catalog. Keeping this catalog current ensures that all relevant considerations are accounted for in future implementations.

Because the Consolidated Gaps, Solutions, and Dependencies Matrix lists gaps, their potential solutions, and dependencies, it is an excellent resource for defining Transition Architectures. These Transition Architectures serve as clear *milestones* on the roadmap to the Target Architecture, each providing measurable business value. The development of Transition Architectures should be based on the Consolidated Gaps, Solutions, and Dependencies Matrix. Figure 12-42 illustrates how the input from the Consolidated Gaps, Solutions, and Dependencies Matrix can be used to visualize the Baseline, Transition, and Target Architectures using a Plateau Planning.

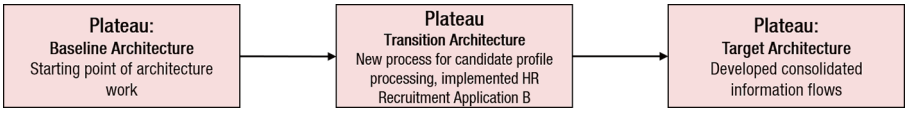


Figure 12-42. *Plateau Planning of Baseline, Transition, and Target Architectures*

In Figure 12-42, the gaps from the business and application domains (see items #1 and #2 in Table 12-57) are visualized using the Transition Architecture Plateau (the middle block). Item #3 in Table 12-57, a gap from the information domain, forms the Target Architecture Plateau.

Table 12-57. *Consolidated Gaps, Solutions, and Dependencies Matrix*

#	Architecture Domain	Gap	Potential Solution(s)	Dependencies
1	Business	New candidate profile processing process	Design new process for candidate profile processing (Business Interaction Matrix, Process Flow Diagram).	Replacement of HR Recruitment Application A
2	Application	New application workflow mechanism	Implement HR Recruitment Application B (Process/ Application Realization Diagram, Application/ Function Matrix).	
3	Information	Consolidated information storage	Develop consolidated information flows (Data Entity/Business Function Matrix, Application/Data Matrix).	

Section 12.8.3 details the creation of an Architecture Roadmap. A Plateau Planning, as illustrated in Figure 12-42, is a useful tool for this process.

Architecture Definition Increments Table: Allows architects to detail a series of Transition Architectures in tabular form. The table outlines the state of the architecture at specific times during its development.

The idea behind creating an Architecture Definition Increments Table (Table 12-58) is to list the projects and their assigned incremental deliverables across the Transition Architectures. Another way to achieve the same result is to create a slightly modified Plateau Planning Diagram (see Figure 12-43).

Table 12-58. *Architecture Definition Increments Table*

Project	Transition Architecture 1	Transition Architecture 2	Transition Architecture 3
Project A: Implement new HR Recruitment Application	Identified missing capabilities	Implemented HR Recruitment Application B	Eliminated HR Recruitment Application A
Project B: Automate capabilities through HR Recruitment Application		Automated capability Inflow Support	Automated capability Personnel Information Management

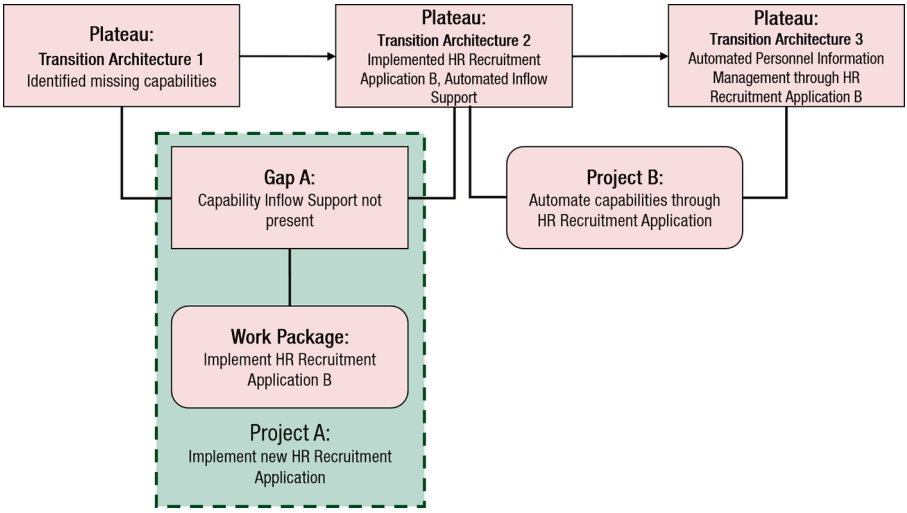


Figure 12-43. *Plateau Planning Diagram showing projects instead of gaps and work packages*

Instead of visualizing the gaps and work packages separately (as shown on the left in Figure 12-43), the project names are used to detail the activities required to close each gap (shown on the right). This is because projects often consist of one or more work packages. In turn, work packages contain the activities to close gaps between architecture states.

For instructions on how to create a Work Package Portfolio Map and an Architecture Roadmap, refer to the next section, "Creating an Architecture Roadmap".

12.8.3 Creating an Architecture Roadmap

The Architecture Roadmap is a pivotal deliverable within the TOGAF Framework, serving as a strategic guide that outlines the transition from the current Baseline Architecture to the desired Target Architecture. This roadmap is incrementally developed throughout phases E and F and informed by readily identifiable roadmap components from phases B, C, and D within the Architecture Development Method, ensuring a comprehensive and cohesive plan.

The Architecture Roadmap not only serves as a planning tool but also as a communication instrument, conveying the strategic direction and planned initiatives to stakeholders across the organization. By providing a visual representation of the transformation journey, it fosters understanding and buy-in from all involved parties. Moreover, the roadmap's iterative development allows for flexibility, enabling adjustments in response to changing business environments or emerging technologies, thereby ensuring the architecture remains aligned with the organization's evolving needs.

By detailing individual work packages (also called initiatives) and their sequencing, the Architecture Roadmap provides a clear timeline for implementing architectural changes, aligning each step with the organization's strategic objectives. It shows the progression from the Baseline Architecture to the Target Architecture.

The Architecture Roadmap highlights the business value of individual work packages at each stage. Transition Architectures necessary to effectively realize the Target Architecture are identified as intermediate steps.

In Phase E (Opportunities and Solutions), the Architecture Roadmap is refined to identify and prioritize work packages that address capability gaps and leverage opportunities for improvement. This phase ensures that each work package is evaluated against business goals, resource availability, and potential risks, facilitating informed decision-making. The outcome is a well-structured plan that balances immediate needs with long-term objectives, setting the stage for effective implementation in subsequent phases.

Before creating the Architecture Roadmap, a diagram such as Figure 12-44 is often created. This figure shows the phases that an architecture may be in (Baseline, Transition, and Target). Each phase – also known as a *plateau* – implies that there is work to be done to get to that phase. The gap between phases is represented by the *gap* entity.

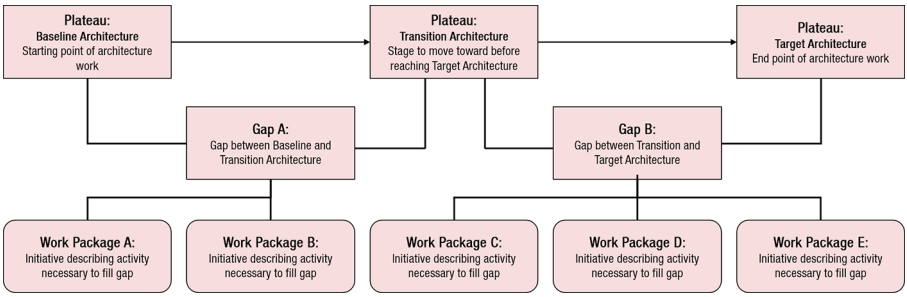


Figure 12-44. *From Baseline through Transition to Target Architecture*

To denote the work required to move from one phase to the next, the work package entity is used. These work packages represent the activities needed to bridge the corresponding gap.

High-level work packages are similar to initiatives. A cluster, set, or grouping of multiple work packages can be considered an initiative. These clusters of work packages often form a project, program, or even a portfolio.

A work package generally does not describe an ongoing activity such as a business process. The subject of the work package is performed once and produces a well-defined end result. This can be a goal or an objective

or the activities necessary to fill the gap between two architecture states. A work package can be used to model tasks within a project, entire projects, programs, or entire portfolios. In an agile context, a work package can be used to model the work done in an agile iteration (e.g., a sprint) or in a higher-level increment [9].

The TOGAF Standard defines a work package as a set of actions aimed at achieving one or more objectives:

A set of actions identified to achieve one or more objectives for the business. A work package can be a part of a project, a complete project, or a program [2].

Conceptually, a work package is similar to a business process in that it consists of a series of related tasks aimed at producing a well-defined end result. In terms of content, a work package can be said to be a *unique and one-time process*; it performs a series of activities that lead to an end result. A work package can be described in much the same way as a process. Figure 12-45 illustrates the similarities (and differences) between a work package and a process.

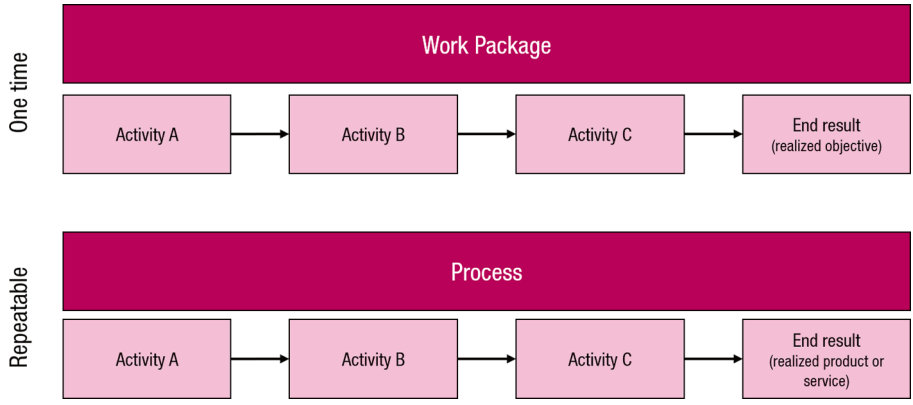


Figure 12-45. *Similarities and differences between work packages and processes*

Work packages translate the previously defined initiatives into concrete, actionable steps. Figure 12-44 illustrates the operation and use of work packages. It shows an overview of (from left to right) the Baseline Architecture, Transition Architecture, and the Target Architecture. Between each set of architecture states (Baseline and Transition, and Transition and Target) is a gap visible. Each of the gaps can be bridged by performing the activities listed in the corresponding work packages.

The TOGAF Standard references the work package entity within the Foundation Enterprise Metamodel. However, this entity is not used in any artifacts.

Therefore, I would like to propose an addition to the TOGAF Standard regarding this point.

It is important to have a thorough understanding of the initiatives (work packages) that make their way into the Architecture Roadmap. For this reason, I suggest adding two new artifacts to be produced during Phase E of the Architecture Development Method.

These are a Work Package Portfolio Map and an Architecture Roadmap.

Work Package Portfolio Map: An important architecture deliverable that serves as input to the Architecture Roadmap is the Work Package Portfolio Map. The associated goals and objectives contained in the Work Package Portfolio Map are elements that recur in the Architecture Roadmap.

A Work Package Portfolio Map consists of an overview of all work packages that are derived from either the organizational strategy or the architecture work at hand. They are usually shown clustered or grouped together. A Work Package Portfolio Map is similar in structure and visualization to an Application Portfolio Catalog. Table 12-59 shows an example of a Work Package Portfolio Map.

Table 12-59. *Work Package Portfolio Map*

Work Package Name	Associated with Objective(s)	Associated with Goal(s)
Work Package A: Fit job profile X into existing job profile	Objective A: Make adjustments to comply with legislation demands by Q3 2028.	Goal A: Implement new regulatory requirements.
Work Package B: Recruit staff for job profile X	Objective A: Make adjustments to comply with legislation demands by Q3 2028.	
	Objective B: Eliminate overlapping job profiles by 2027.	Goal B: Efficiently use human resources.
Work Package C: Distribute updated job profiles	Objective B: Eliminate overlapping job profiles by 2027.	
Work Package D: Update compensation application	Objective C: Re-evaluate job compensation by Q4 2028.	
Work Package E: Update job information	Objective C: Re-evaluate job compensation by Q4 2028.	

A Work Package Portfolio Map is the foundation of the Architecture Roadmap. The goals and objectives contained in this artifact are recurring elements. For governance purposes, it is important to link the initiatives from the Work Package Portfolio Map to the items in the Architecture Roadmap. This allows stakeholders and senior management to track the architecture's progress and performance against the organization's strategy. Without initiatives tied to organizational goals and objectives, this is virtually impossible to measure.

For example, Table 12-59 shows that Work Packages A and B are linked to Objective A, which, in turn, is linked to Goal A. Work Package B is also linked to Objective B. The second goal, Goal B, is composed of two objectives (B and C). Objective B is composed of Work Packages B and C, whereas Objective C has Work Packages D and E linked to it.

If the Work Package Portfolio Map was to be visualized (by creating a Work Package Portfolio Diagram), the result would be as shown in Figure 12-46.

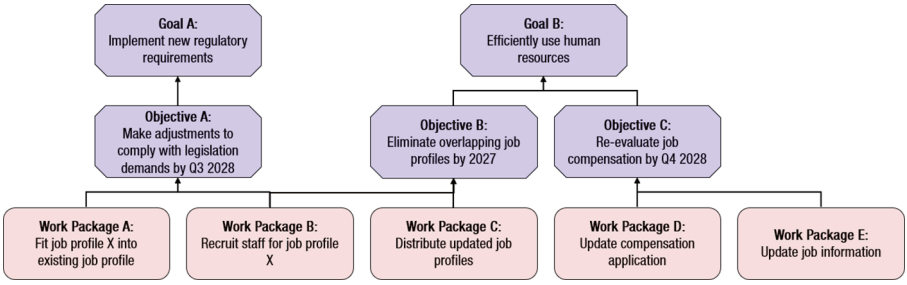


Figure 12-46. *Work Package Portfolio Diagram*

Linking the work packages (or initiatives) to the objectives and goals is an absolute must, because this allows the architecture activities to be tracked back to organizational goals and objectives.

During Phase F (Migration Planning), it is important to map the work packages needed to implement the architecture to the ongoing projects in the organization (refer to Section 12.9.1 for additional information). This prevents things from being done twice and can be used as a kind of consistency check.

To ensure the relationship between work packages and projects, it is advisable to add an extra column to the Work Package Portfolio Map. This column should contain the projects that are related to the work packages.

Architecture Roadmap: An Architecture Roadmap consists of initiatives (or work packages). These entities are the embodiment of the activities that need to be performed to realize an organization's strategy.

Phases B, C, and D of the Architecture Development Method develop architectures for the business, data, application, and technology domains. Each of these architectures creates components that find their way onto the roadmap. The form in which they appear on the roadmap is that of a work package or initiative. Either are plannable entities that contain a *description* of the activity, together with a *start* and *end date*.

Creating an Architecture Roadmap is an important part of developing an Enterprise Architecture. The development of a roadmap enables an organization to align the created initiatives with the defined goals and corresponding objectives. In effect, a roadmap represents the realization of the organization's strategy, visualized in concrete and defined steps.

A roadmap should never be created without linking business goals and objectives to the initiatives [9].

Goals and objectives, key components of the organization's strategy, are defined using SMART[4] criteria. This makes them specific and therefore understandable, realistic (or relevant), and, above all, measurable. The latter is particularly important for translating goals and objectives into initiatives by developing the architecture.

Once the organization's goals and objectives are defined, initiatives can be created by following the phases of the Architecture Development Method. When these initiatives are implemented, they achieve (part of) the objectives. Therefore, it is important to have a clear understanding of the initiatives so that the appropriate governance can be applied at the right time.

As mentioned earlier, initiatives are visualized using the work package entity. A work package contains a description of the initiative and has

[4] Specific, Measurable, Achievable, Realistic, and Timebound (SMART).

a start and end date. Work packages can be associated with additional information such as status, cost, or ownership. The entities are plotted over time in a roadmap using architecture tools.

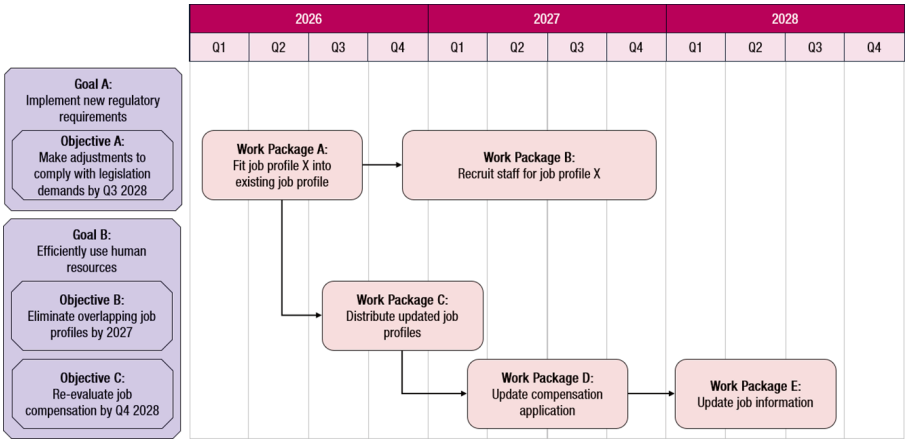


Figure 12-47. *Architecture Roadmap containing work packages*

Using a good architecture tool, it is relatively easy to create an Architecture Roadmap. This artifact shows which work packages are related to which goals or intended objectives. Additionally, owners and (progress) states can be assigned to the work packages. Thus, an Architecture Roadmap can be used to monitor the progress of the implementation of the organization's strategy.

A roadmap detailed with properties provides a wealth of information for an organization. For example, it can be used to visualize the cost of achieving the goals and objectives. By assigning the cost of executing a work package as an attribute to the element, the roadmap can show how this relates to the realization of the objective(s). Having an overview of the costs per work package can help prioritize projects or programs.

An Architecture Roadmap is created by processing the work packages from the Work Package Portfolio Map (see Table 12-59) and plotting them as activities over time. This is done by positioning the goals and/or objectives on the left side of the roadmap (vertically) and the work packages from the Work

Package Portfolio Map in the timeline area (horizontally). The roadmap can be enriched by adding additional properties to the work packages. Examples include owners of the activities (organization units or business roles), potential costs, and progress states (e.g., not started, pending, completed, canceled).

By adding these additional properties to the work packages, a broad spectrum of additional options becomes available. For instance, by adding the business roles as work package owners to the work packages, it becomes possible to visualize who is responsible for what. This can be helpful in monitoring the progress of the organization's goals and objectives. Color views are a powerful way to visually illustrate specific parts of a diagram – in this particular case, the progress of a project. An example of a color view that highlights the progress is shown in Figure 12-48.

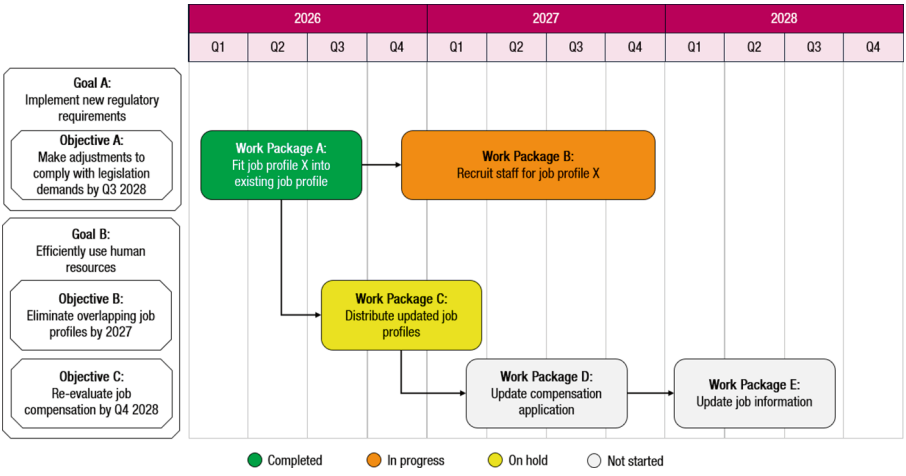


Figure 12-48. *Color view of an Architecture Roadmap*

The primary purpose of an Architecture Roadmap is to be able to plot work packages over time so that the organization has a good picture of what activities are needed to realize the desired Target Architecture. By cleverly using the translation of Transition Architectures into work packages, the intermediate steps can also be visualized. An Architecture Roadmap is a powerful tool in the context of Enterprise Architecture.

Phase E (Opportunities and Solutions) focuses on transforming the architecture vision and Architecture Roadmap into actionable project solutions. This phase identifies potential work packages, including Transition Architectures, and evaluates them against business goals, financial feasibility, and resource availability. By aligning opportunities with the Target Architecture, Phase E establishes a clear connection between strategic objectives and tactical implementation.

A critical aspect of Phase E is prioritizing and sequencing projects based on their value, dependencies, and organizational constraints. The phase ensures that all proposed solutions address identified gaps and risks while optimizing resource use.

Stakeholder engagement plays a central role, as their input ensures alignment with business goals and addresses practical considerations. Phase E concludes with a formal agreement on the prioritized set of projects and solutions, bridging the high-level architecture vision with the practical execution needed to realize the organization's strategic goals.

Phase E (Opportunities and Solutions) is followed by the Migration Planning Phase, where a detailed Architecture Roadmap and schedule for implementing the solutions are developed.



Further Reading

Additional articles on creating a Work Package Portfolio Map and an Architecture Roadmap are available at the following URLs:

https://eawheel.com/wheel/execute/roadmap/work-package-portfolio-map/

https://eawheel.com/wheel/execute/roadmap/architecture-roadmap/

For more information about the Architecture Roadmapping process, visit the following URL:

https://eawheel.com/blog/2024/04/architecture-roadmapping/

12.9. Phase F: Migration Planning

In this phase, the architects focus on creating a detailed plan for implementing the selected solutions. A step-by-step approach for transitioning from the baseline state to the target state architecture is developed. This includes defining Transition Architectures and identifying critical milestones, resources, and timelines. The Migration Planning document provides the roadmap for the transformation journey.

12.9.1 Creating a Migration Planning

The primary objective of Phase F (Migration Planning) is to create a detailed plan for the implementation of the architecture. This involves identifying the projects and initiatives (the aforementioned work packages) that will help the organization move from the Baseline Architecture to the Target Architecture. The aim is to:

- **Prioritize projects** and activities that are part of the transition.
- **Assess dependencies and risks** between projects.
- **Plan resource allocation** (budget, people, technology) for these projects.
- **Develop a migration strategy** that ensures business continuity.

Crucial inputs for this phase are the Target Architecture and the Architecture Roadmap components. These components include the gaps identified between the Baseline and Target Architectures, opportunities for improvement, and solutions. They were developed during phases B–D and were translated to work packages (initiatives) in Phase E. The Architecture Roadmap gets finalized during Phase F (Migration Planning). Other key inputs to this phase are the Implementation Governance Model, which will

guide the architecture governance during the migration process. Last but not least, the insights from previous phases on potential impacts and risks also play an important role. Table 12-60 lists the steps to perform Phase F (Migration Planning).

Table 12-60. *Steps to perform Phase F (Migration Planning)*

Step to Perform	Description of Step	Output Generated by Step
Confirm management framework interactions	Align the Implementation and Migration Plan with an organization's management frameworks.	Updated Implementation and Migration Plan
Assign business value to work packages	Define and measure business value for work packages, identify corresponding projects, assess associated risks, and estimate business value.	Updated Work Package Portfolio Map Business Value and Risk Matrix
Estimate resource requirements, project timings, and availability/ delivery vehicle	Allocate resources and estimate costs for each project increment, distinguishing between capital and operational expenses, and identify cost-saving opportunities.	
Prioritize migration projects through cost/benefit assessment and risk validation	Determine the sequence of projects based on business priorities, technical dependencies, resource availability, and risk factors.	Risk Management Plan

(*continued*)

Table 12-60. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Confirm Architecture Roadmap	Update the Architecture Roadmap, incorporating any necessary Transition Architectures.	Transition Architecture State Evolution Table Finalized Architecture Roadmap
Complete Implementation and Migration Plan	Develop a detailed timeline with clear milestones for each transition stage.	Finalized Implementation and Migration Plan
Complete architecture development cycle	If the Architecture Capability's maturity permits, craft an Implementation Governance Model. Document lessons learned.	

Confirm Management Framework Interactions: Coordinating the Implementation and Migration Plan with an organization's management frameworks is crucial for achieving concrete business outcomes. Typically, four key frameworks must collaborate closely:

- **Business Planning** conceives, directs, and provides the necessary resources for activities aimed at achieving specific business objectives.
- **Enterprise Architecture** structures and contextualizes all organizational activities, delivering tangible business outcomes.

- **Project/Portfolio Management** coordinates, designs, and builds business systems that deliver the desired outcomes.
- **Operations Management** integrates, operates, and maintains deliverables that achieve the business objectives.

The Implementation and Migration Plan will impact the outputs of each of these frameworks and must be reflected within them. During this step, it's essential to understand the organization's frameworks and ensure that the plans are coordinated and summarized within each framework's plans. Identify the people in the organization who are responsible for the frameworks listed above. These will typically be department heads, as well as portfolio or project managers. Present the Implementation and Migration Plan to them, and make sure that the steps described in it are adopted in the other frameworks, where applicable. Conversely, if there are things described in the other frameworks that may affect the Implementation and Migration Plan, they need to be incorporated.

The outcome of this step may be that the Implementation and Migration Plan becomes part of a different plan produced by another framework, with Enterprise Architecture participation.



Understand Existing Frameworks: First, get to know the main management frameworks the organization uses. These include Business Planning, Enterprise Architecture, Project/Portfolio Management, and Operations Management.

Coordinate Plans: Make sure the Implementation and Migration Plan fits well with these frameworks. The steps in the plan should be reflected in theirs, and vice versa. If other frameworks have plans that could affect the Implementation and Migration Plan, be sure to include them.

Engage the Right People: Identify the heads of the departments and project managers responsible for these frameworks. These are the people who need to be involved in making sure the plans line up.

Present the Plan: Share the Implementation and Migration Plan with the department heads and make sure it's aligned with their frameworks. They should adopt the necessary parts of the Implementation and Migration Plan, where applicable.

Update and Finalize: As a result, the Implementation and Migration Plan might be integrated into other framework plans, especially those involving Enterprise Architecture. Ensure everyone is on the same page to move forward smoothly.

Assign Business Value to Work Packages: Begin by defining what constitutes business value within the organization and determining how to measure it. Then, apply this understanding to each project and its increments.

Utilize the work packages (refer to the Work Package Portfolio Map) to identify projects for inclusion in the Implementation and Migration Plan. These projects will be further developed in subsequent steps of Phase F. Adjustments to the Architecture Roadmap and Architecture Definition Document may be necessary to accommodate these projects and their increments. Update the Work Package Portfolio Map by mapping the work packages to the projects.

Next, assign risks to the projects and their increments by consolidating risks identified in the Consolidated Gaps, Solutions, and Dependencies Matrix from Phase E.

Finally, estimate the business value for each project using the Business Value and Risk Matrix (see Section 12.9.2).

Assessing business value effectively involves creating a matrix that evaluates both business value and risk indices. The business value index should encompass criteria such as adherence to principles, financial contribution, strategic alignment, and competitive positioning. Conversely, the risk index should consider factors like project size and complexity, technological aspects, organizational capacity, and potential failure impacts. Assigning appropriate weights to each criterion is essential.



Define Business Value: Start by figuring out what business value means for the organization and how to measure it. Then, apply this to each project and its phases.

Identify Projects: Use the Work Package Portfolio Map to spot which projects should be included in the Implementation and Migration Plan. These projects will be developed further in the next steps of Phase F.

Update Architecture Deliverables: Adjust the Architecture Roadmap and Definition Document to fit the new projects and their stages. Also, update the Work Package Portfolio Map to link the projects with their work packages.

Assess Risks: Look at the risks from the Consolidated Gaps, Solutions, and Dependencies Matrix in Phase E, and apply these to the projects and their phases.

Estimate Business Value: Finally, calculate and visualize the business value for each project using the Business Value and Risk Matrix.

Estimate Resource Requirements, Project Timings, and Availability/Delivery Vehicle: This process entails providing initial cost estimates, which should be distinctly categorized into capital expenditures (to establish the capability) and operational and maintenance costs (to sustain the capability). Identifying opportunities to offset expenses by decommissioning existing systems when delivering new or enhanced capabilities is also crucial. Assigning the required resources to each activity and aggregating them at both the project increment and overall project levels ensures a structured and efficient approach to Enterprise Architecture development. Most of the activities that need to be performed during this step of Phase F will be performed by a project manager or someone in a similar role. It is important to stay involved here.



Estimate Resources: Start by estimating how much the project will cost. Break the costs into two parts: one for the initial setup (capital expenses) and another for ongoing operations (maintenance).

Look for Savings: Think about ways to save money. For example, could using old systems be stopped and replaced by new ones?

Assign Resources: Ensure each task has the right people and resources assigned to it. Tally up all the resources needed for each step of the project.

Involve Project Management: This step is mainly managed by the project manager or someone in charge. Stay involved and make sure everything is going as planned.

Prioritize Migration Projects Through Cost/Benefit Assessment and Risk Validation: This step consists of four sub-steps. The first is to *assess the business value and cost*. This is done by determining the net benefit of Solution Building Blocks delivered by projects. This involves a thorough analysis to ensure that the anticipated benefits justify the associated costs. By doing so, organizations can make informed decisions that align with their strategic objectives.

The second step is about *risk mitigation*. Identifying and mitigating risks is paramount. One of the tools that can aid the architect is creating a comprehensive review of potential risks associated with project deliverables. By proactively addressing these risks, organizations can update project plans with pertinent risk-related insights, ensuring that all potential challenges are accounted for and managed effectively.

Third, *building consensus among stakeholders* is critical to successfully prioritizing projects. The TOGAF Standard provides criteria that incorporate elements from the draft Architecture Roadmap, developed in Phase E, as well as individual stakeholder agendas. This holistic approach ensures that even projects with modest immediate benefits but critical long-term deliverables receive appropriate prioritization.

A side-by-side comparison of the requirements, desires, and concerns of all stakeholders involved can be used to prioritize. In some cases, it may be necessary to re-engage with stakeholders to address some of the concerns. It may also be useful to look at the sum of the wants and needs with the stakeholders and work together to determine the best prioritization. Stakeholders may include the following groups:

- **Architecture Team:** Ensures technical feasibility and alignment with the overall architectural vision
- **Business Stakeholders:** To prioritize projects based on business needs
- **Project Managers:** To ensure that the migration aligns with broader program or project plans
- **Finance and Resource Planners:** To ensure that budgetary and resource constraints are considered

Formal risk assessment review is the essential fourth step. This step involves reviewing and revising the risk assessment, if necessary. It ensures a comprehensive understanding of residual risks associated with project prioritization and the projected funding line, enabling organizations to proceed with confidence. This is where a Risk Management Plan is created. This plan addresses potential risks identified during migration planning, with proposed mitigation strategies. A Risk Management Plan contains at least three basic components:

- **Document Information:** Things like the project name, date, and author of the document, including the version number.
- **Approvals:** A list of the dates, names, and signatures of the people who approved the approach described in the document.
- **Approach:** A detailed description of how risks will be managed. It also includes a summary of the tools that will be used and an outline of the information that will be recorded. Perhaps more importantly, it defines the metrics that will be used to classify risks. Risk tolerance is also described in the document.



Assess Business Value and Costs: Start by evaluating how much value each project brings versus its cost. This involves understanding the expected benefits of each project and comparing them to the costs involved. This helps ensure projects contribute to the broader business goals.

Mitigate Risks: Risk Management is critical. Identify any risks tied to the projects and plan ahead to manage them. It's helpful to create a detailed review to highlight potential challenges and ensure that these risks are properly handled in the project plans. Use the Risk Management technique for this.

Build Stakeholder Consensus: Next, it's essential to get buy-in from all key players. Use criteria based on long-term goals and individual stakeholder needs to guide this process. Regular discussions might be needed to address any concerns and align priorities.

Review Risk Assessment: Finally, revisit and update the risk assessment as needed. This ensures that all risks are accounted for and the necessary mitigation strategies are in place, allowing the organization to move forward with confidence.

Confirm Architecture Roadmap: To ensure a successful implementation, it's essential to update the *Architecture Roadmap*, incorporating any necessary Transition Architectures. This involves reviewing current progress to determine appropriate intervals between Transition Architectures, considering factors such as business value increments, capability enhancements, and associated risks. Once these capability increments are finalized, consolidating deliverables by project will lead to a refined Architecture Roadmap. In practice, this means updating the work packages that were mapped to the goals and objectives in the first version of the Architecture Roadmap. This meticulous process

is crucial for coordinating the development of multiple concurrent architecture instances. Utilizing a *Transition Architecture State Evolution Table* (see Section 12.9.2) can effectively illustrate the proposed states of Domain Architectures at varying levels of detail.

If the implementation approach has evolved due to the confirmation of implementation increments, it's imperative to update the Architecture Definition Document accordingly. This update may involve setting project objectives and aligning projects and their deliverables with the Transition Architectures.



Update the Architecture Roadmap: Begin by reviewing and updating the Architecture Roadmap. Add any necessary Transition Architectures.

Review Progress: Assess how things are going and decide the right gaps for each Transition Architecture, considering business needs, new capabilities, and potential risks.

Finalize Changes: Once the needed improvements are clear, group the deliverables by project. This helps refine the roadmap.

Track Progress: Update work packages that align with the original roadmap goals.

Use Artifacts: A helpful artifact like the Transition Architecture State Evolution Table can show architecture changes at different stages.

Complete Implementation and Migration Plan: A thorough capture of all external dependencies is essential, ensuring they are incorporated into the plan. Additionally, a comprehensive assessment of resource availability is conducted to guarantee that the necessary assets are in place for successful implementation. Detailed project plans are often embedded within the overarching Implementation and Migration Plan, providing a clear roadmap for execution.



Identify All External Dependencies: List everything that could affect the project and make sure it's included in the plan.

Check Resource Availability: Ensure all needed resources (people, tools, etc.) are ready to go.

Create a Detailed Project Plan: Develop a clear roadmap for how things will happen, including key milestones.

Complete Architecture Development Cycle: Transitioning from architecture development to realization is a pivotal step to be performed. At this juncture, and if the Architecture Capability's maturity permits, crafting an Implementation Governance Model becomes essential. Documenting lessons learned during architecture development is crucial; these insights should feed into the governance processes in Phase H, thereby enhancing the Architecture Capability.

The Architecture Roadmap and the Implementation and Migration Plan should mirror the detail level of the Architecture Definition Document from phases B, C, and D. If subsequent phases demand more granularity, it indicates a shift to a different architectural tier. In such cases, initiating another Architecture Development Method cycle at a more detailed level might be necessary.



Architecture Development Cycle: Moving from architecture creation to real-world application is a crucial step. At this stage, if the team's skills allow, it's time to develop a plan for overseeing implementation (governance model).

Document Insights: Keep track of key lessons learned during development. These should be used to improve future governance processes in Phase H, which will make the architecture stronger.

Details Matter: The Architecture Roadmap and Implementation and Migration Plan should be as detailed as the Architecture Definition Documents from earlier phases (B, C, D). If more detail is needed later on, that's a sign to start a new cycle at a more detailed level.

In short, during Phase F (Migration Planning), the Transition Architectures are further developed. These are intermediate architectures that serve as a stepping stone between the Baseline and Target Architectures. They help manage risk and ensure the organization is making incremental progress. Phase F focuses on creating a detailed and practical implementation plan (the Implementation and Migration Plan) to guide the organization through its architectural transformation. It ensures that the transition is phased, manageable, and aligned with business needs, minimizing risks and maximizing value.

The Implementation and Migration Plan serves as input to Phase G (Implementation Governance), where the execution of the plan will be governed, monitored, and adjusted as needed.

12.9.2 Phase F (Migration Planning) Artifacts

Phase F of the Architecture Development Method focuses primarily on creating an Implementation and Migration Plan. Output from Phase E serves as input for Phase F. Activities include assessing the dependencies,

costs, and benefits of various migration projects within the organization. Several artifacts greatly aid in realizing the Implementation and Migration Plan.

Table 12-61 lists the artifacts that can be created during the execution of Phase F (Migration Planning).

Table 12-61. *Artifacts created during Phase F (Migration Planning)*

ADM Phase	Artifacts	Used Entities
Phase F: Migration Planning	Transition Architecture State Evolution Table	Work package, organization unit, function, business service, process, logical/physical application component, data entity, logical/physical technology component
	Business Value and Risk Matrix	Work package, risk

Transition Architecture State Evolution Table: As mentioned in the previous section, it is essential to include any Transition Architectures in the Architecture Roadmap to ensure a successful architecture implementation. In order to do this, the current progress needs to be reviewed to determine appropriate intervals (and corresponding actions) between Transition Architectures. In practice, this means updating the work packages that were mapped to the goals and objectives in the first version of the Architecture Roadmap. This process is crucial for coordinating the development of multiple concurrent architecture instances. Utilizing a *Transition Architecture State Evolution Table* can effectively illustrate the proposed states of domain or Segment Architectures at varying levels of detail.

Figure 12-49 shows the proposed Solution Building Blocks in the cells under the Transition and Target Architecture headings. In this case, HR Recruitment Application A, HR Recruitment Application B, and Database

Management System are the three Solution Building Blocks used in the example. The reason they are called Solution Building Blocks rather than Architecture Building Blocks is because they all describe the specific applications in detail. Architecture Building Blocks would refer to high-level descriptions of products or processes.

Domain	Service	Transition Architecture 1	Transition Architecture 2	Target Architecture
Applications	Application Services	HR Recruitment Application A (replace)	HR Recruitment Application B (transition)	HR Recruitment Application B (new)
Technology	Database Management Services	Database Management System (retain)	Database Management System (retain)	Database Management System (retain)

To be replaced In transition In place (new or retained)

Figure 12-49. *Transition Architecture State Evolution Table*

Remember the difference between Architecture Building Blocks and Solution Building Blocks? They were described in Chapter 7, Sections 7.5.2.4 and 7.5.2.5, respectively.

Architecture Building Blocks typically describe a required capability and shape the specification of Solution Building Blocks. For example, an organization may require a Human Resources Management capability that is supported by a number of Solution Building Blocks, such as an organization unit, a business service, an application, and a data entity.

The Architecture Building Block does not specify a particular product, process, or service, so it remains high-level. Solution Building Blocks, on the other hand, do.

The Transition Architecture State Evolution Table shows the Transition Architectures in the top row. To the left, the corresponding architecture domain or segment is shown, along with the service it affects. For example, looking at the first row in Figure 12-49, it is evident that HR Recruitment Application A needs to be replaced by HR Recruitment Application B. This observation is recognized as the first transition to be made and is therefore called Transition Architecture 1.

In Transition Architecture 2, HR Recruitment Application B is introduced to the landscape. It requires the application services mentioned in the Service column to be transitioned from HR Recruitment Application A. Finally, in Transition Architecture 3, the new application (HR Recruitment Application B) is fully implemented and operational, and HR Recruitment Application A no longer exists in the landscape.

Looking at the second row of the Transition Architecture State Evolution Table, it is clear that the Database Management System exists in the Baseline Architecture (because it is retained during Transition Architecture 1). It also exists in Transition Architectures 2 and 3, which means it will also exist in the Target Architecture. The database management services supported HR Recruitment Application A until it was replaced. Later, during Transition Architectures 2 and 3, they support HR Recruitment Application B.

A Transition Architecture State Evolution Table can be created by making a table with two main columns labeled *Domain* and *Service*. These columns serve as the basis for categorizing the different domains and their related services within the architecture. For each relevant Transition Architecture, add an additional column to the right of the *Service* column. Label these columns according to the respective Transition Architecture names or identifiers. This setup allows for documenting changes specific to each Transition Architecture.

In the *Domain* column, list each pertinent architecture domain (e.g., Business, Data, Application, Technology). In the *Service* column adjacent to each domain, specify the corresponding service or component associated with that domain.

For each service listed, move across the row to the columns representing the Transition Architectures. In each cell, provide a brief description of the planned changes or developments for that service within the context of its respective Transition Architecture. This could include modifications or upgrades (in transition), deprecations (to be removed), or the introduction of new functionalities (new or retained).

Repeat this process for every domain and its associated services that are part of the project scope. This comprehensive approach ensures that all aspects of the architecture are considered and documented.

A Transition Architecture State Evolution Table provides a lot of information about the architecture work to be done. It demands a lot of the architect(s) who create the table, as it must be done meticulously for each project. The pros outweigh the cons because using an artifact such as the Transition Architecture State Evolution Table clearly details the necessary steps to be performed per architecture domain (or segment) and service for each Transition Architecture.

Business Value and Risk Matrix: Effective assessment of business value can be achieved by creating what the TOGAF Standard calls a Business Value and Risk Matrix. The artifact has two dimensions: a *business value* index on the y-axis and a *risk* index on the x-axis. The business value index includes criteria such as compliance, strategic alignment, and competitive position, while the risk index includes factors such as size and complexity, technology, organizational capacity, and impact of failure. Assigning individual weights to each criterion is critical.

Establishing decision criteria *before* populating the diagram is essential. Senior management should develop and approve the indices, criteria, and their respective weights to ensure alignment with organizational goals.

The Business Value and Risk Matrix in Figure 12-50 shows that projects A, B, C, and D deliver the highest business value. Projects D and H pose the greatest risk. However, since Project H delivers less business value than Project D, it is more likely to be canceled.

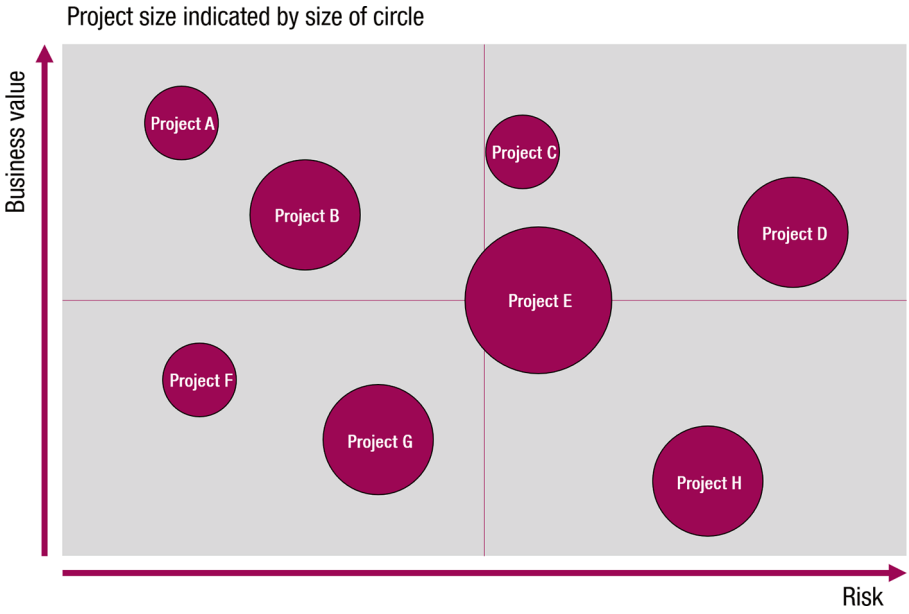


Figure 12-50. *Business Value and Risk Matrix*

The project size depends on the number of work packages associated with the project. Therefore, it is important to update the Work Package Portfolio Map with the projects. The Work Package Portfolio Map artifact (see Chapter 12, Section 12.8.3) can then be used to determine the project size.

The colors indicate the presence of (potential) risks.

The Business Value and Risk Matrix is used in Phase F (Migration Planning) to assign business value to work packages.

This matrix is created by identifying the criteria that determine a project's contribution to an organization. Common factors include compliance with regulations, alignment with strategic goals, and enhancement of competitive position.

It's also important to determine the factors that assess a project's potential challenges. These may include project size and complexity, technology involved, organizational capacity, and potential failure impact.

Recognize that not all criteria carry equal importance. Assign appropriate weights to each factor to reflect its significance in the assessment.

Engage senior management to review and approve the defined indices and criteria and their respective weights. This ensures the assessment aligns with the organization's goals and priorities.

Then, plot each project in the matrix based on its evaluated business value and risk. This visual representation will help compare projects and make informed decisions. Use colors to indicate whether a project is at risk, potentially at risk, or not at risk with regard to execution.

12.10. Phase G: Implementation Governance

Implementation Governance is about establishing the mechanisms to oversee the execution of the architecture project. It is where the architectural vision transitions into operational reality. Phase G involves defining the organizational structure, roles, and responsibilities to ensure that the architecture is implemented as intended. Architects work closely with project management teams to monitor progress and address any deviations from the plan. The result is a Governance Framework that helps manage risk and ensures alignment with the architecture vision. This phase ensures that the solutions developed align with the defined architecture and adhere to established standards. It's the stage where meticulous planning meets execution, and governance plays a pivotal role in overseeing this transformation.

12.10.1 Establishing Architecture Governance

The primary objectives of Phase G include *ensuring adherence to the established architecture, managing any changes* during implementation, and *continually assessing progress* to make necessary adjustments to meet the organization's objectives.

Key activities include establishing an Architecture Governance Framework that defines roles and responsibilities, monitoring the progress of implementation projects, and promptly addressing any issues that arise. This structured approach ensures that the implementation is aligned with the organization's strategic goals and the architectural blueprint.

Setting up and implementing an Architecture Governance Framework is explained in more detail in Chapter 8. Phase G (Implementation Governance) leverages this. Table 12-62 lists the steps to perform Phase G.

Table 12-62. *Steps to perform Phase G (Implementation Governance)*

Step to Perform	Description of Step	Output Generated by Step
Confirm scope and priorities	Identify and address gaps within the existing architecture. Review the Work Package Portfolio Map to determine which initiatives lead to optimal value extraction.	Architecture Contract
Identify deployment resources and skills	Identify the tools needed and the skills to use them.	
Guide development of solutions deployment	Map out the scope of the project. Define the strategic requirements, as well as the various change requests needed to implement the architecture.	Change requests

(*continued*)

Table 12-62. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Perform architecture compliance reviews	Regularly assess the implementation activities to ensure that they are consistent with the architectural standards and principles.	Compliance assessments
Implement business and IT operations	Implement business and IT service delivery, skills development, and training.	Architecture-compliant solutions
Perform post-implementation review	Examine the end result of the architecture work. If there are deviations, submit change requests to correct them.	Publication of reviews

Confirm Scope and Priorities: This step identifies and addresses gaps within the existing architecture. It involves a thorough gap analysis to pinpoint deficiencies and determine the necessary Solution Building Blocks to bridge these gaps. Solution Architects play a pivotal role in this process, meticulously defining how each Solution Building Block will be implemented. It's essential to recognize that multiple projects may target similar capabilities; therefore, Solution Architects must ensure optimal value extraction from these initiatives by leveraging existing investments. The Work Package Portfolio Map again plays an important role in providing insight into the initiatives.

To implement this first step, it's important to engage with key stakeholders and ensure alignment between the architectural plans and the development team's priorities. Set up a collaborative meeting or workshop with the development management team to confirm scope and priorities. This is essential because it fosters direct communication, where all parties can align their views and expectations. Before the meeting, review any prior

architectural work, including the architecture vision, business requirements, and the broader deployment objectives. This gives context for the discussion and ensures everyone is on the same page regarding strategic goals.



Confirm the Scope and Priorities: Start by identifying gaps in the current architecture. This involves a detailed gap analysis to find where improvements are needed. Once the gaps are clear, figure out what Solution Building Blocks are required to fill them. Solution Architects lead this effort. Keep in mind that different projects might target the same goals, so it's crucial to extract the most value by leveraging past investments. The Work Package Portfolio Map can provide useful insights here.

Align with Stakeholders: It's important to make sure the architectural plans match the development team's priorities. Set up a workshop or meeting with key stakeholders to confirm scope and priorities. This direct communication helps ensure everyone is on the same page. Before the meeting, review existing architectural documents like the architecture vision and business requirements to give everyone context and align expectations with strategic goals.

Identify Deployment Resources and Skills: In the context of project resources, it's essential to ensure that development teams are well versed in the overall Enterprise Architecture deliverables and the expectations specific to their development and implementation projects. This includes identifying the system development methods required for solutions development.

The use of modeling language tools will go a long way in getting everyone involved to speak the same language.

Evaluate the readiness of the development team to implement the architecture. This includes assessing their current skill sets and competencies, tools, and processes and determining if any gaps need to be addressed before proceeding. Identify any dependencies between the architecture deployment and other ongoing initiatives or projects. Use the updated Work Package Portfolio Map for this. Make sure to understand if any existing work must be completed or aligned with the deployment.



Identify Skills and Resources: Before jumping into development, make sure the team understands the big picture of Enterprise Architecture. This includes knowing the expected deliverables and the development methods that fit the project. Using modeling tools helps everyone get on the same page and communicate clearly.

Check Readiness: Assess the team's current skills, competencies, tools, and processes. Look for gaps that could prevent them from successfully implementing the architecture. If there are any skill or tool shortages, address them before moving forward.

Identify Dependencies: Check if the architecture rollout depends on other ongoing projects. Use an updated Work Package Portfolio Map to track these dependencies. Be clear on any existing tasks that must align with the deployment to avoid delays.

Guide Development of Solutions Deployment: To perform this step, it is necessary to formulate a project recommendation. This should be done for each individual implementation. To do this, the scope of the project should be mapped out and accompanied with an impact analysis.

It's important to clarify what will and won't be included. Ensure to define clear boundaries for what will be part of the deployment in this phase. Include features, functionalities, systems, and business processes.

Engage with the stakeholders to determine which features or systems should be prioritized. This might involve evaluating which components will deliver the most immediate business value or address the highest-priority business needs.

Together with the development management team, create a priority list. This can be based on business urgency, technical dependencies, available resources, or potential risks. The *Business Value and Risk Matrix* comes in handy at this point (refer to Section 12.9.2).

The strategic requirements (from an architectural perspective) should be defined, as well as the various change requests needed to implement the architecture. This means managing any changes or deviations from the original architecture plan. Performing change management involves evaluating the impact of changes and ensuring that they are appropriately documented and approved.

Be sure to document and confirm both the functional and non-functional scope clearly. Share this document with stakeholders to ensure there are no misunderstandings about what will be deployed.

Update the Architecture Repository with the latest architecture work.



Create a Project Recommendation: Start by outlining the project, including what's in and out of scope. Be sure to clearly identify what features, systems, and processes are involved.

Engage with Stakeholders: Collaborate with key stakeholders to determine the most important features and systems to focus on based on business needs.

Prioritize Tasks: Work with the architecture development team to create a list of tasks, ranking them by business urgency, technical dependencies, and risks.

Set Strategic and Change Management Requirements: Define the key requirements for the project and anticipate any needed changes to the original plan. Ensure changes are evaluated, documented, and approved.

Document Scope and Share: Clearly document both the functional and non-functional scope of the deployment, and share with stakeholders for alignment.

Update the Architecture Repository: Don't forget to update the Architecture Repository with the latest architecture work for consistency and clarity.

Perform Architecture Compliance Reviews: As the architecture is implemented, it is important to keep track of whether or not it is going according to plan. One of the things that needs to be done during this step is to perform an ongoing review of each building block from an architecture compliance perspective. Compliance checking involves regularly assessing the implementation activities to ensure that they are consistent with the architectural standards and principles established in earlier phases. In other words, it ensures that the building blocks are being implemented in the agreed manner. If not, a new change request must be issued to address the situation.

Define what success looks like for this phase, whether it's specific performance metrics, completion of key deliverables, or meeting certain quality standards. Agree on metrics for success. This will help ensure that everyone is aligned on the desired outcomes.

Before proceeding, confirm that all stakeholders agree with the defined scope, priorities, and resources. In other words, get them to formally sign off on the work to be done. All architecture compliance audits are reviewed and coordinated with relevant stakeholders. This is crucial for ensuring there's shared ownership and accountability for the deployment. Document all the decisions made during the meeting/workshop and create a formal deployment plan. This should include the scope, priorities, milestones, timelines, resource plans, and success metrics.

Finally, tracking the progress of the implementation against established metrics and performance indicators to ensure that it meets time, cost, and quality objectives is another important part of this step. This particular part is described in more detail in Section 12.10.2.



Review the Architecture's Progress: As the architecture is being built, regularly check if it aligns with the initial plan. This includes reviewing each component (building block) for compliance with the standards set in earlier stages.

Define Success: Set clear goals for this phase, whether it's meeting performance targets, completing key tasks, or ensuring quality. Agree on these success metrics to keep everyone on the same page.

Get Stakeholder Approval: Before moving forward, confirm that all stakeholders agree with the plan, priorities, and resources. They should formally approve the work to be done.

Document and Plan: After the stakeholder meeting, document the decisions made and create a formal plan. This should include the scope, milestones, timelines, resources, and success measures.

Track Progress: Regularly monitor the project to ensure it's meeting its goals in terms of time, cost, and quality. Keep an eye on the metrics to ensure everything stays on track.

Implement Business and IT Operations: The next step in Phase G is to execute the deployment projects. These include the implementation of business and IT service delivery, skills development and training, and the publication of communication documentation. It is important to publish new baseline architectures to the Architecture Repository and update other affected repositories, such as operational configuration management stores (such as a CMDB, if one exists).



Deploy Business and IT Operations: The first thing to do is to roll out the deployment projects. This includes setting up how the business and IT services will be delivered, plus ensuring the teams have the right skills through training.

Publish Documentation: Next, it's key to share any important communication materials. This includes publishing updated architectures to the Architecture Repository. Make sure to also update other systems that manage configurations, like a CMDB, if there is one.

Keep Everything Updated: Lastly, don't forget to regularly check and refresh all related documents and repositories to stay on top of any changes.

Perform Post-implementation Review: Once the previously conceived architecture has been implemented, it is important to complete the work. To do this, post-implementation reviews should be performed. These reviews examine the end result of the architecture work. If there are deviations from the original Implementation and Migration Plan, change requests should be submitted to correct the deviations. At this point, Phase H (Architecture Change Management) is initiated.

The actual completion of the work is done by publishing the reviews performed and formally closing the projects. The involvement of project managers is desirable for this last step.

One other important factor to consider throughout Phase G is Risk Management. This technique is used to identify and mitigate risks associated with the implementation to minimize disruptions and ensure successful delivery. Risk Management is further detailed in Chapter 14, Section 14.8.



Do a Post-implementation Review: Once the architecture is set up, take time to evaluate how everything turned out.

Check for Differences: Compare the final result to the original plan. If there are discrepancies, request changes to fix them.

Start Architecture Change Management: This marks the beginning of Phase H (Architecture Change Management).

Wrap Up the Project: Finalize everything by sharing the review results and formally closing the project. It's helpful if project managers are involved in this step.

Phase G (Implementation Governance) serves as the bridge between meticulously crafted designs and their real-world deployment. It serves as a framework for overseeing the architecture implementation, ensuring compliance, managing changes, and engaging stakeholders effectively. It is essential for achieving the intended outcomes of the architectural vision laid out in the previous phases of the Architecture Development Method.

Through effective governance and control mechanisms, organizations can confidently move forward, knowing that their architectural vision is being realized in a controlled and aligned manner. Implementation governance is critical for ensuring that the architecture delivers the desired business value and that any issues are addressed promptly.

In essence, Phase G is where the rubber meets the road. It helps to minimize risks, manage stakeholder expectations, and ensure alignment with the organization's strategic goals.

12.10.2 Architecture Performance Monitoring

Monitoring the performance of the implementation of the architecture – in particular the implementation of the Target Architecture – can be done by creating dashboards and/or graphs. These tools allow not only the architects but all stakeholders to monitor and adjust the architecture development as needed as the implementation progresses.

There are two main areas that should be addressed when measuring progress. These are tracking the completion of work packages and measuring the achievement of goals and objectives.

12.10.2.1. Measuring Progress in Completing Work Packages

During the development of the architecture, several work packages need to be executed and completed in order to realize the Target Architecture. Most of the work packages consist of one or more activities that need to be performed.

It is necessary to track the progress of these activities to get a good idea of the progress of the architecture development. This can be achieved by measuring the completion of the work packages containing the activities. In order to help determine their progress, a percentage can be assigned to each activity within a work package. Executing all activities normally leads to the completion of the work package. The method of assigning percentages should be applied to each individual activity for which progress is to be measured.

Let's say that the execution of a work package involves four separate activities. Each activity can be assigned one-fourth of the total, leading up to 25% per activity.

Once these four activities have been successfully performed, the completion of the work package has been realized for 100% (see Figure 12-51). This is of course a very basic approach as it assumes that a proportional percentage is assigned to all activities of the work package to be completed.

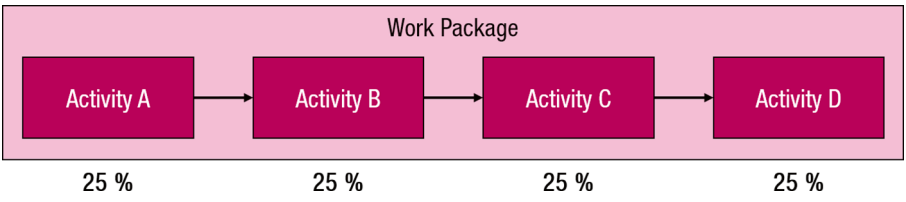


Figure 12-51. *Basic method for assigning percentages to activities*

A more refined method involves assigning percentages based on, for example, the duration of each activity. When the more refined method (Figure 12-52) is used, for example, assigning a percentage based on the duration of each activity, the distribution of percentages is often quite different. To illustrate, consider a work package consisting of four activities, A, B, C, and D. Activity A takes one hour to complete. Activities B and C each take two hours. Activity D has a lead time of three hours.

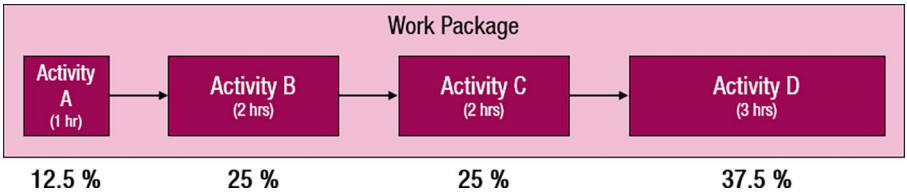


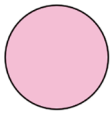
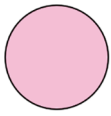
Figure 12-52. *More refined method of assigning percentages to activities*

Using the more refined method of assigning percentages in proportion to the duration of the activities, the percentages for activities A–D would be 12.5, 25, 25, and 37.5%, respectively. Thus, assigning percentages based on the more refined method provides a more detailed picture of progress than using the basic concept of proportional percentages.

When the two methods are applied to a work package in which three of the four activities are completed, the basic method yields a rate of 75%. The more refined method shows that completion of the first three activities accounts for 62.5%. The latter method, therefore, provides a more realistic picture of the progress being made.

Another common technique for charting the progress of individual work packages is to use percentages in the form of pie charts. Table 12-63 shows an example of progress per work package. Table 12-63 lists the five work packages that were shown earlier (see Figure 12-44 and Figure 12-47 in Section 12.8.3). Based on the values 0, 25, 50, 75, and 100%, small pie charts are used to illustrate the progress of *each individual work package*.

Table 12-63. *Progress Summary per Work Package*

Work Package	Description	Progress
Work Package A	Fit job profile X into existing job profile.	
Work Package B	Recruit staff for job profile X.	
Work Package C	Distribute updated job profiles.	
Work Package D	Update compensation application.	
Work Package E	Update job information.	

A third way to track progress is to use spider or bar charts. These types of charts use percentages to plot the current state against the desired end state, 100%. Which form of visualization is chosen depends largely on personal preference. None of the forms described is more or less useful than the others.

12.10.2.2. Measuring Achievement of Goals and Objectives

Aside from measuring the progress of the completion of individual work packages and their activities, it is also very useful to track the progress being made on the realization of the organization's goals and objectives.

As mentioned in Section 12.8.3, the Architecture Roadmap contains an overview of the initiatives that need to be deployed in order to realize the Target Architecture. The initiatives are related to the organization's goals and objectives.

Using the same approach as described for charting progress for each work package, progress toward goals and objectives can also be charted.

To do this, it is important to assign a percentage to the progress of each initiative (work package). By using the approach described in Section 12.10.2, a progress percentage can be assigned to each individual work package. Add the percentages of all the work packages related to a specific goal or objective, and divide by the number of work packages. The resulting number can be used as the average progress percentage per goal or objective.

As an example, suppose an Architecture Roadmap contains two goals to be realized (see Figure 12-53). Goal A has two work packages (A and B) assigned to it. Goal B is linked to work packages C, D, and E. Work Package A is 100% complete and Work Package B is 75% complete. Goal B is related to Work Package C, which is only 50% complete and currently on hold. The activities belonging to work packages D and E have not yet started, so the percentage for these work packages is 0%.

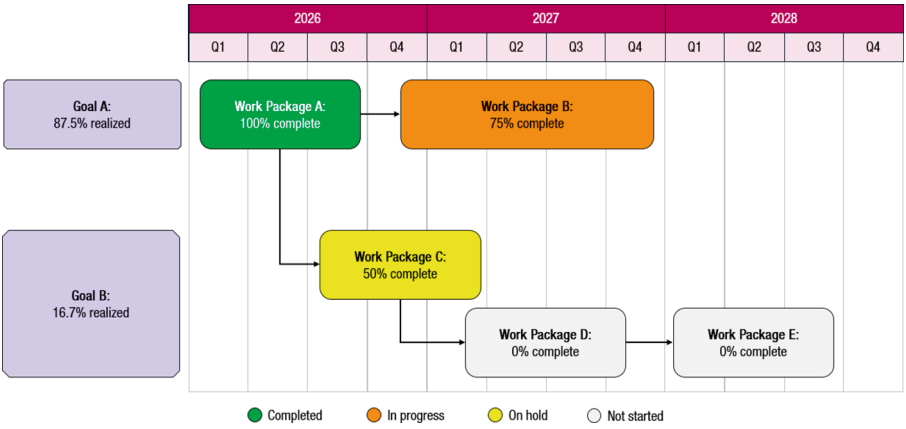


Figure 12-53. *Architecture Roadmap showing work package progress percentages*

The result of averaging work packages A and B is 87.5% (100% + 75% / 2). The same calculation for work packages C, D, and E yields 16.7% (50% + 0% + 0% / 3). Thus, goals A and B are 87.5% and 16.7% realized, respectively. Figure 12-54 illustrates the above example.

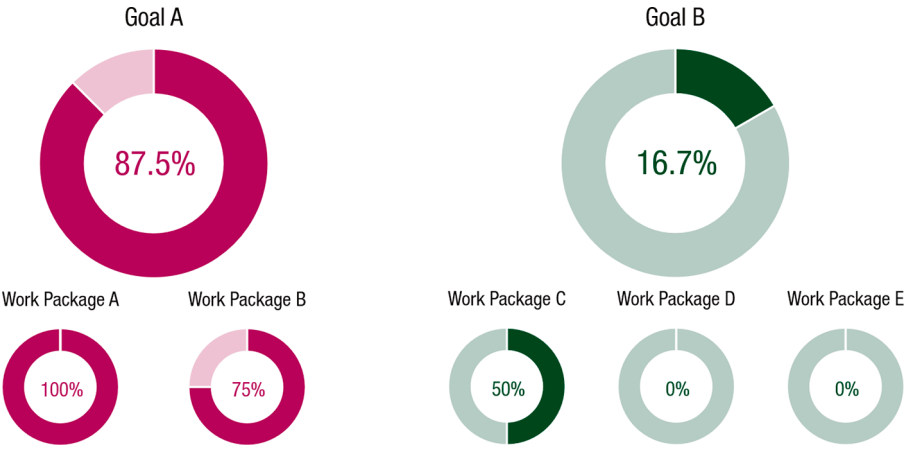


Figure 12-54. *Goal realization progress*

Understanding the progress of both the completion of work packages and the realization of organizational goals and objectives is important to both the architecture team and senior management. The method described here is relatively simple. However, it helps to quickly understand the progress of both concepts.

12.11. Phase H: Architecture Change Management

Architecture is not static, and the environment in which it operates evolves over time. The Architecture Change Management Phase is concerned with managing changes to the architecture throughout its lifecycle. It includes assessing the impact of changes, defining procedures for making changes, and ensuring that the architecture remains aligned with business goals.

Phase H (Architecture Change Management) of the Architecture Development Method is critical because it ensures that the architecture remains relevant and aligned with business needs as they evolve over time.

12.11.1 Managing Architecture Changes

The key objectives of the Architecture Change Management Phase are to *continuously assess changes* in the organizational strategy, business environment, and technology landscape that may impact the architecture. It is essential to ensure that there are processes and governance structures in place to handle proposed changes to the architecture effectively. Therefore, it is necessary to evaluate the potential impact of changes on existing architecture components, ensuring that modifications align with the overall architectural vision.

When implementing approved changes, it is important to facilitate the process while minimizing disruption to ongoing operations. Stakeholders need to be kept informed about changes to the architecture, ensuring that they understand the implications and benefits.

After implementing changes, it is vital to review the architecture to validate that it continues to meet business requirements and that the changes have been successfully integrated. Table 12-64 lists the steps to perform Phase H (Architecture Change Management).

Table 12-64. *Steps to perform Phase H (Architecture Change Management)*

Step to Perform	Description of Step	Output Generated by Step
Establish value realization process	Implement a change management process to govern modifications to the architecture in order to align changes with business value and maintain architectural integrity.	Architecture Compliance Report
Deploy monitoring tools	Deploy tools to monitor the business and technology changes that could impact the architecture, as well as the maturity of the Architecture Capability.	
Manage risks	Identify, classify, and mitigate risk(s) before business transformation.	Risk Management Plan
Provide analysis for architecture change management	Analyze information from monitoring tools, stakeholder feedback, and performance reviews.	

(*continued*)

Table 12-64. (*continued*)

Step to Perform	Description of Step	Output Generated by Step
Develop change requirements to meet performance targets	Identify and define necessary modifications to the architecture.	Changes to the architecture framework and principles (minor changes) New Request for Architecture Work (major changes)
Manage governance process	Develop policies and guidelines that outline how changes should be proposed, reviewed, and approved.	
Activate the process to implement change	Create a structured rollout plan with timelines, dependencies, and milestones.	Architecture Contract

Establish Value Realization Process: Establishing a robust value and change management process is essential for governing the evolution of an Enterprise Architecture post-deployment. This process delineates the specific scenarios that permit modifications to the Enterprise Architecture or its components and outlines the procedures for implementing such changes. Additionally, it defines the conditions that trigger a new cycle of the Architecture Development Method to develop a refreshed architecture.

The architecture change management process is intricately linked to the organization's Architecture Governance Framework (see Chapter 8) and the management of the Architecture Contract between the architecture team and business stakeholders. During Phase H, it is imperative for the

governance body to establish clear criteria to assess whether a change request necessitates a simple architecture update or the initiation of a new Architecture Development Method cycle.

An Architecture Compliance Report plays a crucial role in this process by evaluating whether proposed changes adhere to the current architecture. In cases of non-compliance, exemptions may be granted if justified appropriately. For changes with significant architectural impact, a comprehensive strategy should be devised to manage their effects.

An Architecture Compliance Report should state whether the change is compliant to the current architecture. If it is non-compliant, an exemption may be granted with valid rationale. If the change has high impact on the architecture, then a strategy to manage its impact should be defined [20].

Providing universal guidelines for establishing these criteria is challenging due to the varying risk appetites of organizations. The iterative nature of the Architecture Development Method allows an organization to continue to mature its governance body. Each time the Architecture Development Method is used, the organization is able to take a small step forward toward a higher maturity level (the use of maturity models is explained in Chapter 15). Over time, this leads to the development of criteria that are tailored to the organization's specific needs.



Define When to Change: Clearly outline the situations that allow for modifications to the Enterprise Architecture or its parts.

Set Up Procedures: Develop step-by-step methods for making these changes when the defined situations occur.

Know When to Restart: Identify the conditions that require starting a new cycle of the Architecture Development Method to create an updated architecture.

Deploy Monitoring Tools: This step entails deploying tools to monitor the business and technology changes that could impact the architecture, as well as the maturity of the Architecture Capability. In practice, tools such as a change management application are often used for this in combination with project management methods such as PRINCE2 or service management methods such as ITIL. To monitor the maturity of the Architecture Capability, maturity models provide a convenient method.



Choose the Right Tools: Select tools that can track changes in the business and technology environments. Change management applications are commonly used for this purpose.

Integrate with Management Methods: Combine these tools with established project management methods like PRINCE2, or service management frameworks such as ITIL, to ensure comprehensive oversight.

Assess Architecture Maturity: Utilize maturity models to evaluate and monitor the development of the Architecture Capability. These models provide a structured way to measure progress and identify areas for improvement.

Manage Risks: Embarking on an architecture or business transformation inherently involves risks. Identifying, classifying, and mitigating these risks before initiation is crucial for effective tracking throughout the transformation process.

Continuous risk monitoring is performed during Phase H with respect to changes to the architecture. The Architecture Change Management Phase identifies risks associated with new requirements, technologies, or business changes. It implements risk response mechanisms as part of change management. Risk Management is further detailed in Chapter 14, Section 14.8.



Monitor Continuously: Keep a constant watch on the transformation process, as new risks can emerge unexpectedly, such as those from mergers or acquisitions.

Establish Governance: Set up a governance framework to formally accept and manage risks, ensuring they are addressed appropriately throughout the transformation.

Adapt to Changes: During the Architecture Change Management Phase, identify risks associated with new requirements, technologies, or business changes, and implement appropriate responses as part of change management.

Provide Analysis for Architecture Change Management: During this step, it is important to gather information from monitoring tools, stakeholder feedback, and performance reviews. Based on the information gathered, various scenario analyses and impact assessments can be performed to understand the potential impact of the changes to the architecture.

Collaboration with key stakeholders is very important. Work with business units, IT, and external partners to validate the results of the analyses. Then present the findings in a format that decision-makers can use to drive action. Useful formats include executive summaries and heat maps.

Remember that change analysis is not a one-time activity. It needs to be an ongoing process. Therefore, it is important to create feedback loops. This will allow the organization to establish a mechanism where architectural changes are continually reviewed and improved.



Gather Information: Begin by collecting data from various sources such as monitoring tools, stakeholder feedback, and performance reviews. This comprehensive information forms the foundation for informed decision-making.

Perform Scenario Analyses and Impact Assessments: Utilize the gathered data to conduct various scenario analyses and impact assessments. This helps in understanding the potential effects of proposed changes on the architecture and operations.

Collaborate with Key Stakeholders: Engage with business units, IT teams, and external partners to validate the results of your analyses. Their insights and perspectives are invaluable in ensuring the accuracy and feasibility of proposed changes.

Present Findings to Decision-Makers: Compile the analysis into actionable insights and present them in formats that facilitate decision-making. Executive summaries and heat maps are effective tools for conveying complex information succinctly.

Establish Continuous Feedback Loops: Recognize that change analysis is not a one-time activity. Implement mechanisms for ongoing review and improvement of architectural changes. Continuous feedback loops enable the organization to adapt proactively to emerging challenges and opportunities.

Develop Change Requirements to Meet Performance Targets: This step refers to identifying and defining the necessary modifications to the Enterprise Architecture to ensure it continues to support the organization's performance goals. It involves evaluating how well the current architecture meets business and IT performance targets. Ensuring that the proposed changes align with business strategies and deliver measurable value is another thing to consider.

An important input to this step are the *gap analysis results* that were created during the Architecture Development Method phases B, C, and D. The gaps described can be used to indicate what changes are needed to comply with the proposed architecture.

Changes may be made to the architecture framework itself, including the Architecture Principles. If this is the case, it may be necessary to update the Enterprise Metamodel and/or the Content Framework so that the architecture developed from that point forward reflects these changes. These modifications are often referred to as *minor changes*. If larger or more complex changes to the architecture are required, a new Request for Architecture Work must be created. This document is used to define and describe the *major changes* that trigger a new cycle of the Architecture Development Method.

Essentially, developing change requirements ensures that the architecture remains agile, continuously evolving to meet business and technical performance expectations.

Too often, organizations are left with an Enterprise Architecture that fit yesterday's organization, but may not provide sufficient capabilities to meet the needs of today's and tomorrow's organization.

In many cases, the architecture still fits, but the underlying solutions may not, and some changes are required. The Enterprise Architect must be aware of these change requirements and view them as an essential part of the ongoing renewal of the architecture [20].



Use Gap Analysis Results: Use the results from previous analyses (specifically from phases B, C, and D of the Architecture Development Method) to pinpoint discrepancies between the current architecture and desired outcomes.

Identify Necessary Changes: Based on the gap analysis, determine the specific modifications required to align the architecture with performance targets. Ensure these changes are in harmony with the organization's business strategies and are expected to deliver measurable benefits.

Update Architecture Framework: If the identified changes affect foundational elements like Architecture Principles, update relevant documents such as the Enterprise Metamodel and Content Framework to reflect these adjustments.

Document Major Changes: For significant or complex modifications, prepare a Request for Architecture Work document. This formalizes the proposed changes and may initiate a new cycle of the Architecture Development Method.

Manage Governance Process: To manage the change governance process, it is advisable to define who approves changes to the architecture and what criteria will be used to evaluate them. A necessary step is to develop policies or guidelines that outline how changes should be proposed, reviewed, and approved. Involve the Architecture Board in this process, as it can help to monitor changes. Governance should always be aligned with organizational standards, regulatory requirements, and best practices. Use scorecards or dashboards to track governance effectiveness and make necessary adjustments.



Identify Decision-Makers: Determine who will be responsible for approving changes to the architecture. This could be an individual or a group, such as an Architecture Board.

Set Evaluation Criteria: Define clear criteria that will be used to assess proposed changes. This ensures that every change is evaluated consistently and objectively.

Develop Policies and Guidelines: Create comprehensive policies or guidelines that outline the procedures for proposing, reviewing, and approving changes. This provides a structured approach and sets clear expectations for all stakeholders.

Engage the Architecture Board: Involve the Architecture Board in the governance process. Their role is to oversee and monitor changes, ensuring that each modification aligns with the organization's strategic objectives and architectural principles.

Align with Standards and Regulations: Ensure that the governance process adheres to organizational standards, complies with regulatory requirements, and incorporates industry best practices. This alignment helps maintain consistency and legal compliance.

Monitor and Adjust: Utilize tools like scorecards or dashboards to track the effectiveness of the governance process. Regularly review these metrics and make necessary adjustments to improve and adapt to evolving organizational needs.

Activate the Process to Implement Change: Once all the necessary changes to the architecture have been formally approved, it is essential to create a structured rollout plan with timelines, dependencies, and milestones. This will help ensure that all stakeholders (executives, IT teams, end users) understand the changes and their role in the process. Communicating with key stakeholders has proven to be essential in a number of different areas, and this is no exception. Communication is key.

Implementing changes is best done in a phased approach. This will almost always minimize disruption and allow for adjustments if needed. Be sure to continually track the results of the change and refine the implementation approach as needed.



Get Approval for Changes: Before initiating any modifications, ensure that all proposed changes have received formal approval from the relevant decision-makers.

Create a Detailed Rollout Plan: Develop a comprehensive plan that outlines the timeline, identifies dependencies, and sets clear milestones. This plan will serve as a roadmap for the implementation process.

Inform All Stakeholders: Communicate the details of the upcoming changes to everyone involved, including executives, IT teams, and end users. Clearly explain their roles and responsibilities to ensure alignment and understanding.

Implement Changes in Phases: Adopt a phased approach to roll out the changes. This strategy helps minimize disruptions and provides opportunities to make adjustments as necessary.

Monitor and Adjust: Continuously track the outcomes of each phase. Use the insights gained to refine the implementation strategy and address any issues promptly.

Each of the steps described in this phase requires continuous engagement, data-driven decision-making, and an iterative mindset. The key is not just to define a process but to ensure it remains flexible, responsive, and aligned with evolving business needs.

In short, Phase H (Architecture Change Management) is about identifying the drivers for change. This means identifying internal and external factors that require change, such as shifts in business strategy, technological advances, or regulatory requirements. It is essential to formulate change proposals that detail the rationale, expected benefits,

and impact of the changes. The proposed changes must then go through a thorough analysis process. This is where risk, cost, and impact on existing architectures are determined.

The next step is to follow established governance processes to approve or reject the proposed changes, always involving relevant stakeholders. A detailed plan for implementing the approved changes is then developed. This plan must include timelines, resources, and responsibilities. During the implementation of the changes, it is imperative to ensure effective communication and coordination among the teams. After the changes have been implemented, the results must be monitored and feedback gathered. The feedback is necessary to make informed adjustments as needed.

The importance of architecture change management is that it enables the organization to adapt quickly to market or technology changes, ensuring that the architecture can support evolving business needs. Change management provides a method for systematically evaluating and managing change. As a result, organizations can reduce the risks associated with architecture changes.

And, just as importantly, architecture change management helps maintain alignment between the architecture and business goals, ensuring that the architecture delivers maximum business value.

12.12. Requirements Management

The Requirements Management Phase plays a vital role in ensuring that the architecture development process aligns with the business needs and stakeholder expectations.

As the name already suggests, the Requirements Management Phase is all about identifying requirements. This entails gathering and documenting requirements from stakeholders, which ensures a comprehensive understanding of their needs and expectations. These requirements then need to be managed so they can be made available when other phases of the Architecture Development Method call on them. It is important to

establish a system for tracking and managing requirements throughout the entire Architecture Development Method cycle. This includes categorizing, prioritizing, and versioning requirements as needed.

Requirements need to be continuously refined and clarified based on feedback and evolving business needs. This may involve engaging with stakeholders to validate and adjust requirements as necessary. Make sure to verify that the requirements align with the overall architecture vision and objectives by assessing how changes in requirements might impact the architecture. Sharing requirements with relevant stakeholders will ensure transparency and buy-in. Clear communication helps to manage expectations and reduce misunderstandings.

12.12.1 Capture, Manage, and Refine Requirements

Although not considered a separate phase, Requirements Management is a continuous activity that runs throughout the Architecture Development Method. It involves *managing and maintaining requirements* and ensuring that they are properly addressed during each phase. As the architecture evolves, new requirements may emerge, and existing requirements may change. Effective requirements management ensures that the final architecture meets all the necessary criteria.

The Requirements Management Phase is designed to capture, manage, and refine the requirements that inform and drive the architecture development process throughout the Architecture Development Method cycle. This ensures that the architecture is aligned with the business goals and objectives, something that cannot be stressed enough.

The key objective of this phase is to make sure that the requirements management process is available and applied during all phases of the Architecture Development Method. This phase ensures that all requirements that are identified during an Architecture Development Method cycle – or even during a specific phase of a cycle – are thoughtfully managed.

Table 12-65 lists all the steps to be performed during the Requirements Management Phase and during other phases of the Architecture Development Method.

Table 12-65. *Steps to Perform Requirements Management*

No.	Requirements Management Phase Step	Architecture Development Method Phase Step
1		Identify and document requirements.
2	Establish baseline requirements and document them in the Requirements Repository.	
3	Monitor baseline requirements.	
4		Identify new and changed requirements.
5	Identify changed requirements and record priorities.	
6		Assess impact of changed requirements on current and previous phases.
7		Implement requirements arising from Phase H.
8	Update the Architecture Requirements Repository with information relating to the changes requested.	
9		Implement change in the current phase.
10		Assess and revise gap analysis for past phases.

1. **Identify and Document Requirements:** Phase A (Architecture Vision) outlines the architecture to be developed. All requirements from the previous phase are recorded in the Requirements Repository of the Architecture Repository (see Figure 12-55).

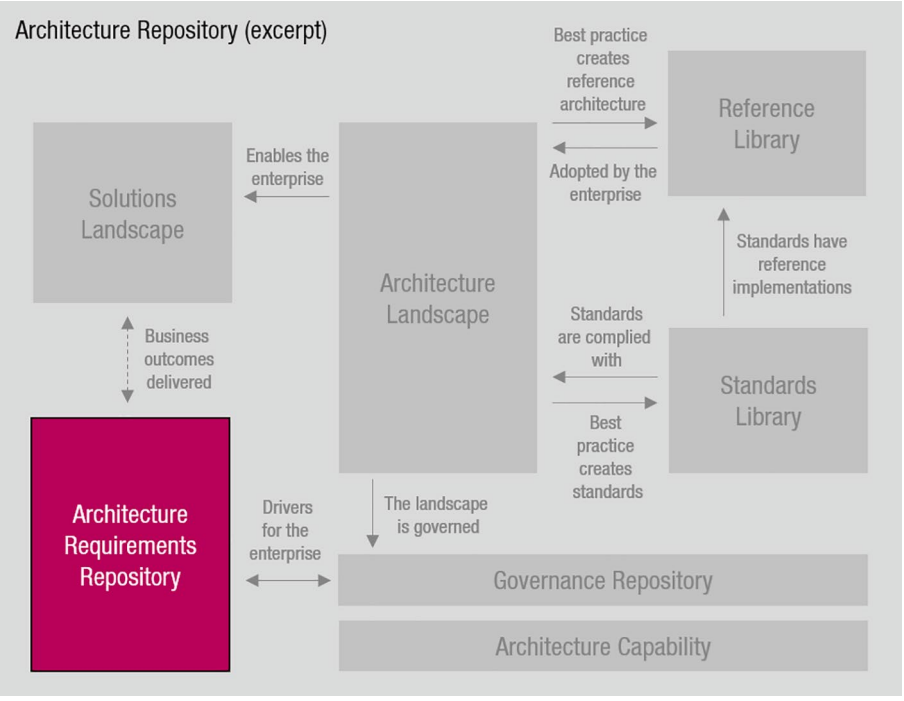


Figure 12-55. *Excerpt of the Architecture Repository, showing the Requirements Repository*

2. **Establish Baseline Requirements:** Performing one or more iterations of the Architecture Development Method during phases B, C, and D identifies the requirements for the Business, Data, Application, and Technology Architectures to be created. During Phase E (Opportunities and Solutions), the

interoperability requirements for the architecture segments are identified. All of the requirements from the above phases are also recorded in the Requirements Repository of the Architecture Repository (see Figure 12-55).

3. **Monitor Baseline Requirements:** This step involves the continuous tracking and management of the established requirements that serve as a foundation for the architecture. This process ensures that the requirements remain relevant, valid, and aligned with business objectives throughout the architecture development lifecycle. Requirements are actively tracked to detect any changes due to evolving business needs, regulations, or technological advancements.

4. **Identify New and Changed Requirements:** Based on the priorities that have been set, requirements should be added, or existing requirements should be changed. As the priorities change, the requirements should reflect those changes. It is a constant game of re-evaluating and updating requirements as business priorities are adjusted.

5. **Identify Changed Requirements and Record Priorities:** When moving through the Architecture Development Method, it is important to identify changed requirements. They must be properly prioritized by the architect working on a particular phase. This involves validating not only the

priorities but also the requirements with the relevant stakeholders. Be aware of any conflicts that may arise. They must be managed through the phases of the Architecture Development Method to a successful conclusion.

6. **Assess Impact of Changed Requirements:** If requirements change, either during the current phase or the previous phase, their impact must be assessed. It could mean that a change needs to be implemented. This change could be implemented during the current iteration of the Architecture Development Method or postponed to a later iteration.

7. **Implement Requirements Arising from Phase H:** During Phase H, any predefined changes to the architecture are implemented. In some cases, this may result in the creation of new or updated requirements. If these requirements are then validated and prioritized, they can be made available to new iterations of the Architecture Development Method and used in future architecture changes.

8. **Update the Architecture Requirements Repository:** After all requirements changes have been prioritized and validated, the Architecture Requirements Repository must be updated to contain the correct requirements.

9. **Implement Change in the Current Phase:** This is the point at which all changes to the architecture are implemented. At this point, no specific action is required from the Requirements Management Phase.

10. **Assess and Revise Gap Analysis for Past Phases:** During the gap analysis performed in phases B–D, requirements known as *gap requirements* (identifying the gaps between the Baseline and Target Architectures) may have been identified. This step ensures that they are addressed, documented, and recorded in the Architecture Requirements Repository and that the Target Architecture is revised accordingly.

Managing requirements is essential during architecture development because it ensures that the architecture remains aligned with business objectives, reduces risks, and supports successful project execution.

12.12.2 Requirements Management Artifacts

The Requirements Management Phase recognizes only a single essential deliverable (see Table 12-66). This is a comprehensive list of documented requirements, the Requirements Catalog. This catalog is often accompanied by a requirements management plan that outlines how requirements will be tracked and managed. Together, these two outputs ensure that updates to the architecture vision and models are based on refined requirements.

Table 12-66. *Artifacts created during Requirements Management*

ADM Phase	Artifacts	Used Entities
Requirements Management	Requirements Catalog	Requirement, objective, goal, assumption, constraint, gap

Requirements Catalog: The Requirements Catalog captures things that the organization needs to do to meet its objectives. It is therefore extremely important to link requirements to business goals or objectives whenever possible. This ensures that requirements can always be traced back to the realization of business goals based on organizational strategy.

Requirements generated from architecture engagements are typically implemented through change initiatives identified and scoped during Phase E (Opportunities and Solutions). Requirements can also be used as a quality assurance tool to ensure that a particular architecture is fit for purpose (i.e., the architecture can meet all identified requirements). An example of a Requirements Catalog is shown in Table 12-67.

Table 12-67. *Requirements Catalog*

Requirement	Goal
A centralized digital feedback platform needs to be made available that enables real-time collection and routing of patient satisfaction and safety-related feedback to appropriate departments.	Improve patient satisfaction. Enhance patient safety.
Clinical workflows must be digitized and integrated into the hospital's EHR system, allowing automated role-based task assignments.	Optimize clinical workflows. Increase staff productivity.
A cost-tracking dashboard must be implemented that links departmental spending to patient outcomes, enabling quarterly reviews of cost-effectiveness.	Reduce operational costs.

Requirements management is not a one-time activity. It is continuous throughout the Architecture Development Method cycle. As the architecture evolves, requirements may change, and ongoing management ensures that the architecture remains relevant and effective.

The Requirements Management Phase in the Architecture Development Method is essential for aligning the architecture with business goals, ensuring that stakeholder needs are met, and maintaining the integrity of the architecture throughout its lifecycle.

12.13. The Architecture Development Method in Short

A lot happens during the implementation of the Architecture Development Method. This is not surprising since it is the heart of the TOGAF Standard. Although the Architecture Development Method is called an approach, it is really a logical method that groups key activities to clarify their relationships and information flow. Each phase of the Architecture Development Method consists of a number of steps (process steps) that *can* be performed.

I say *can* here deliberately, because it is certainly not the case that all the steps of the Architecture Development Method must be implemented. This is something that is often misunderstood.

One of the most common complaints or (negative) comments about the use of the TOGAF Standard is that many architects assume that everything in the Architecture Development Method has to be implemented. This is completely wrong and misguided.

The whole idea behind the TOGAF Standard – and therefore the Architecture Development Method – is to provide tools and a

description of all the actions that can potentially be performed when the organization or architecture work requires or needs them.

Tailoring the framework is paramount, and it is not for nothing that the very first chapter of the TOGAF Standard refers directly to tailoring.

I would therefore like to reiterate that the Architecture Development Method in no way prescribes that all the steps must be performed. Even the order of the steps is not fixed.

In fact, the Architecture Development Method provides an iterative approach (see Chapter 13.3) that can be run over and over again, where skipping steps (e.g., when using an agile approach) is actually supported and even encouraged.

This chapter explained in detail what can happen during each phase of the Architecture Development Method. To make things easier, here is a brief summary of each phase and the corresponding steps to be performed, including a short explanation of each step.

Preliminary Phase:

- **Define the Enterprise:** By defining the organization, the scope of the impacted organization is determined, allowing architects to create an understanding of the organization and its structure.
- **Identify Key Drivers and Organizational Context:** Pinpointing the critical factors that influence the architecture, such as key drivers and the organizational context.

- **Manage Relationships Between Frameworks:** Understanding how various management frameworks interact with each other and creating a framework to manage the architecture activities.
- **Assess Enterprise Architecture Maturity:** Evaluating where the organization stands in its architectural development.
- **Establish the Enterprise Architecture Team:** Establishing an architecture team to perform the architecture work.
- **Set Architecture Principles:** Defining the principles that will guide all architecture-related efforts.
- **Choose and Tailor the Architecture Framework:** Identifying and tailoring the TOGAF Standard, Architecture Development Method, and the Content Framework.

Phase A (Architecture Vision):

- **Identify Stakeholders and Determine Requirements:** Identify stakeholders and requirements to allow for the architecture and related work to be communicated and aligned.
- **Assess Readiness for Business Transformation:** Evaluate the organization's change readiness, guiding the architecture's scope and highlighting areas requiring attention.
- **Elaborate on Business Goals and Architecture Principles:** Validate and refine the business goals and Architecture Principles identified in the previous phase.

- **Create a High-Level Architecture Vision:** Develop a high-level description of the architecture solution that will meet business requirements and outline how it aligns with the overall business strategy.

Phase B (Business Architecture),
Phase C (Information Systems Architectures),
Phase D (Technology Architecture):

- **Select Reference Models, Viewpoints, and Tools:** Choose relevant models and viewpoints from the Architecture Repository to match the business goals and stakeholder interests.
- **Develop Baseline and Target Business Architecture Description:** Describe the current and future Business, Data, Application, and Technology Architecture to a degree that supports the architecture vision, building on the Architecture Repository's resources as needed.
- **Perform Gap Analysis:** Confirm internal consistency, identify any gaps between baseline and target, and validate that models align with business objectives.
- **Define Candidate Roadmap Components:** Using the baseline, target, and gap analysis, build an initial roadmap to guide future actions and prioritization across the remaining architecture phases.
- **Resolve Impacts Across the Architecture Landscape:** Assess the Business, Data, Application, and Technology Architecture's wider impacts across the organization and its alignment with existing or planned projects.

- **Conduct Formal Stakeholder Review:** Revisit the architecture's initial motivation and make any necessary refinements to ensure it's fit to support other architecture domains.
- **Finalize the Business, Data, Application, and Technology Architecture:** Standardize building blocks, fully document them, and check alignment with business goals. Finalize documentation within the Architecture Repository.
- **Create the Architecture Definition Document:** Complete all relevant sections in the Architecture Definition Document, adding models, reports, and graphics as needed to capture a clear architectural view.

Phase E (Opportunities and Solutions):

- **Determine/Confirm Key Corporate Change Attributes:** Align the architecture with an organization's business culture. This involves creating an Implementation Factor Catalog and assessing the transition capabilities of organizational units.
- **Determine Business Constraints for Implementation:** Identify business drivers that could compromise the Target Architecture. If there are any constraints, try to resolve them.
- **Review and Consolidate Gap Analysis Results from Phases B–D:** Consolidate and integrate gap analysis results from the Business, Information Systems, and Technology Architectures.

- **Consolidate and Reconcile Interoperability Requirements:** Identify and address constraints that may hinder interoperability.
- **Refine and Validate Dependencies:** Analyze dependencies by grouping related activities.
- **Confirm Readiness and Risk for Business Transformation:** Review the Business Transformation Readiness Assessment to inform the Architecture Roadmap and implementation strategy.
- **Formulate Implementation and Migration Strategy:** Develop an Implementation and Migration Strategy by selecting a strategic approach and applying methodologies to effectively implement the Target Architecture while mitigating identified risks.
- **Identify and Group Major Work Packages:** Assess and address gaps in business capabilities in order to enable the organization to effectively plan and implement necessary changes aligned with strategic objectives.
- **Identify Transition Architecture(s):** Identify Transition Architecture(s) to provide structured, value-driven steps toward achieving the Target Architecture, tailored to the organization's implementation strategy and capacity for change.
- **Create Architecture Roadmap and Implementation and Migration Plan:** Create a structured approach to transition from Baseline to Target Architectures through a detailed Architecture Roadmap and Implementation and Migration Plan.

Phase F (Migration Planning):

- **Confirm Management Framework Interactions:** Align the Implementation and Migration Plan with an organization's management frameworks.
- **Assign Business Value to Work Packages:** Define and measure business value for work packages, identify corresponding projects, assess associated risks, and estimate business value.
- **Estimate Resource Requirements, Project Timings, and Availability/Delivery Vehicle:** Allocate resources and estimate costs for each project increment, distinguishing between capital and operational expenses, and identify cost-saving opportunities.
- **Prioritize Migration Projects Through Cost/Benefit Assessment and Risk Validation:** Determine the sequence of projects based on business priorities, technical dependencies, resource availability, and risk factors.
- **Confirm Architecture Roadmap:** Update the Architecture Roadmap, incorporating any necessary Transition Architectures.
- **Complete Implementation and Migration Plan:** Develop a detailed timeline with clear milestones for each transition stage.
- **Complete Architecture Development Cycle:** If the Architecture Capability's maturity permits, craft an Implementation Governance Model. Document lessons learned.

Phase G (Implementation Governance):

- **Confirm Scope and Priorities:** Identify and address gaps within the existing architecture. Review the Work Package Portfolio Map to determine which initiatives lead to optimal value extraction.
- **Identify Deployment Resources and Skills:** Identify the tools needed and the skills to use them.
- **Guide Development of Solutions Deployment:** Map out the scope of the project. Define the strategic requirements, as well as the various change requests needed to implement the architecture.
- **Perform Architecture Compliance Reviews:** Regularly assess the implementation activities to ensure that they are consistent with the architectural standards and principles.
- **Implement Business and IT Operations:** Implement business and IT service delivery, skills development, and training.
- **Perform Post-implementation Review:** Examine the end result of the architecture work. If there are deviations, submit change requests to correct them.

Phase H (Architecture Change Management):

- **Establish Value Realization Process:** Implement a change management process to govern modifications to the architecture in order to align changes with business value and maintain architectural integrity.

- **Deploy Monitoring Tools:** Deploy tools to monitor the business and technology changes that could impact the architecture, as well as the maturity of the Architecture Capability.
- **Manage Risks:** Identify, classify, and mitigate risk(s) before business transformation.
- **Provide Analysis for Architecture Change Management:** Analyze information from monitoring tools, stakeholder feedback, and performance reviews.
- **Develop Change Requirements to Meet Performance Targets:** Identify and define necessary modifications to the architecture.
- **Manage Governance Process:** Develop policies and guidelines that outline how changes should be proposed, reviewed, and approved.
- **Activate the Process to Implement Change:** Create a structured rollout plan with timelines, dependencies, and milestones.

Requirements Management:

- **Identify and Document Requirements:** Record all requirements from the Architecture Vision Phase in the Requirements Repository of the Architecture Repository.
- **Establish Baseline Requirements:** Document the baseline requirements created during phases B, C, D, and E in the Requirements Repository of the Architecture Repository.

- **Monitor Baseline Requirements:** Continuously track and manage the established requirements.
- **Identify New and Changed Requirements:** Add or change new and existing requirements.
- **Identify Changed Requirements and Record Priorities:** Identify and prioritize changed requirements. This step involves validating the requirements with the relevant stakeholders. Resolve any conflicts that may arise.
- **Assess Impact of Changed Requirements:** Changes to requirements might implicate a change to the architecture. Determine whether to implement new changes during the current or later iterations of the Architecture Development Method.
- **Implement Requirements Arising from Phase H:** Make new or updated requirements available for future architecture changes.
- **Update the Architecture Requirements Repository:** Record all changed, prioritized, and validated requirements in the Architecture Requirements Repository.
- **Implement Change in the Current Phase:** No specific action is required from the Requirements Management Phase.
- **Assess and Revise Gap Analysis for Past Phases:** Identify gap requirements and revise the Target Architecture as needed.

The summary of the steps per phase of the Architecture Development Method are visualized in Figure 12-56.

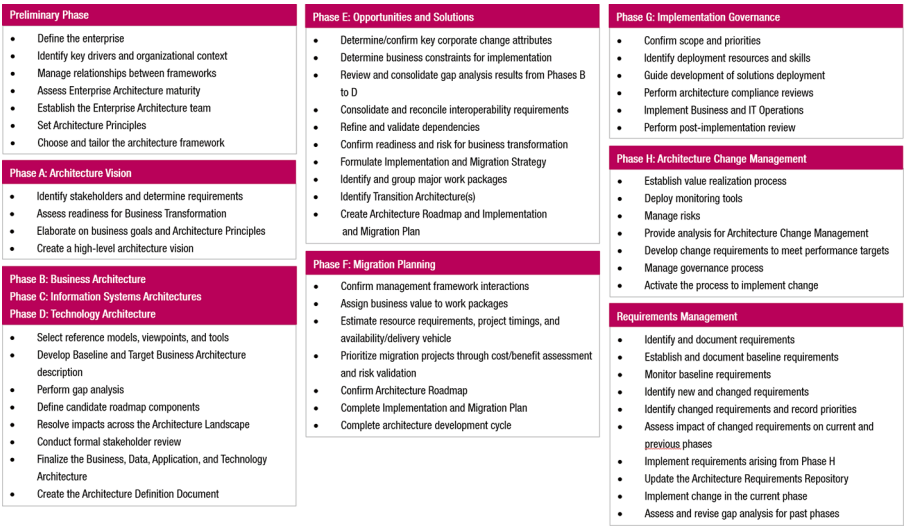


Figure 12-56. *Summary of the steps of the Architecture Development Method*

A large, poster-size version of Figure 12-56 can be downloaded from https://eawheel.com/books/media/.

12.14. Summary

This chapter has covered the Architecture Development Method, the heart of the TOGAF Standard. Often confused with a process, the Architecture Development Method is a logical method that groups key activities to clarify their relationships and information flow. It is a key component of the TOGAF Standard as it provides a comprehensive and iterative approach for developing and managing Enterprise Architectures.

The Architecture Development Method consists of ten phases, each with its own distinct objectives:

- **Preliminary Phase:** The first phase of the Architecture Development Method has two unique objectives. One is to *determine the Architecture Capability* that the organization believes it needs, and the other is to *establish it* in that form.
- **Architecture Vision Phase:** The main goal of the Architecture Vision Phase is to *establish a high-level vision of the architecture* and to *ensure stakeholder alignment*. It focuses on creating a shared understanding of the architecture's goals and ensuring that the business requirements are clearly articulated and agreed upon by all stakeholders.
- **Phase B (Business Architecture), Phase C (Information Systems Architectures), Phase D (Technology Architecture):** The objectives of phases B, C, and D are twofold. The first one is to *craft the Target Architecture* to outline how the organization should function to meet business objectives. The second objective is to *identify potential components for the Architecture Roadmap*, pinpointing the gaps between Baseline and Target Architectures that must be bridged.
- **Phase E (Opportunities and Solutions):** This phase contains three key objectives. The first is to *identify the work packages* required to realize the Target Architecture. The second objective is to *determine whether an incremental approach is required* to achieve

the first objective. The third and final key objective is to *define the building blocks* needed to complete the Target Architecture.

- **Phase F (Migration Planning):** The primary objective of Phase F (Migration Planning) is to *create a detailed plan* for the implementation of the architecture.
- **Phase G (Implementation Governance):** The primary objectives of Phase G include *ensuring adherence to the established architecture*, *managing any changes* during implementation, and *continually assessing progress* to make necessary adjustments to meet the organization's objectives.
- **Phase H (Architecture Change Management):** The key objectives of the Architecture Change Management Phase are to *continuously assess changes* in the organizational strategy, business environment, and technology landscape that may impact the architecture.
- **Requirements Management Phase:** Although not considered a separate phase, Requirements Management is a continuous activity that runs throughout the Architecture Development Method. It involves *managing and maintaining requirements* and ensuring that they are properly addressed during each phase.

The Architecture Development Method describes all the actions one *can* take, not what *must* be taken. It also does not prescribe the order of these steps. The TOGAF Standard explicitly states that the Architecture Development Method is to be tailored so that it (the steps and its content) fits an organization's needs.

The Architecture Development Method delivers its iterative approach by providing a step-by-step guide for creating and maintaining architecture artifacts and ensuring alignment with business goals.

This chapter has reviewed all the phases of the Architecture Development Method. Each phase has been explained in detail, and examples of each artifact to be created have been provided.